Computing Metacells - Iterative Process

1. Setup

In this vignette we will demonstrate how to compute metacells starting with brand-new data, with minimal or no reliance on previous analysis of similar data. This is in contrast to the one-pass vignette, where magically all our decisions were correct and we could immediately compute the final result (in practice, we simply applied there the final results from the iterative process below).

In the real world, the actual process used will fall between these two extremes. That is, by using prior knowledge that applies to the specific data set, some of the decisions we make below will be correct from the start, but most likely, some still need to be revised, andn require some iterations to finalize.

To remove doubt, the results presented here are **not** suitable for use in any serious analysis. This is an example of the process, nothing more.

We'll start with importing the python libraries we'll be using and setup some global configuration. Just importing takes a few seconds, mainly because Python's importing of C++ extensions with a large number of functions is inefficient.

In [1]:
```
%%time
import anndata as ad              # For reading/writing AnnData files
import matplotlib.pyplot as plt   # For plotting
import metacells as mc            # The Metacells package
import numpy as np                # For array/matrix operations
import pandas as pd               # For data frames
import os                         # For filesystem operations
import seaborn as sb              # For plotting
import scipy.sparse as sp         # For sparse matrices
import shutil                     # for filesystem operations
from math import hypot            # For plotting
from typing import *              # For type annotations
```

```
CPU times: user 6.59 s, sys: 9.98 s, total: 16.6 s
Wall time: 5.82 s
```

Then, configure the packages we use. Feel free to tweak as necessary.

In [2]:
```
%%time

# Use SVG for scalable low-element-count diagrams.
%config InlineBackend.figure_formats = ["png"]

# A matter of personal preference.
sb.set_style("white")
```

```
# Running operations on an inefficient layout can make code **much** slower.
# For example, summing the columns of a row-major matrix.
# By default this will just be a warning.
# We set it to be an error here to make sure the vignette does not lead you
#
# Note that this only affects the Metacells package.
# Numpy will happily and silently take 100x longer for running such ineffici
# At least, there's no way I can tell to create a warning or error for this;
# also, the implementation for "inefficient" operations could be *much* fast
#
# The workaround in either case is to explicitly re-layout the 2D matrix bef
# This turns out to be much faster, especially when the matrix can be reused
# Note that numpy is also very slow when doing matrix re-layout,
# so the metacells package provides a function for doing it more efficiently
#
# Sigh.
mc.ut.allow_inefficient_layout(False)
```

```
CPU times: user 6.09 ms, sys: 0 ns, total: 6.09 ms
Wall time: 5.99 ms
```

Out[2]:
```
True
```

Since the process here is iterative, we'll use a separate directory to hold the results of each iteration, to keep things organized. In this example, we'll have 4 iterations in all, as well as the final results.

In [3]:
```
%%time
shutil.rmtree("../output/iterative", ignore_errors=True)
shutil.rmtree("../mcview/iterative", ignore_errors=True)
os.makedirs("../output/iterative/iteration-1/figures", exist_ok=True)
os.makedirs("../output/iterative/iteration-2", exist_ok=True)
os.makedirs("../output/iterative/iteration-3", exist_ok=True)
os.makedirs("../output/iterative/iteration-4", exist_ok=True)
os.makedirs("../output/iterative/final", exist_ok=True)
```

```
CPU times: user 3.58 ms, sys: 2.93 s, total: 2.93 s
Wall time: 11.3 s
```

2. Reading the data

Our input here is the "full" data. This data already isn't the "raw" data. That is, we expect it to have gone through preliminary processing using basic cell-oriented tools. For example, removing "doublet" cells, cells which are actually empty wells, etc. The specifics of such pre-processing may be different for each data set, and are outside the scope of this vignette.

Reading this data takes a ridiculously long time necause the AnnData implementation, in its infinite wisdom, does not use memory-mapping. When we switch to using `DAF` things would be much better and faster.

In [4]:
```
%%time
full = ad.read_h5ad("../blobs/hca_bm.full.h5ad")
```

```
mc.ut.top_level(full)
mc.ut.set_name(full, "hca_bm.full")
print(f"Full: {full.n_obs} cells, {full.n_vars} genes")
```

```
Full: 378000 cells, 33694 genes
CPU times: user 382 ms, sys: 27.4 s, total: 27.8 s
Wall time: 48.7 s
```

3. Cleaning the data

Having this "full" data, we will demonstrate performing some additional "universal" steps to extract the "clean" data out of it. Importantly, all we will do here is select which subset of the data to use; we will not perform any corrections (such as batch corrections) here. Such corrections are better applies based on the resulting metacell model, as this will allow cell-type-specific analysis to be done.

3.1 Decisions

Cleaning up the data (whether in pre-processing or using the steps below) requires us to make decisions, based on prior biological knowledge and experience with the pipeline. Also, none of the decisions is final. As demonstrated here, we typically iterate the analysis, revisiting and refining prior decisions, until we obtain a "good" final result. This is in contrast to the one-pass vignette where we assume all decisions are "correct" from the start.

3.1.1 Excluding cells by UMIs count

The first decision we need to make is "how many UMIs does a clean cell contain". Cells with "too few" UMIs indicate very low-quality sampling and might be just empty droplets. Cells with "too many" UMIs might be doublets. The thresholds to use depend on the technology and the dataset, and the specifics of any pre-processing done, if any. The method we use here is to look at the distribution of total UMIs per cell, and pick "reasonable" thresholds for exclusions based on it.

This operation is identical to the matching one in the one-pass vignette.

In [5]:
```
PROPERLY_SAMPLED_MIN_CELL_TOTAL = 800
PROPERLY_SAMPLED_MAX_CELL_TOTAL = 20000
```

We can visualize the cell UMIs threshold decision and adjust it accordingly:

In [6]:
```
%%time
total_umis_per_cell = mc.ut.get_o_numpy(full, "__x__", sum=True)
plot = sb.displot(total_umis_per_cell, log_scale=(10, None))
plot.set(xlabel="UMIs", ylabel="Density", yticks=[])

plot.refline(x=PROPERLY_SAMPLED_MIN_CELL_TOTAL, color="darkgreen")
plot.refline(x=PROPERLY_SAMPLED_MAX_CELL_TOTAL, color="crimson")
```

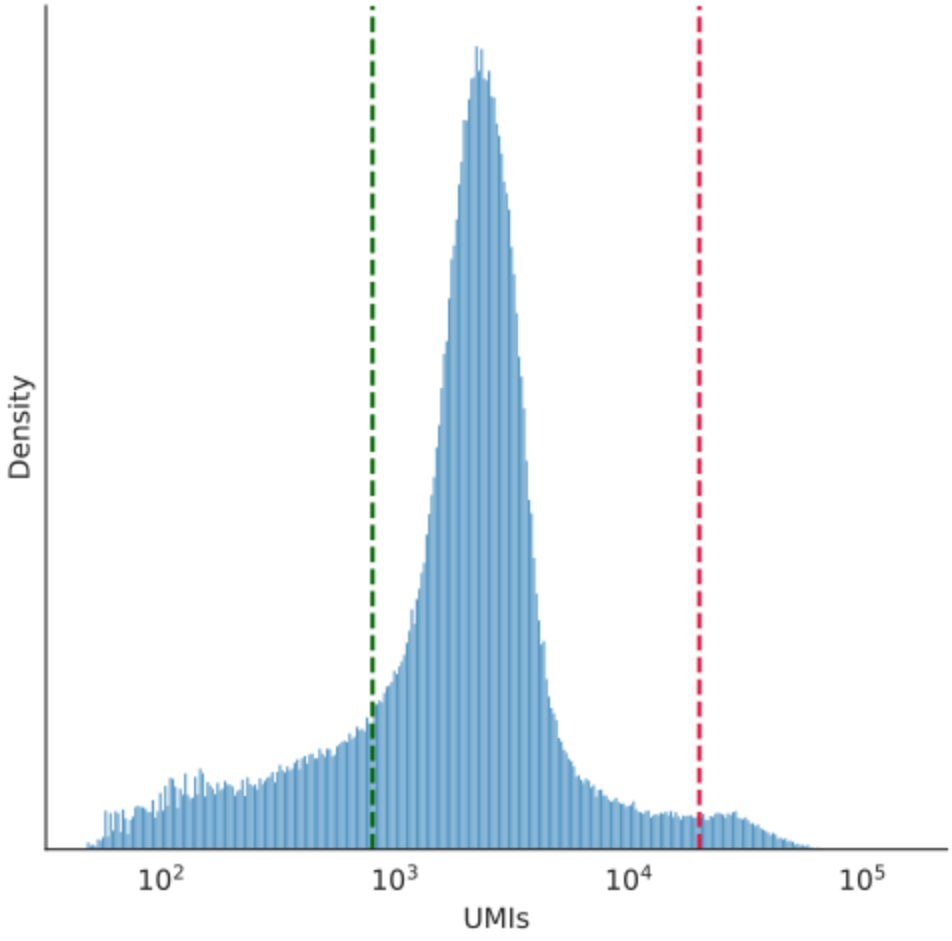
```
plt.savefig("../output/iterative/iteration-1/figures/cell_total_umis.svg")

too_small_cells_count = np.sum(total_umis_per_cell < PROPERLY_SAMPLED_MIN_CE
too_large_cells_count = np.sum(total_umis_per_cell > PROPERLY_SAMPLED_MAX_CE

total_umis_per_cell = mc.ut.get_o_numpy(full, name="__x__", sum=True)
too_small_cells_percent = 100.0 * too_small_cells_count / full.n_obs
too_large_cells_percent = 100.0 * too_large_cells_count / full.n_vars

print(
    f"Will exclude {too_small_cells_count} ({too_small_cells_percent:.2f}%%)
    f" with less than {PROPERLY_SAMPLED_MIN_CELL_TOTAL} UMIs"
)
print(
    f"Will exclude {too_large_cells_count} ({too_large_cells_percent:.2f}%%)
    f" with more than {PROPERLY_SAMPLED_MAX_CELL_TOTAL} UMIs"
)
```

```
Will exclude 66232 (17.52%%) cells with less than 800 UMIs
Will exclude 8672 (25.74%%) cells with more than 20000 UMIs
CPU times: user 5.2 s, sys: 575 ms, total: 5.77 s
Wall time: 5.14 s
```



3.1.2 Excluding genes by name

The next decision we need to make is which genes to exclude from the data, by their names. The poster children for this are mytochondrial genes and strong sex-chromosome genes.

Later on in the analysis we will revisit this, and exclude additional genes. This is in contrast to the one-pass vignette where we get right the list from the start. Realistically, one would start with a list that was used for similar data in the past, and possibly tweak it later as needed.

In [7]:
```
EXCLUDED_GENE_NAMES = ["XIST", "MALAT1"]  # Sex-specific genes.
EXCLUDED_GENE_PATTERNS = ["MT-.*"]        # Mitochondrial.
```

We'll instruct the package to exclude the genes we have chosen. By default, the code will also automatically exclude a few additional genes, specifically genes with very high variance that are not correlated with any other gene (`bursty_lonely_gene` , see the documentation for details). Such genes are useless for grouping cells together; even worse, including them is actively harmful as they cause cells to appear to be deviants (have a gene in which they are very different from the rest of the cells in the same metacell), and thereby be marked as outliers.

In [8]:
```
%%time
mc.pl.exclude_genes(
    full,
    excluded_gene_names=EXCLUDED_GENE_NAMES,
    excluded_gene_patterns=EXCLUDED_GENE_PATTERNS,
    random_seed=123456,
)
```

```
set hca_bm.full.var[bursty_lonely_gene]: 0 true (0%) out of 33694 bools
set hca_bm.full.var[properly_sampled_gene]: 27277 true (80.96%) out of 33694
bools
set hca_bm.full.var[excluded_gene]: 6432 true (19.09%) out of 33694 bools
CPU times: user 38.6 s, sys: 19.5 s, total: 58.1 s
Wall time: 43.5 s
```

3.1.3 Excluding cells by high excluded gene UMIs

The next decision we need to make is which cells to exclude due to containing too many UMIs in the excluded genes. If a cell contains "too many" excluded (mainly mytochondrial) gene UMIs, this may indicate a badly sampled cell, leading to very skewed results. Again, the exact threshold depends on both the technology and the dataset. Here we resort to looking at the distribution of the fraction of excluded genes in each cell, and manually picking the threshold.

We start by computing the total UMIs of excluded genes in each cell. We only need to do this once (as long as we don't change the list of excluded genes).

In [9]:
```
%%time
mc.tl.compute_excluded_gene_umis(full)
```

```
set hca_bm.full.obs[excluded_umis]: 378000 float32s
```

```
CPU times: user 5.02 s, sys: 7.27 s, total: 12.3 s
Wall time: 12.3 s
```

Next we'll pick a maximal fraction of excluded UMIs in each cell.

In [10]:
```
PROPERLY_SAMPLED_MAX_EXCLUDED_GENES_FRACTION = 0.25
```

We can visualize the cell excluded genes fraction threshold decision and adjust it accordingly:

In [11]:
```
%%time
excluded_umis_fraction_regularization = 1e-3  # Avoid 0 values in log scale
excluded_umis_per_cell = mc.ut.get_o_numpy(full, "excluded_umis")
excluded_umis_fraction_per_cell = excluded_umis_per_cell / total_umis_per_ce

excluded_umis_fraction_per_cell += excluded_umis_fraction_regularization
plot = sb.displot(excluded_umis_fraction_per_cell, log_scale=(10, None))
excluded_umis_fraction_per_cell -= excluded_umis_fraction_regularization

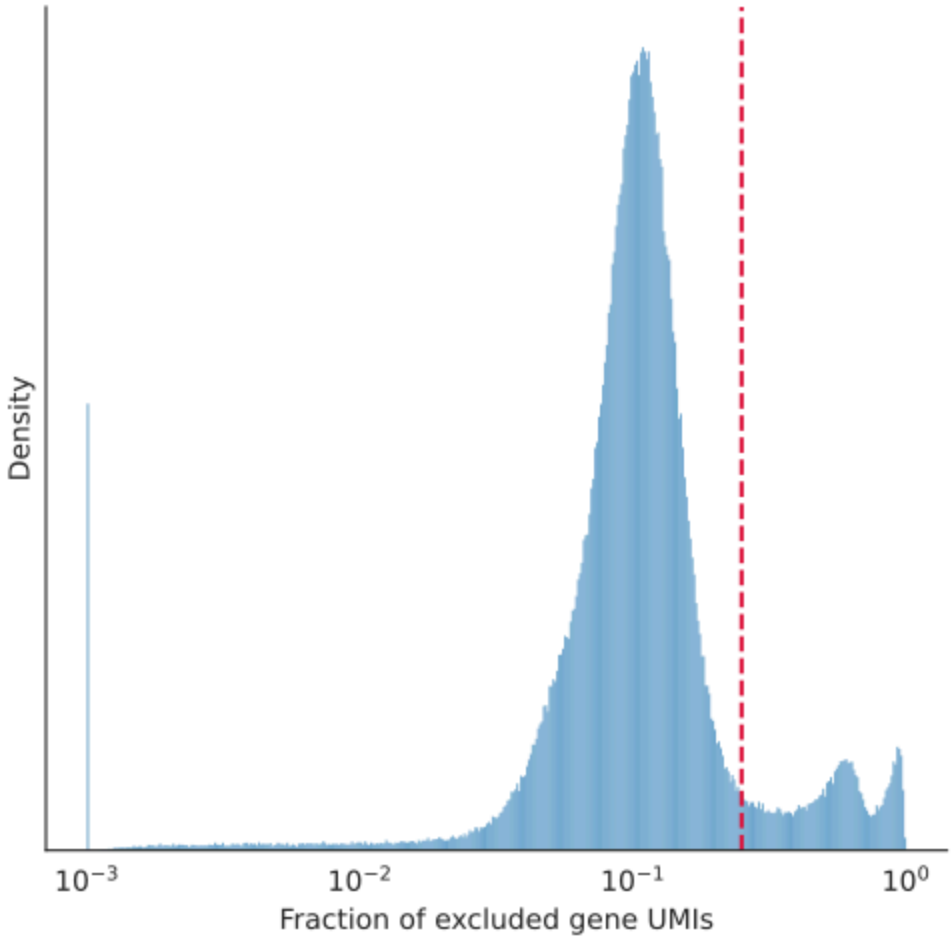
plot.set(xlabel="Fraction of excluded gene UMIs", ylabel="Density", yticks=[
plot.refline(x=PROPERLY_SAMPLED_MAX_EXCLUDED_GENES_FRACTION, color="crimson"

plt.savefig("../output/iterative/iteration-1/figures/cell_excluded_umis_frac

too_excluded_cells_count = np.sum(
    excluded_umis_fraction_per_cell > PROPERLY_SAMPLED_MAX_EXCLUDED_GENES_FR
)
too_excluded_cells_fraction = too_excluded_cells_count / len(total_umis_per_

print(
    f"Will exclude {too_excluded_cells_count} ({100 * too_excluded_cells_fra
    f" with more than {100 * PROPERLY_SAMPLED_MAX_EXCLUDED_GENES_FRACTION:.2
)
```

```
Will exclude 36306 (9.60%) cells with more than 25.00% excluded gene UMIs
CPU times: user 1.28 s, sys: 616 ms, total: 1.9 s
Wall time: 1.45 s
```



We can now instruct the package to exclude the cells we have chosen.

In [12]:
```
%%time
mc.pl.exclude_cells(
    full,
    properly_sampled_min_cell_total=PROPERLY_SAMPLED_MIN_CELL_TOTAL,
    properly_sampled_max_cell_total=PROPERLY_SAMPLED_MAX_CELL_TOTAL,
    properly_sampled_max_excluded_genes_fraction=PROPERLY_SAMPLED_MAX_EXCLUD
)
```

```
set hca_bm.full.obs[properly_sampled_cell]: 297890 true (78.81%) out of 3780
00 bools
set hca_bm.full.obs[excluded_cell]: 80110 true (21.19%) out of 378000 bools
CPU times: user 4.23 ms, sys: 267 µs, total: 4.5 ms
Wall time: 3.58 ms
```

3.2 Extract the clean data

Having decided on which cells and genes to exclude, we can now extract the clean data out of the full dataset.

In [13]:
```
%%time
clean = mc.pl.extract_clean_data(full, name="hca_bm.iteration-1.clean")
```

```
mc.ut.top_level(clean)
print(f"Clean: {clean.n_obs} cells, {clean.n_vars} genes")
```

```
set hca_bm.iteration-1.clean.obs[full_cell_index]: 297890 int32s
set hca_bm.iteration-1.clean.var[full_gene_index]: 27262 int32s
Clean: 297890 cells, 27262 genes
CPU times: user 26.7 s, sys: 22.1 s, total: 48.8 s
Wall time: 48.8 s
```

3.3 Save the data

It is good practice keep the pristine input data in case we'll need to do something else with it. We'll therefore save a copy of the "full" data with the manual annotations we have collected, as well as the "clean" data we extracted from it. We do this prior to computing metacells, which will add further annotations, so we'll be able to easily go back to the unannotated versions of the data if (realistically, when) we revise our decisions.

Using AnnData, this is a full copy of the data; when we'll switch to DAF we'll be able to just store the additional annotations we made, which would take practically no space and be much faster.

In [14]:
```
%%time
full.write_h5ad("../output/iterative/iteration-1/hca_bm.full.h5ad")
full = None  # Alow it to be gc-ed
```

```
CPU times: user 665 ms, sys: 28.6 s, total: 29.3 s
Wall time: 1min 22s
```

In [15]:
```
%%time
clean.write_h5ad("../output/iterative/iteration-1/hca_bm.clean.h5ad")
```

```
CPU times: user 313 ms, sys: 21.5 s, total: 21.8 s
Wall time: 1min 4s
```

4. Compute the 1st iteration metacells

We'll rename our data to "cells" to distinguish it from the metacells we are about to compute. We'll be adding annotations to this data (most importantly, the metacell each cell belongs to), and will later save it separately from the "clean" data we saved above.

In [16]:
```
%%time
cells = clean
clean = None  # Allow it to be gc-ed
mc.ut.set_name(cells, "hca_bm.iteration-1.cells")
print(f"Iteration 1: {cells.n_obs} cells, {cells.n_vars} genes")
```

```
Iteration 1: 297890 cells, 27262 genes
CPU times: user 33 µs, sys: 54 µs, total: 87 µs
Wall time: 79.6 µs
```

4.1 Decisions

Even though we have the clean cells data, we can't compute metacells for it before making a few more decisions.

4.1.2 Lateral genes

A crucial decision when running metacells is the list of genes are lateral, that is, should **not** be used to group cells together. The poster child for this are cell-cycle genes. These genes are strong and any clustering algorithm will therefore prefer to group together cells in the same cell-cycle state, at the expense of mixing up (reasonably close) other cell states, which are what we are actually interested in. Note that lateral genes are still used in deviant cell detection, that is, each lateral gene should still have a consistent level in all the cells in each metacell.

Since our data is brand-new, we don't have a good comprehensive list of lateral genes for it, so we'll need to come up with an initial one, with the expectation we'll be adding more genes down the line during the iterative process.

To create this initial list, we'll start with a very short list of "base" lateral genes, using biological prior knowledge. The list below is not universal (it is specific to human cells, for one thing), but it is a start (cell cycle, stress, ribosomal). We'll then expand it with correlated genes.

Realistically, one would start with a list that was used for similar data in the past, skip the correlated genes expannsion, and possibly tweak it later as needed. However we show here the process staring with no similar data at all, so this (admittedly tedious) step is required.

In [17]:
```
BASE_LATERAL_GENE_NAMES = [
    "AURKA", "MCM3", "MCM4", "MCM7", "MKI67", "PCNA", "RRM2", "SMC4", "TPX2"
    "FOS", "HSP90AB1", "TXN",
]
BASE_LATERAL_GENE_PATTERNS = ["RP[LS].*"]  # Ribosomal
```

In [18]:
```
%%time

# We'll reuse this through the iterations.
# It is just a thin wrapper for mark_lateral_genes,
# and optionally also shows the results.
def update_lateral_genes(
    *,
    names: List[str] = [],
    patterns: List[str] = [],
    op: str = "set",
    show: bool = True
) -> None:
    mc.pl.mark_lateral_genes(
        cells,
```

```
        lateral_gene_names=names,
        lateral_gene_patterns=patterns,
        op=op
    )

    if not show:
        return

    lateral_genes_mask = mc.ut.get_v_numpy(cells, "lateral_gene")
    lateral_gene_names = set(cells.var_names[lateral_genes_mask])

    print(sorted([
        name for name in lateral_gene_names
        if not name.startswith("RPL") and not name.startswith("RPS")
    ]))

    print(f"""and {len([
        name for name in lateral_gene_names if name.startswith("RPL") or nam
    ])} RP[LS].* genes""")

update_lateral_genes(names=BASE_LATERAL_GENE_NAMES, patterns=BASE_LATERAL_GE
```

```
set hca_bm.iteration-1.cells.var[lateral_gene]: 115 true (0.4218%) out of 27
262 bools
['AURKA', 'FOS', 'HSP90AB1', 'MCM3', 'MCM4', 'MCM7', 'MKI67', 'PCNA', 'RRM
2', 'SMC4', 'TPX2', 'TXN']
and 103 RP[LS].* genes
CPU times: user 18.6 ms, sys: 337 µs, total: 18.9 ms
Wall time: 18.4 ms
```

Since our initial list was very partial, we would like to extend it to include any genes highly-correlated with them (as long as this makes sense, biologically speaking).

Manually looking at all 20-30,000 genes to see whether they are correlated with the base genes is obviously not practical. We therefore provide a function that tries to reduce the amount of genes we need to look at, by computing correlations between genes and clustering them to "related gene modules". We can then examine only such modules that contain base lateral genes (or that are correlated with such modules).

When we do this, we conservatively include any genes that we *might* want to add to the lateral genes list (that is, lean heavily in favor of "false positive" errors). We expect the vast majority of the candidates we look at to *not* be added as lateral genes; what is more important is that we do not miss any gene that we *will* want to add.

In [19]:
```
%%time
mc.pl.relate_to_lateral_genes(cells, random_seed=123456)
```

```
set hca_bm.iteration-1.cells.var[lateral_genes_module]: 24299 outliers (89.1
3%) and 2963 grouped (10.87%) out of 27262 int32 elements with 76 groups wit
h mean size 38.99
set hca_bm.iteration-1.cells.varp[lateral_genes_similarity]: csr_matrix 2726
2 X 27262 float32s (8779369 > 0, 1.181%)
```

```
CPU times: user 5min 21s, sys: 2.36 s, total: 5min 23s
Wall time: 16.9 s
```

This correlates all the genes with each other, and clusters them. This is still a lot of data, so let's condense it to the correlation between the overall related gene modules (clusters).

In [20]:
```
%%time
base_lateral_genes_mask = mc.ut.get_v_numpy(cells, "lateral_gene")
base_lateral_gene_names = set(cells.var_names[base_lateral_genes_mask])

module_per_gene = mc.ut.get_v_series(cells, "lateral_genes_module")
base_lateral_gene_modules = np.unique(module_per_gene.values[base_lateral_ge
base_lateral_gene_modules = set(base_lateral_gene_modules[base_lateral_gene_

genes_per_module = np.unique(module_per_gene.values, return_counts=True)[1][
similarity_of_modules = mc.ut.get_vv_proper(cells, "lateral_genes_similarity
similarity_of_modules = mc.ut.sum_groups(similarity_of_modules, module_per_g
similarity_of_modules = mc.ut.to_layout(similarity_of_modules, layout="colum
similarity_of_modules = \
    mc.ut.sum_groups(similarity_of_modules, module_per_gene.values, per="col
similarity_of_modules /= genes_per_module[:, np.newaxis] * genes_per_module[

module_names = [
    f"(*) {gene_module}" if gene_module in base_lateral_gene_modules else st
    for gene_module in range(np.max(module_per_gene.values) + 1)
]
similarity_of_modules = pd.DataFrame(similarity_of_modules, index=module_nam
```

```
CPU times: user 10.1 ms, sys: 1min 57s, total: 1min 57s
Wall time: 2min 4s
```

We can visualize this to use as a coarse guide when looking into the specific modules:

In [21]:
```
%%time
size = similarity_of_modules.shape[0]
if size > 50:
    sb.set(font_scale=50 / size)
size = size * 0.15 + 1
cm = sb.clustermap(
    similarity_of_modules,
    figsize=(size, size),
    vmin=0, vmax=0.5,
    xticklabels=True, yticklabels=True,
    dendrogram_ratio=0.1,
    cmap="YlGnBu",
)
cm.fig.suptitle("Gene Modules Summary", fontsize=10)

plt.savefig("../output/iterative/iteration-1/figures/genes_modules_correlati

plt.show()
sb.set(font_scale=1.0)
```




```
CPU times: user 4.21 s, sys: 13.5 s, total: 17.7 s
Wall time: 12.7 s
```

We now visually inspect each of the potentially relevant gene modules, to decide on which additional genes to list as lateral. We'll generate a disk file for the figure of each module for offline inspection, but only show here in the notebook the gene modules that contain lateral genes, or are potentially correlated with other gene module(s) that contain lateral genes. These would still be "quite a few" modules to go through. For each one, we need to decide whether we want to ignore the module, add all its genes as lateral, or possibly add just a few of its genes as lateral. This depends on biological knowledge, and often requires looking at some resources (such as genecards or even published papers) to make the choice.

In [22]:
```
%%time
SHOW_CORRELATED_MODULES = 0.2 # Show non-lateral modules if correlated to la
MIN_SIMILARITY_TO_SHOW = 0.2 # Show modules only if there's at least this co

base_lateral_gene_modules = np.unique(module_per_gene.values[base_lateral_ge
```

```
base_lateral_gene_modules = base_lateral_gene_modules[base_lateral_gene_modu
similarity_of_genes = mc.ut.get_vv_frame(cells, "lateral_genes_similarity")

for gene_module in range(np.max(module_per_gene) + 1):
    module_genes_mask = module_per_gene.values == gene_module
    similarity_of_module = similarity_of_genes.loc[module_genes_mask, module
    similarity_of_module.index = similarity_of_module.columns = [
        "(*) " + name if name in base_lateral_gene_names else name
        for name in similarity_of_module.index
    ]

    mask = similarity_of_module.values.copy()
    np.fill_diagonal(mask, 0.0)
    max_value = np.max(mask)
    show_in_notebook = gene_module in base_lateral_gene_modules or max_value

    similarity_to_laterals = similarity_of_modules.iloc[gene_module, base_la
    similar_lateral_modules_mask = similarity_to_laterals >= SHOW_CORRELATED
    similar_lateral_modules = base_lateral_gene_modules[np.where(similar_lat
    if gene_module not in base_lateral_gene_modules and len(similar_lateral_
        show_in_notebook = False

    prefix = "(*) " if gene_module in base_lateral_gene_modules else ""
    suffix = ", ".join([
        str(similar_lateral_module)
        for similar_lateral_module
        in similar_lateral_modules
        if similar_lateral_module != gene_module
    ])
    if suffix != "":
        suffix = " ~ " + suffix
    title = f"{prefix}Gene Module {gene_module}{suffix}"

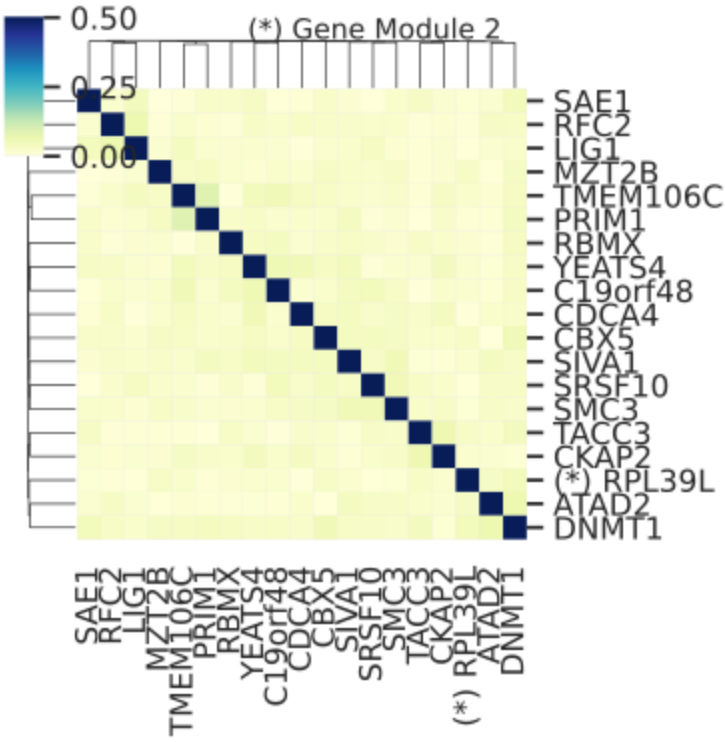
    if len(similar_lateral_modules) > 0:
        with_lateral_modules = set(similar_lateral_modules)
        with_lateral_modules.add(gene_module)
        with_lateral_modules = sorted(with_lateral_modules)
        if len(with_lateral_modules) > 1:
            similarity_with_module = \
                similarity_of_modules.iloc[with_lateral_modules, :].iloc[:,

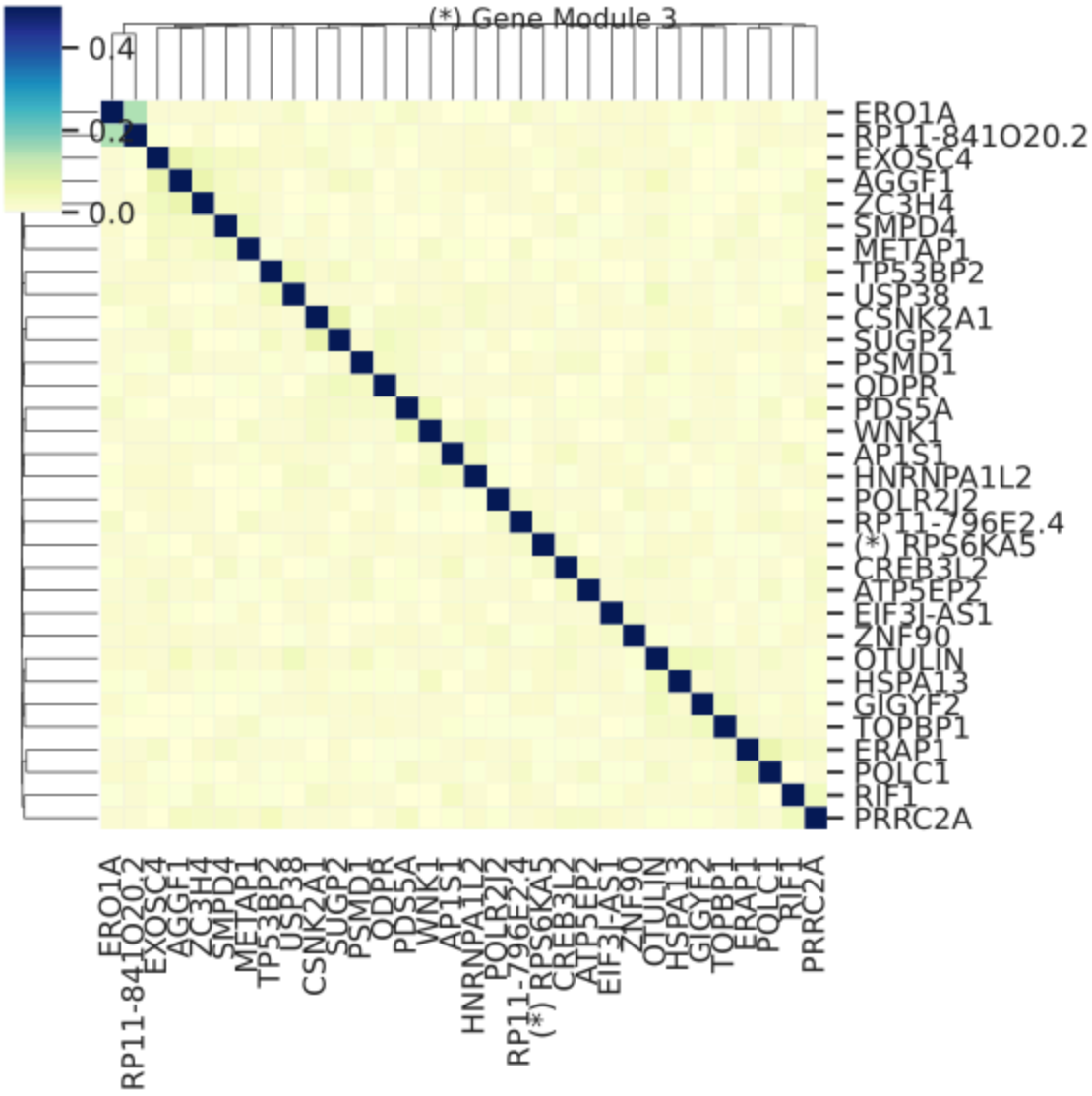
            size = similarity_with_module.shape[0]
            if size > 50:
                sb.set(font_scale=50 / size)
            size = size * 0.15 + 1
            cm = sb.clustermap(
                similarity_with_module,
                figsize=(size, size),
                vmin=0, vmax=0.5,
                xticklabels=True, yticklabels=True,
                dendrogram_ratio=0.1,
                cmap="YlGnBu",
            )
            cm.fig.suptitle(title, fontsize=10)
            plt.savefig(f"../output/iterative/iteration-1/figures/genes_modu
            if show_in_notebook:
```

```
                plt.show()
            else:
                plt.clf()

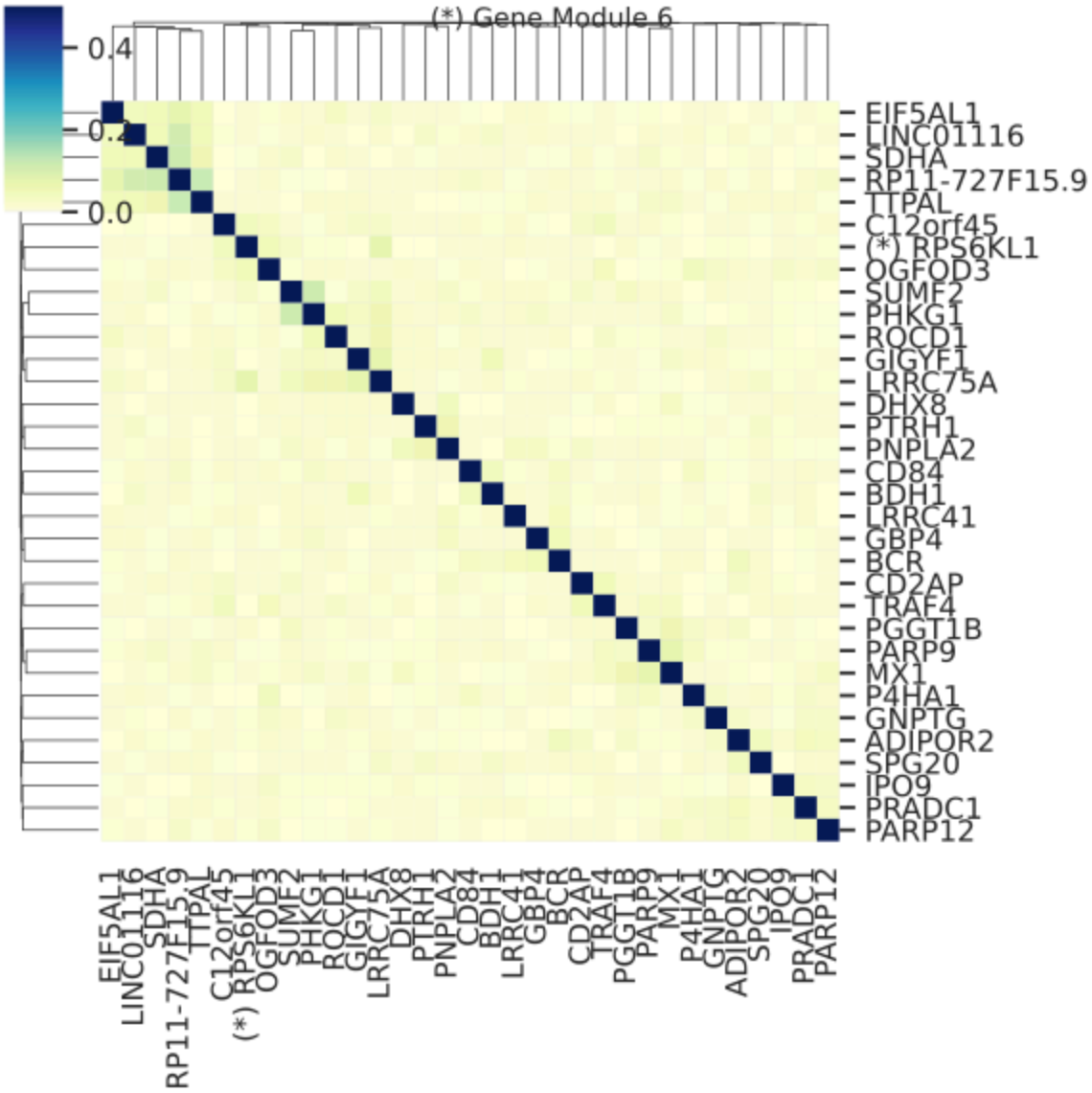
    size = similarity_of_module.shape[0]
    if size > 50:
        sb.set(font_scale=50 / size)
    size = size * 0.15 + 1
    cm = sb.clustermap(
        similarity_of_module,
        figsize=(size, size),
        vmin=0, vmax=0.5,
        xticklabels=True, yticklabels=True,
        dendrogram_ratio=0.1,
        cmap="YlGnBu",
    )
    cm.fig.suptitle(title, fontsize=10)
    plt.savefig(f"../output/iterative/iteration-1/figures/genes_module_{gene
    if show_in_notebook:
        plt.show()
    else:
        plt.clf()
```

```
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<Figure size 180x180 with 0 Axes>
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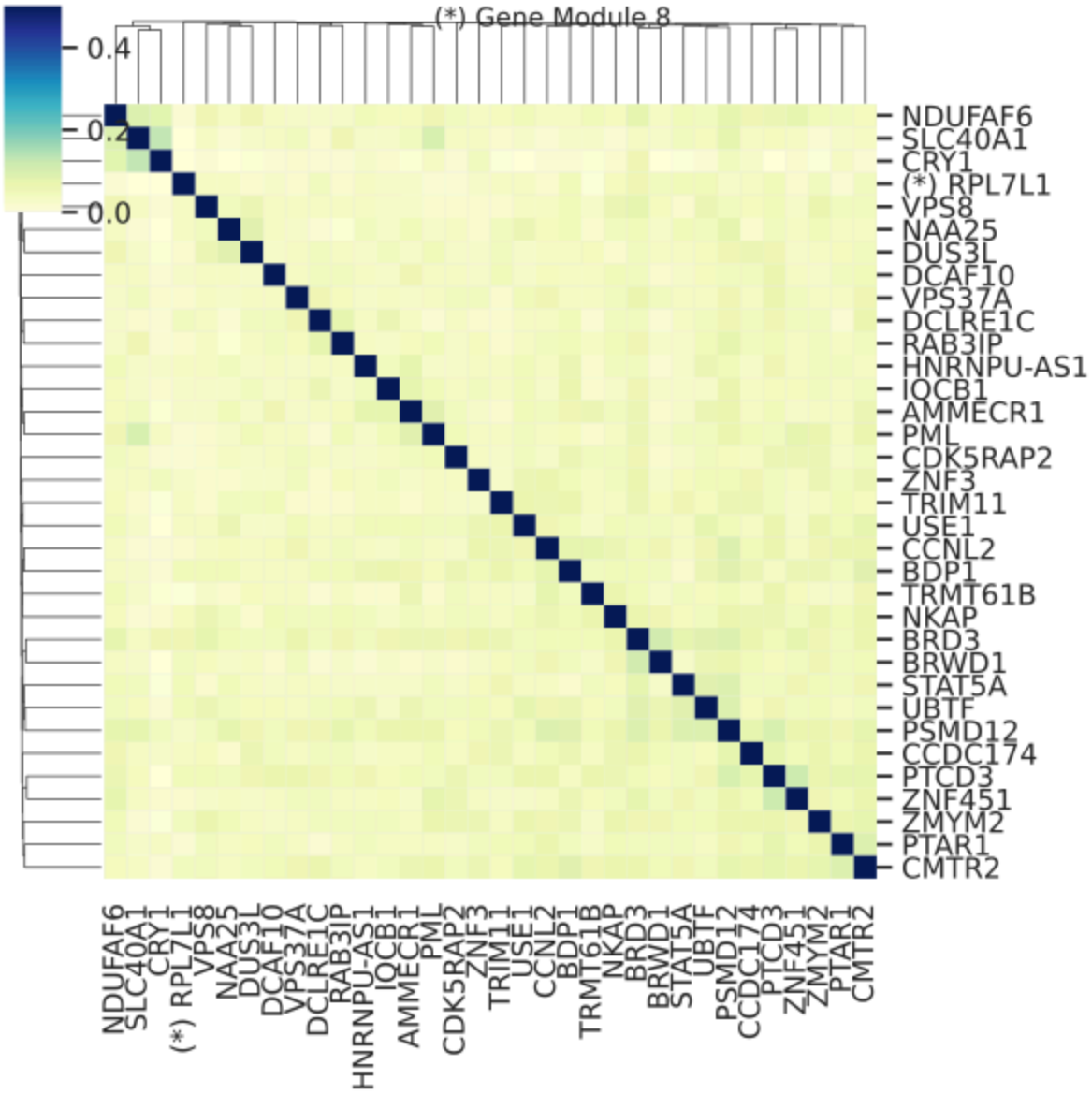


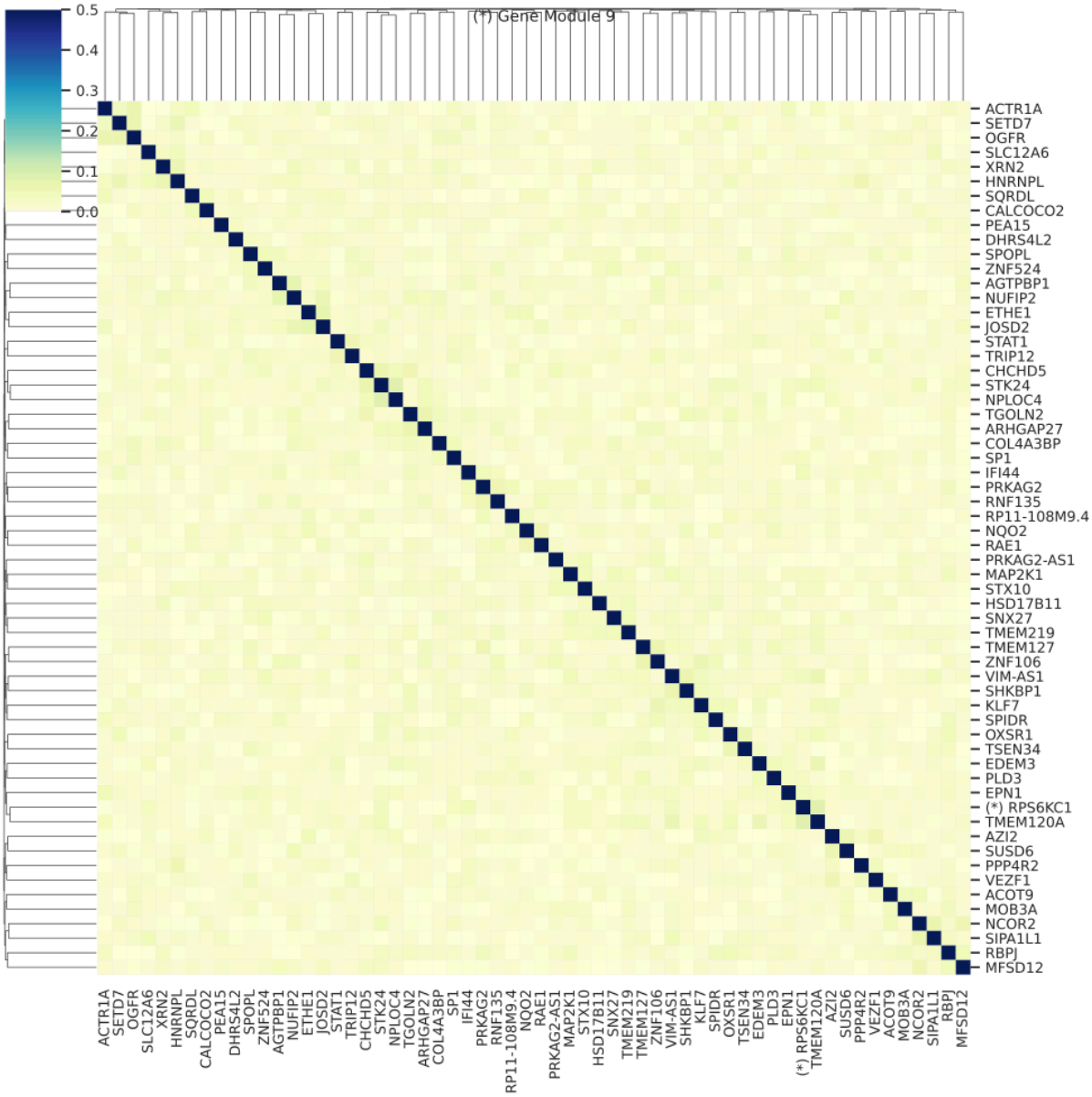


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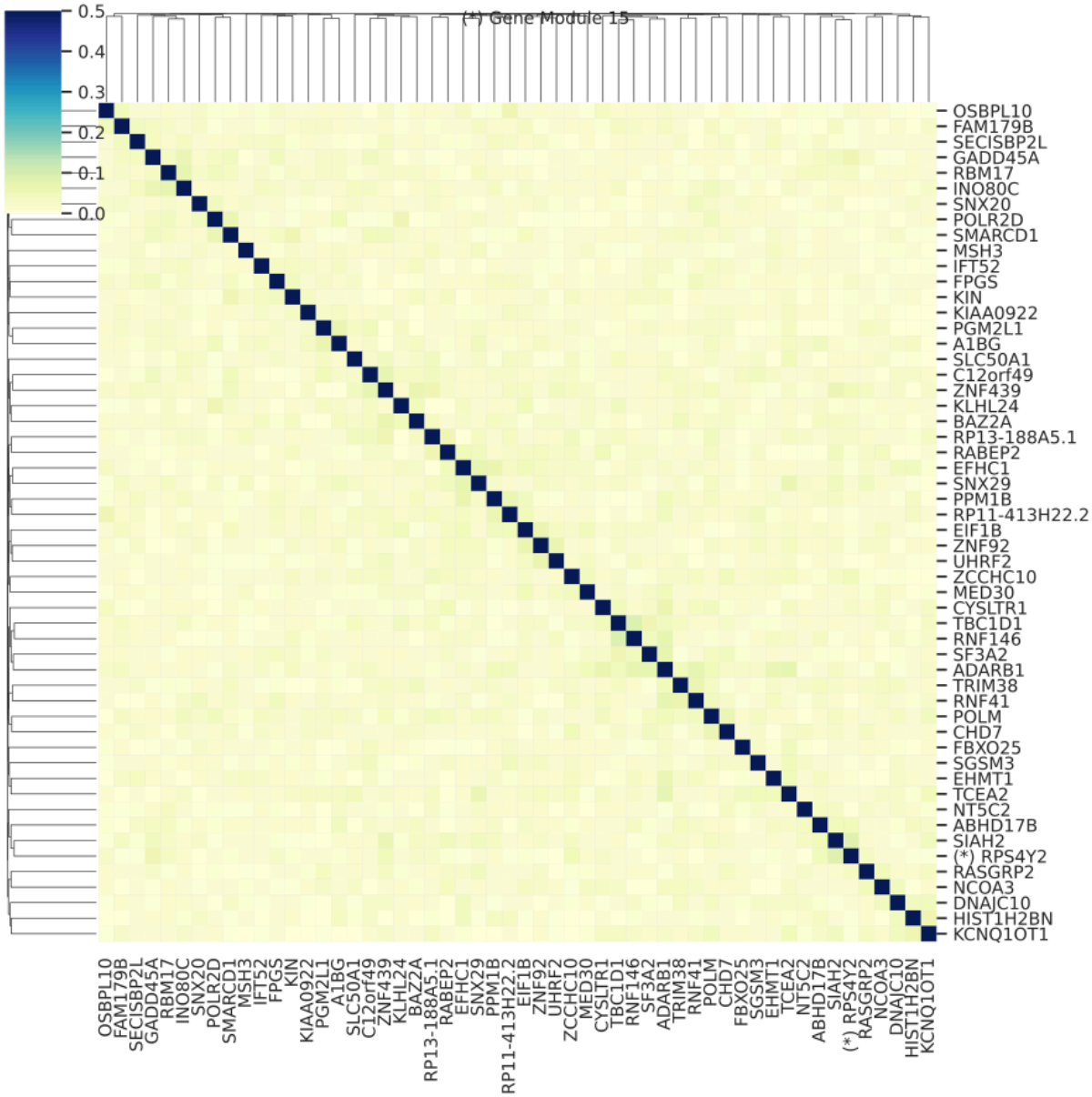


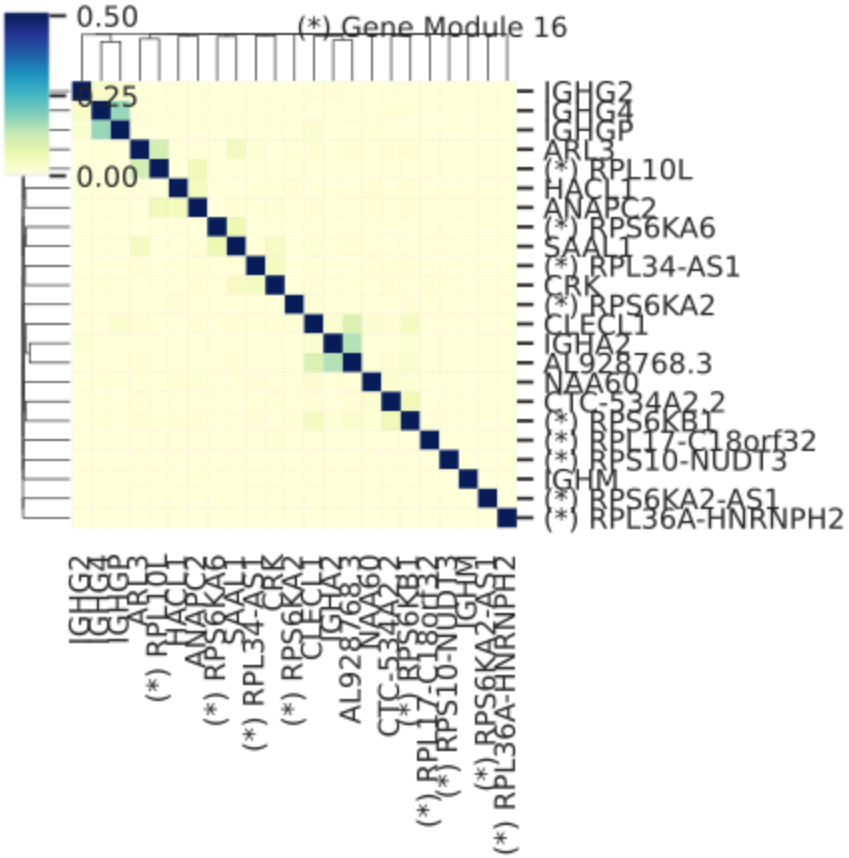
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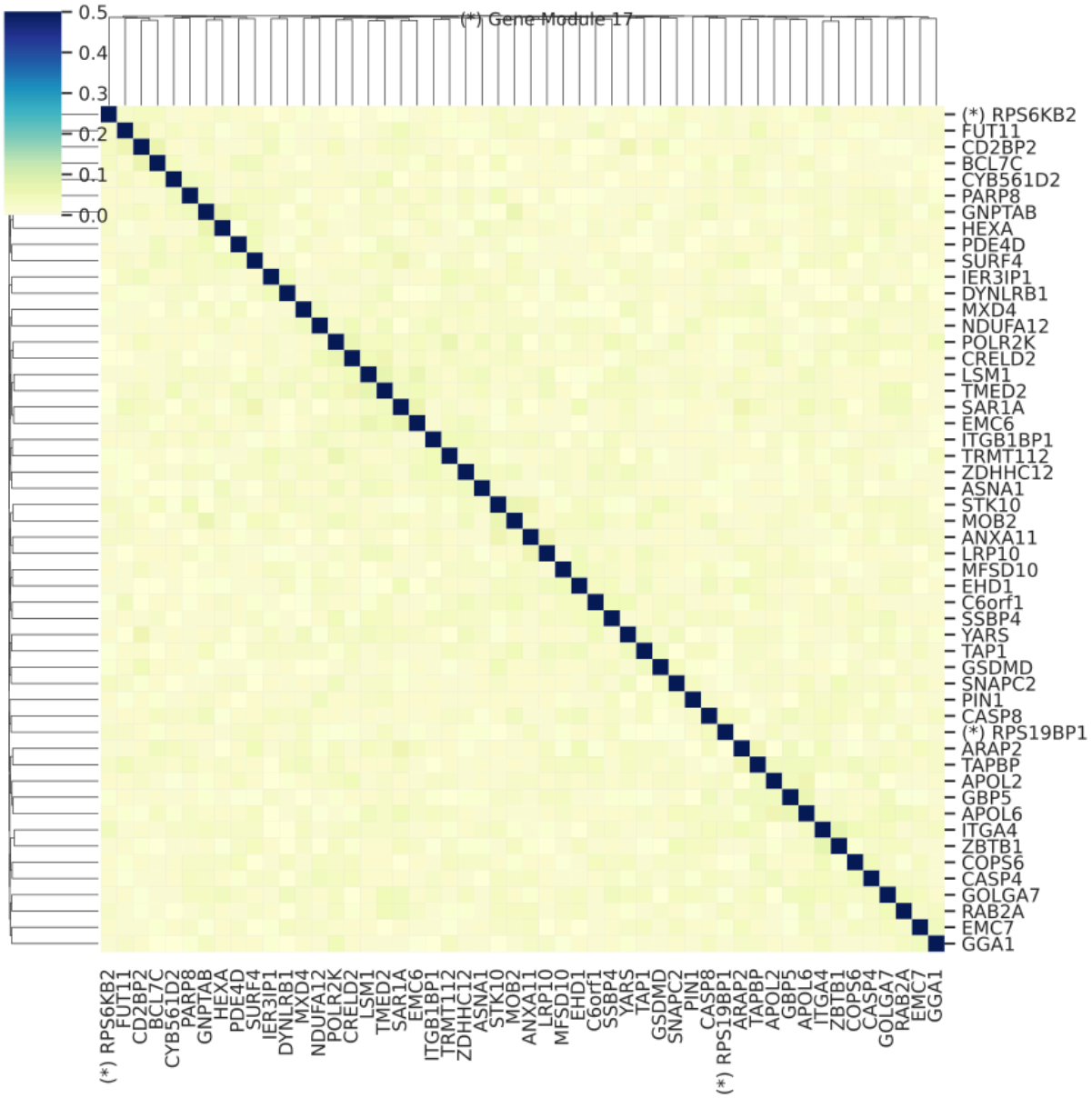

(*) Gene Module 8
0.4
0.2
0.0
NDUFAF6
SLC40A1
CRY1
(*) RPL7L1
VPS8
NAA25
DUS3L
DCAF10
VPS37A
DCLRE1C
RAB3IP
HNRNPU-AS1
IQCB1
AMMECR1
PML
CDK5RAP2
ZNF3
TRIM11
USE1
CCNL2
BDP1
TRMT61B
NKAP
BRD3
BRWD1
STAT5A
UBTF
PSMD12
CCDC174
PTCD3
ZNF451
ZMYM2
PTAR1
CMTR2



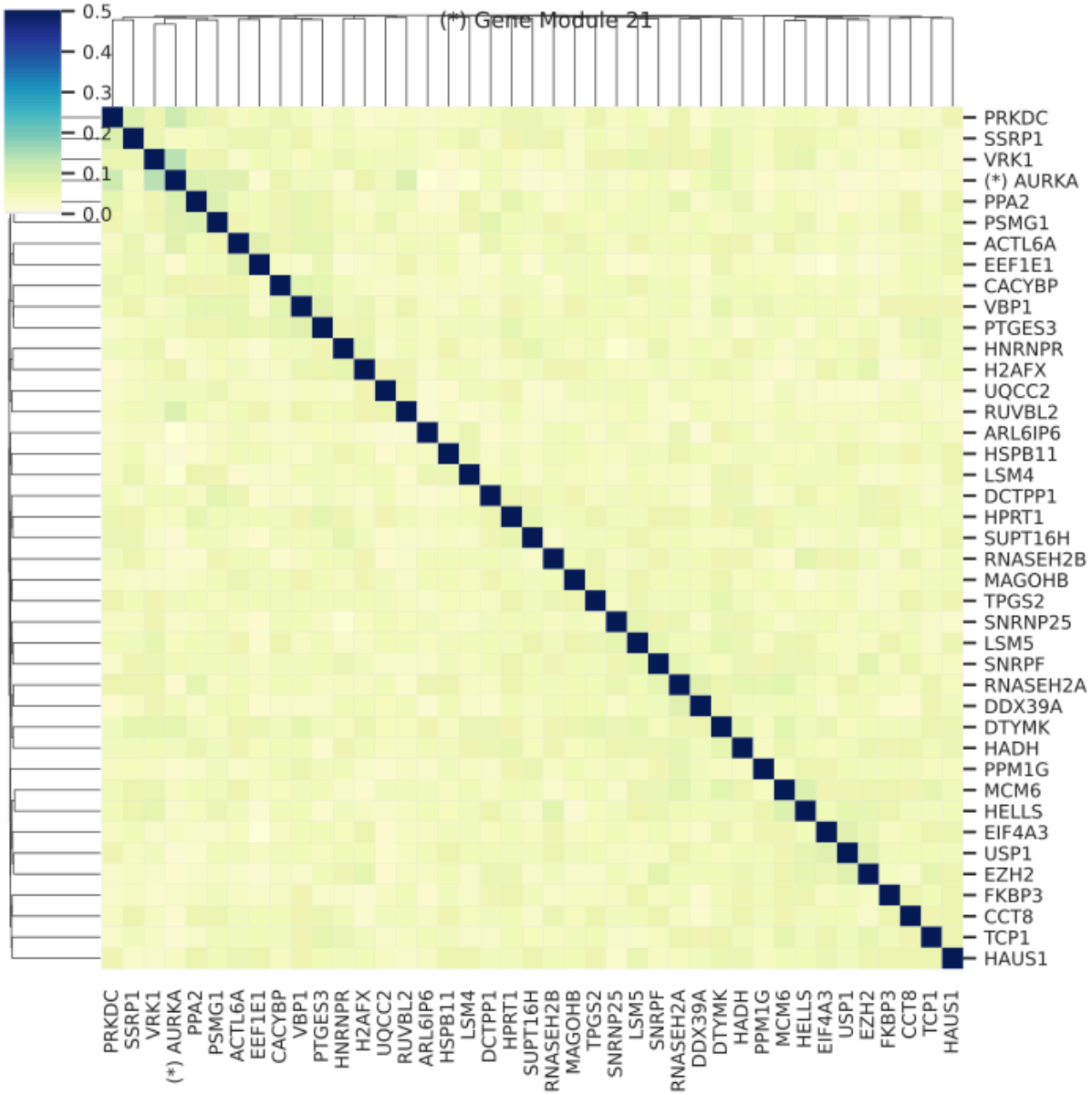
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<Figure size 676.8x676.8 with 0 Axes>
<Figure size 633.6x633.6 with 0 Axes>
<Figure size 471.6x471.6 with 0 Axes>


(*) Gene Module 15
0.5
0.4
0.3
0.2
0.1
0.0
OSBPL10
FAM179B
SECISBP2L
GADD45A
RBM17
INO80C
SNX20
POLR2D
SMARCD1
MSH3
IFT52
FPGS
KIN
KIAA0922
PGM2L1
A1BG
SLC50A1
C12orf49
ZNF439
KLHL24
BAZ2A
RP13-188A5.1
RABEP2
EFHC1
SNX29
PPM1B
RP11-413H22.2
EIF1B
ZNF92
UHRF2
ZCCHC10
MED30
CYSLTR1
TBC1D1
RNF146
SF3A2
ADARB1
TRIM38
RNF41
POLM
CHD7
FBXO25
SGSM3
EHMT1
TCEA2
NT5C2
ABHD17B
SIAH2
(*) RPS4Y2
RASGRP2
NCOA3
DNAJC10
HIST1H2BN
KCNQ1OT1

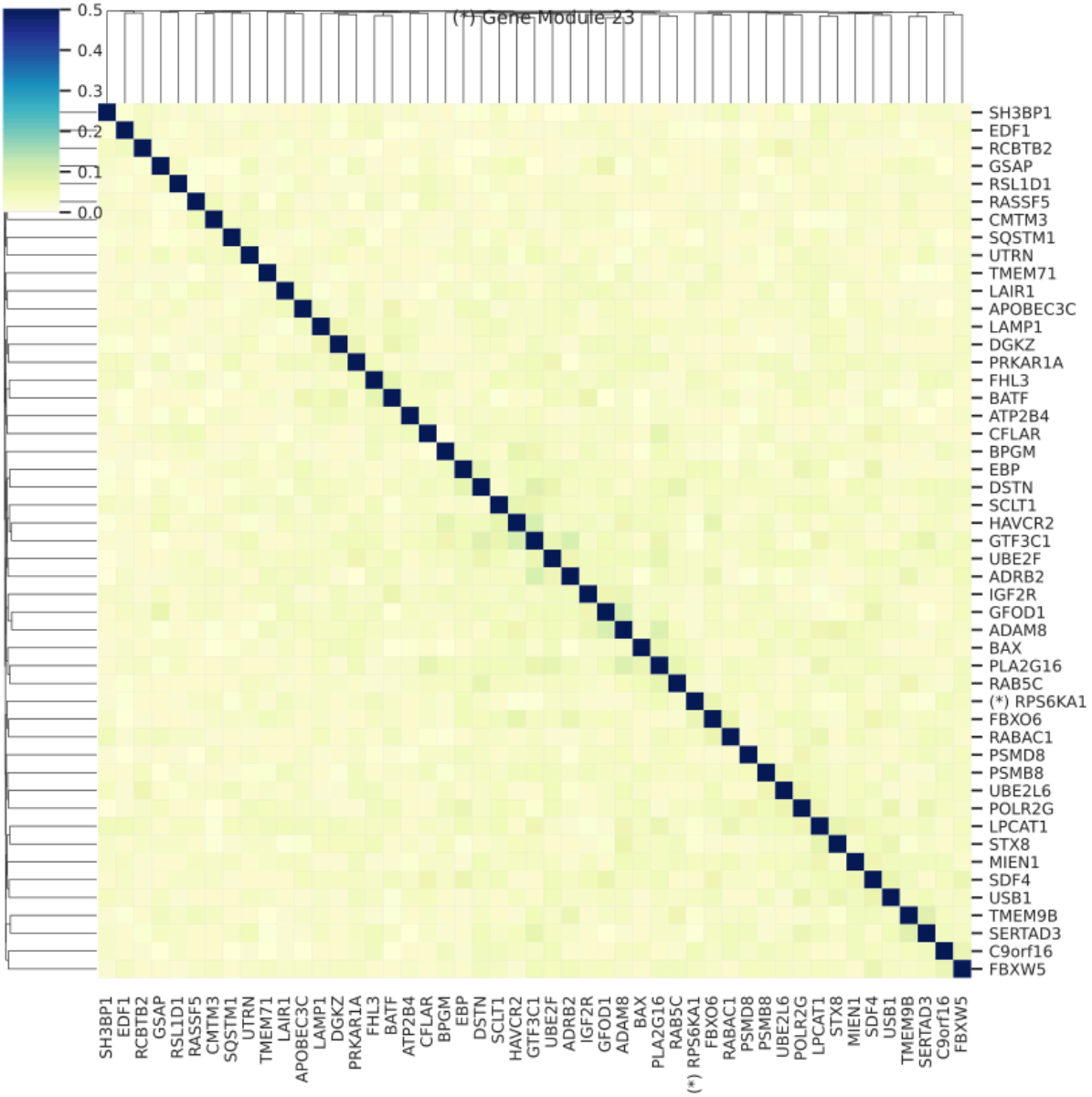

0.50
0.25
0.00
(*) Gene Module 16
IGHG2
IGHG4
IGHGP
ARL3
(*) RPL10L
HACL1
ANAPC2
(*) RPS6KA6
SAAL1
(*) RPL34-AS1
CRK
(*) RPS6KA2
CLECL1
IGHA2
AL928768.3
NAA60
CTC-534A2.2
(*) RPS6KB1
(*) RPL17-C18orf32
(*) RPS10-NUDT3
IGHM
(*) RPS6KA2-AS1
(*) RPL36A-HNRNPH2

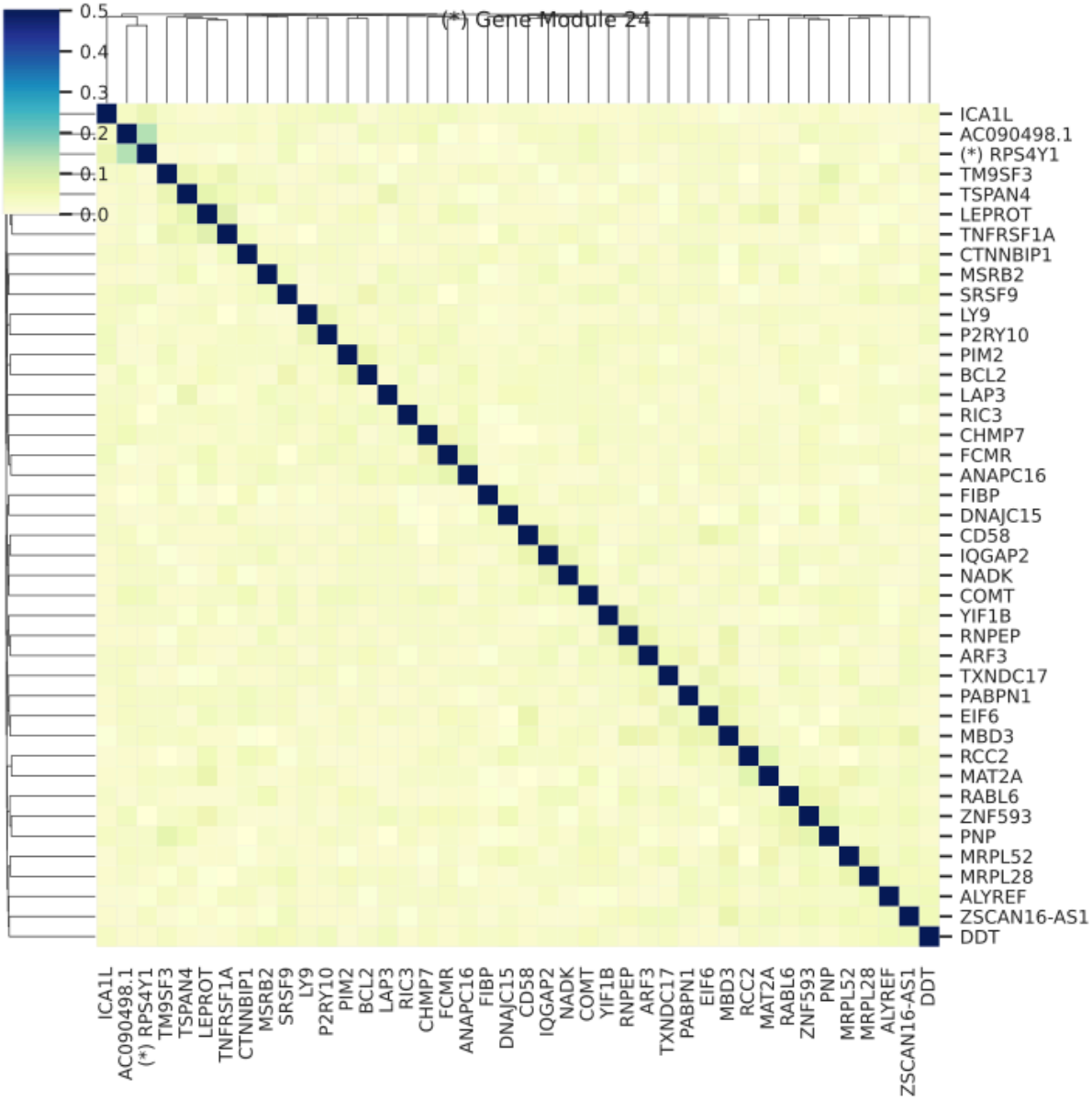


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<Figure size 763.2x763.2 with 0 Axes>

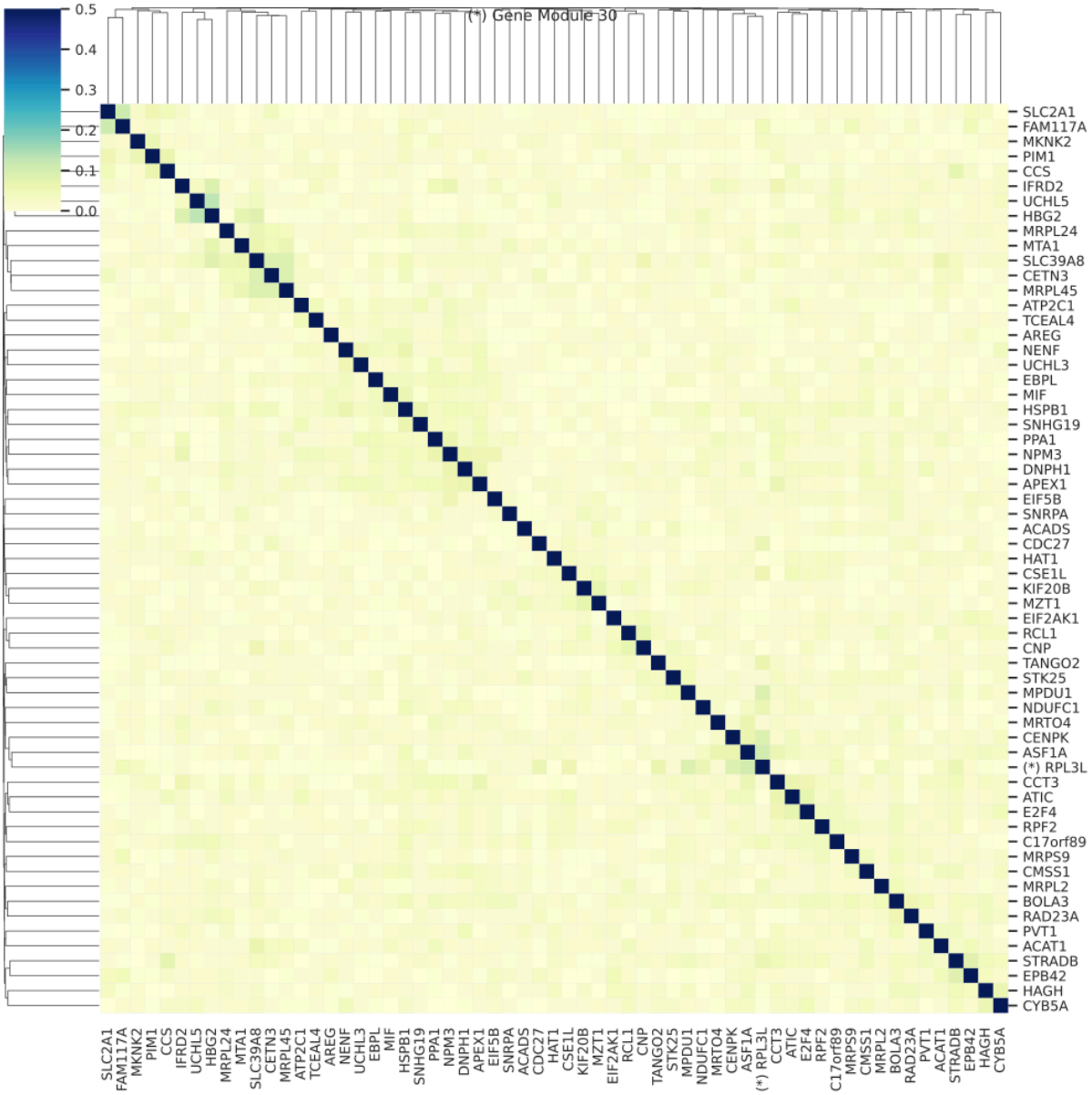


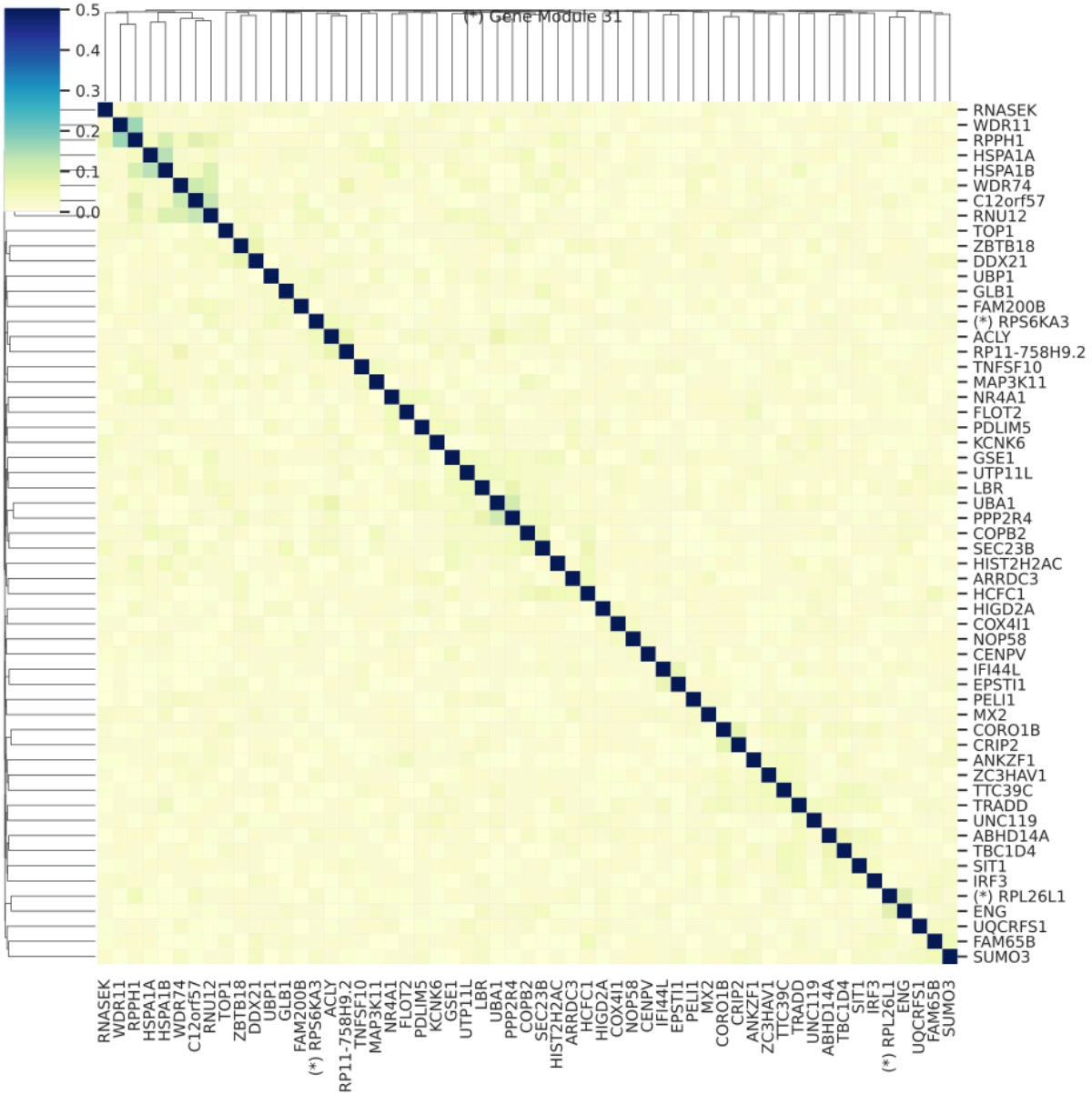
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(*) Gene Module 23
0.5
0.4
0.3
0.2
0.1
0.0
SH3BP1
EDF1
RCBTB2
GSAP
RSL1D1
RASSF5
CMTM3
SQSTM1
UTRN
TMEM71
LAIR1
APOBEC3C
LAMP1
DGKZ
PRKAR1A
FHL3
BATF
ATP2B4
CFLAR
BPGM
EBP
DSTN
SCLT1
HAVCR2
GTF3C1
UBE2F
ADRB2
IGF2R
GFOD1
ADAM8
BAX
PLA2G16
RAB5C
(*) RPS6KA1
FBXO6
RABAC1
PSMD8
PSMB8
UBE2L6
POLR2G
LPCAT1
STX8
MIEN1
SDF4
USB1
TMEM9B
SERTAD3
C9orf16
FBXW5

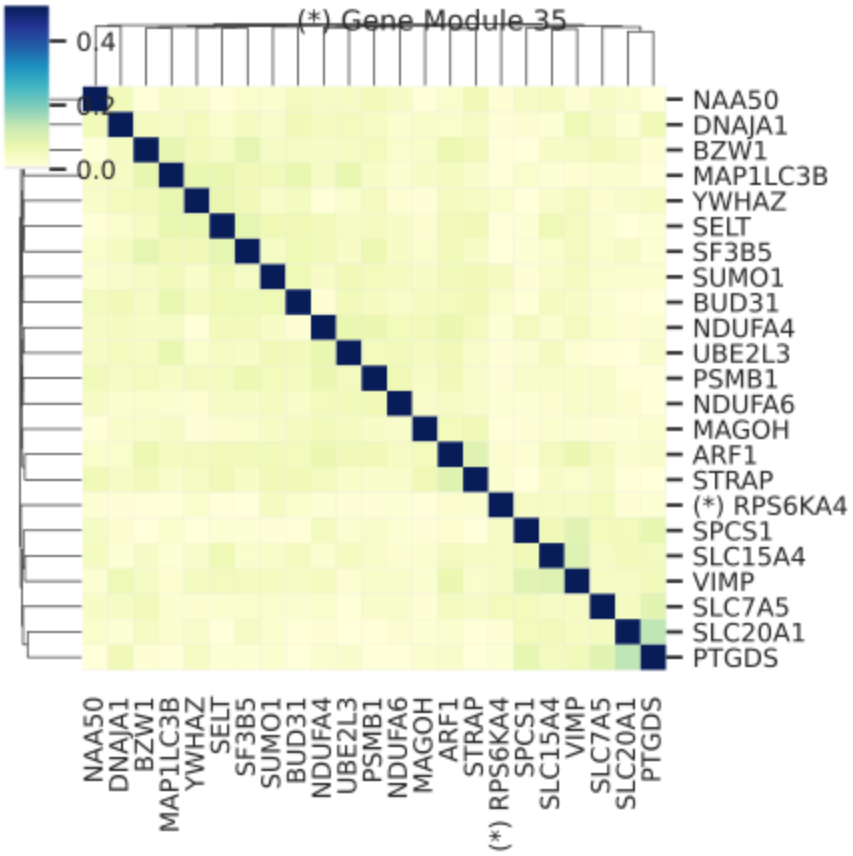


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<Figure size 147.6x147.6 with 0 Axes>
<Figure size 644.4x644.4 with 0 Axes>
<Figure size 612x612 with 0 Axes>
<Figure size 482.4x482.4 with 0 Axes>

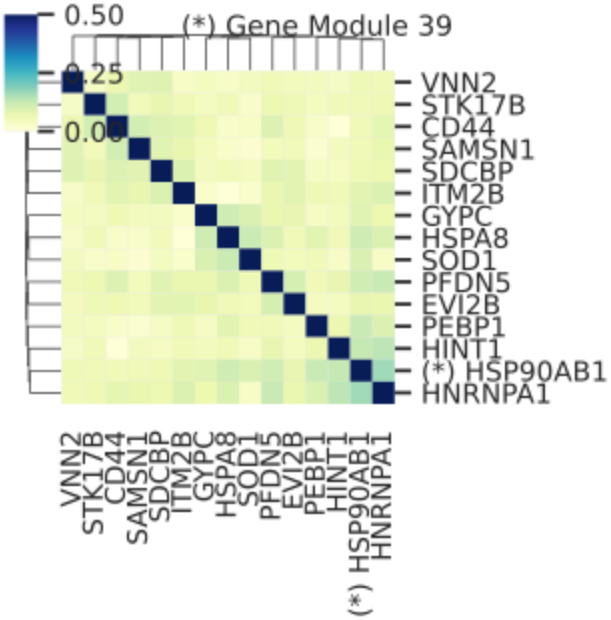

(*) Gene Module 30
0.5
0.4
0.3
0.2
0.1
0.0
SLC2A1
FAM117A
MKNK2
PIM1
CCS
IFRD2
UCHL5
HBG2
MRPL24
MTA1
SLC39A8
CETN3
MRPL45
ATP2C1
TCEAL4
AREG
NENF
UCHL3
EBPL
MIF
HSPB1
SNHG19
PPA1
NPM3
DNPH1
APEX1
EIF5B
SNRPA
ACADS
CDC27
HAT1
CSE1L
KIF20B
MZT1
EIF2AK1
RCL1
CNP
TANGO2
STK25
MPDU1
NDUFC1
MRTO4
CENPK
ASF1A
(*) RPL3L
CCT3
ATIC
E2F4
RPF2
C17orf89
MRPS9
CMSS1
MRPL2
BOLA3
RAD23A
PVT1
ACAT1
STRADB
EPB42
HAGH
CYB5A



<Figure size 720x720 with 0 Axes>
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<Figure size 601.2x601.2 with 0 Axes>

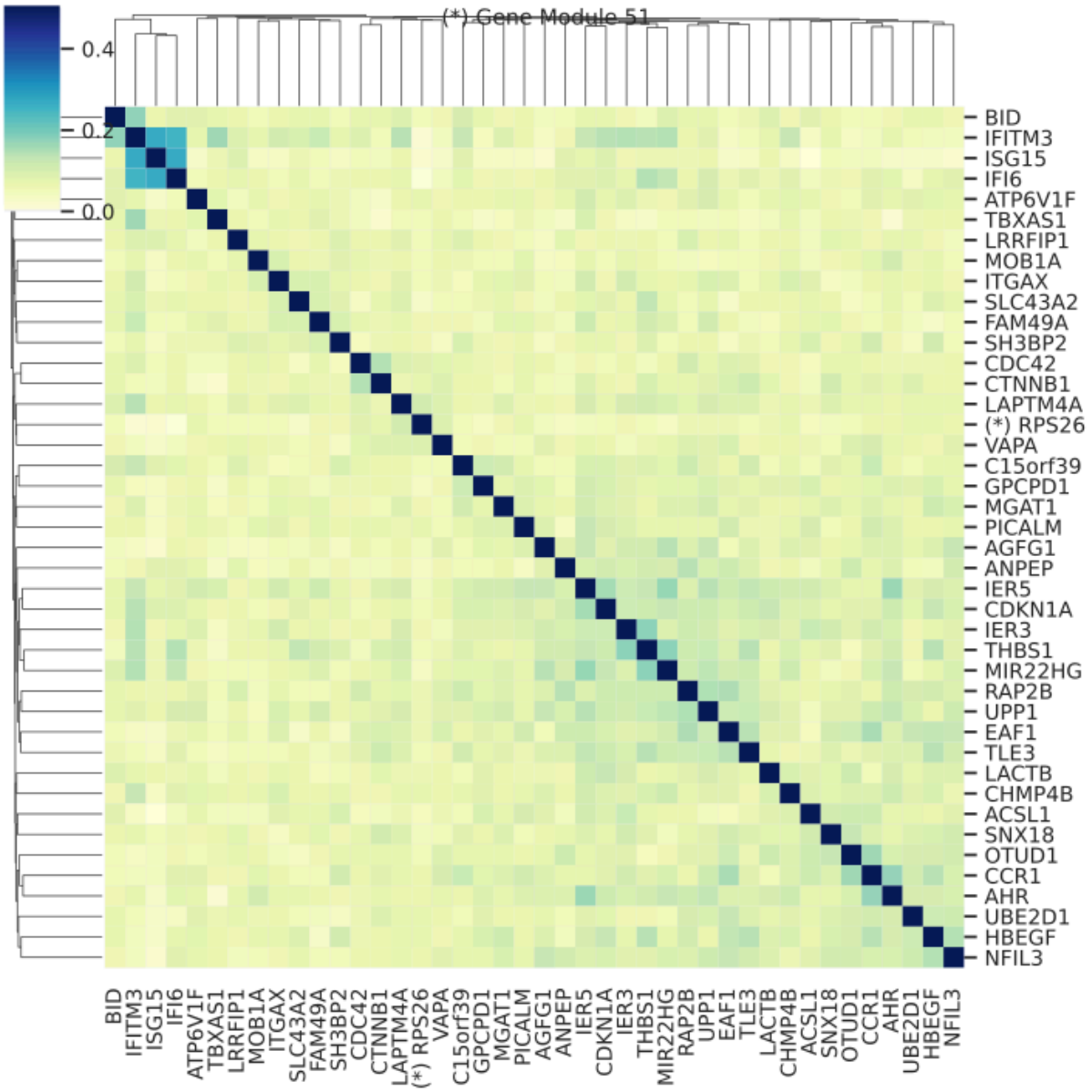


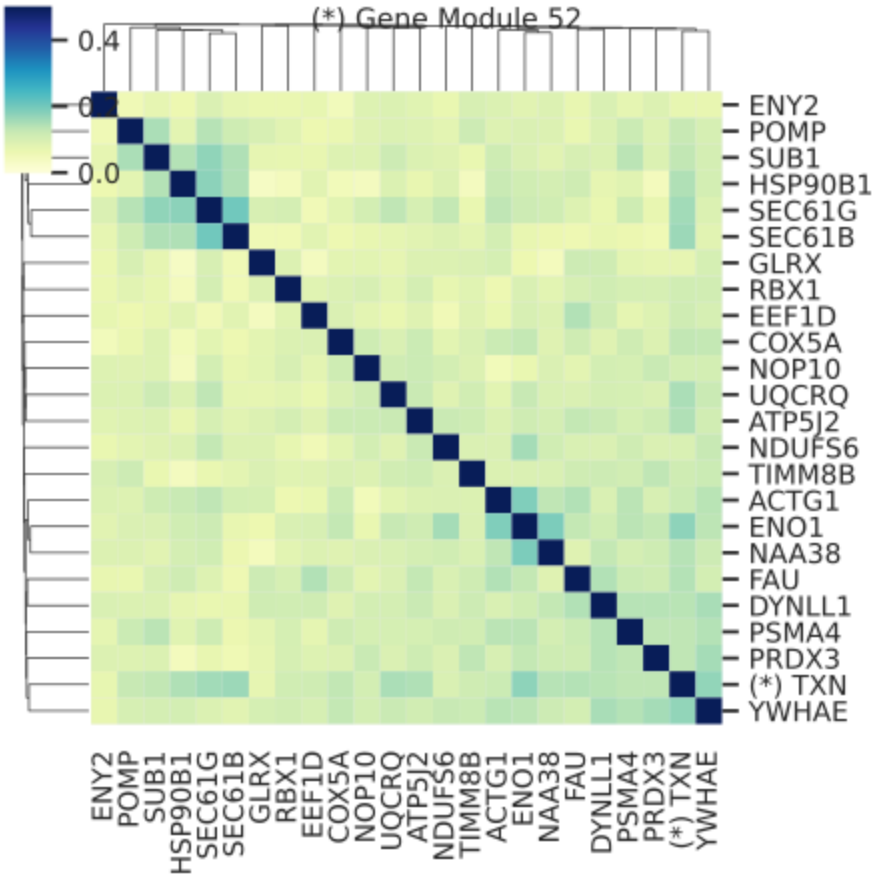
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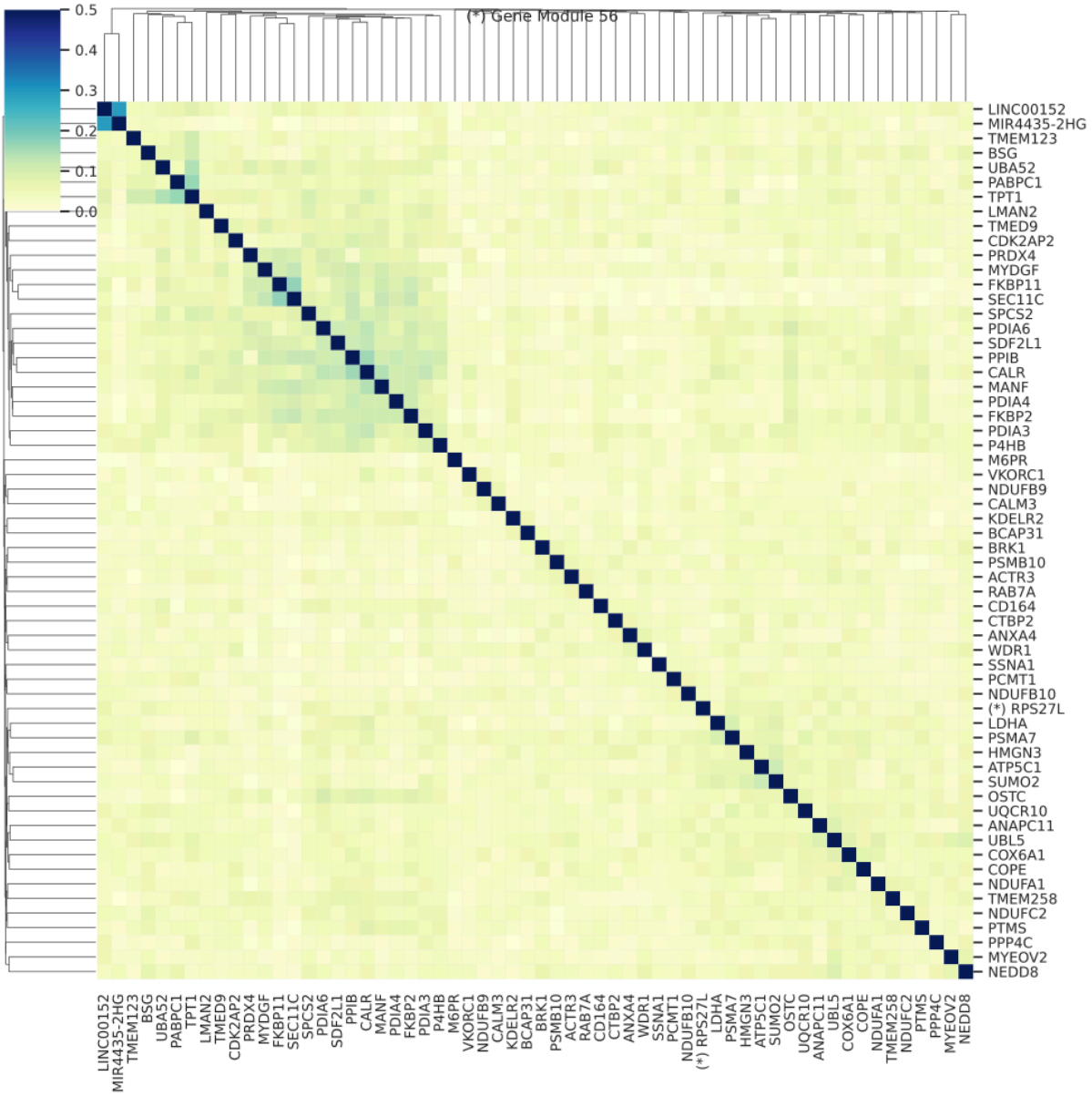
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<Figure size 277.2x277.2 with 0 Axes>
<Figure size 190.8x190.8 with 0 Axes>
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<Figure size 612x612 with 0 Axes>
<Figure size 255.6x255.6 with 0 Axes>

<Figure size 363.6x363.6 with 0 Axes>
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<Figure size 579.6x579.6 with 0 Axes>

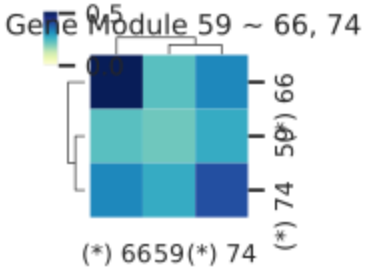


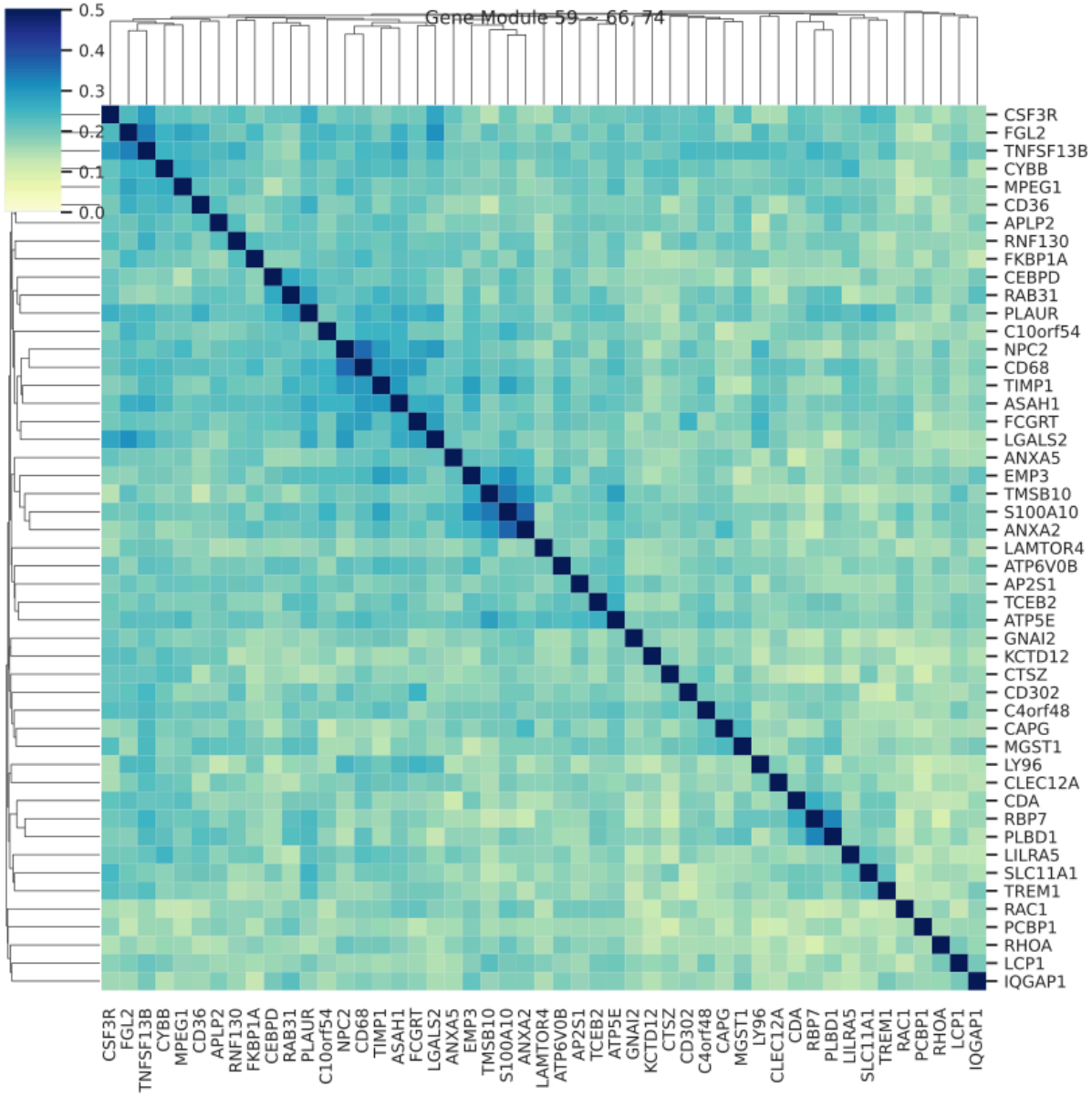


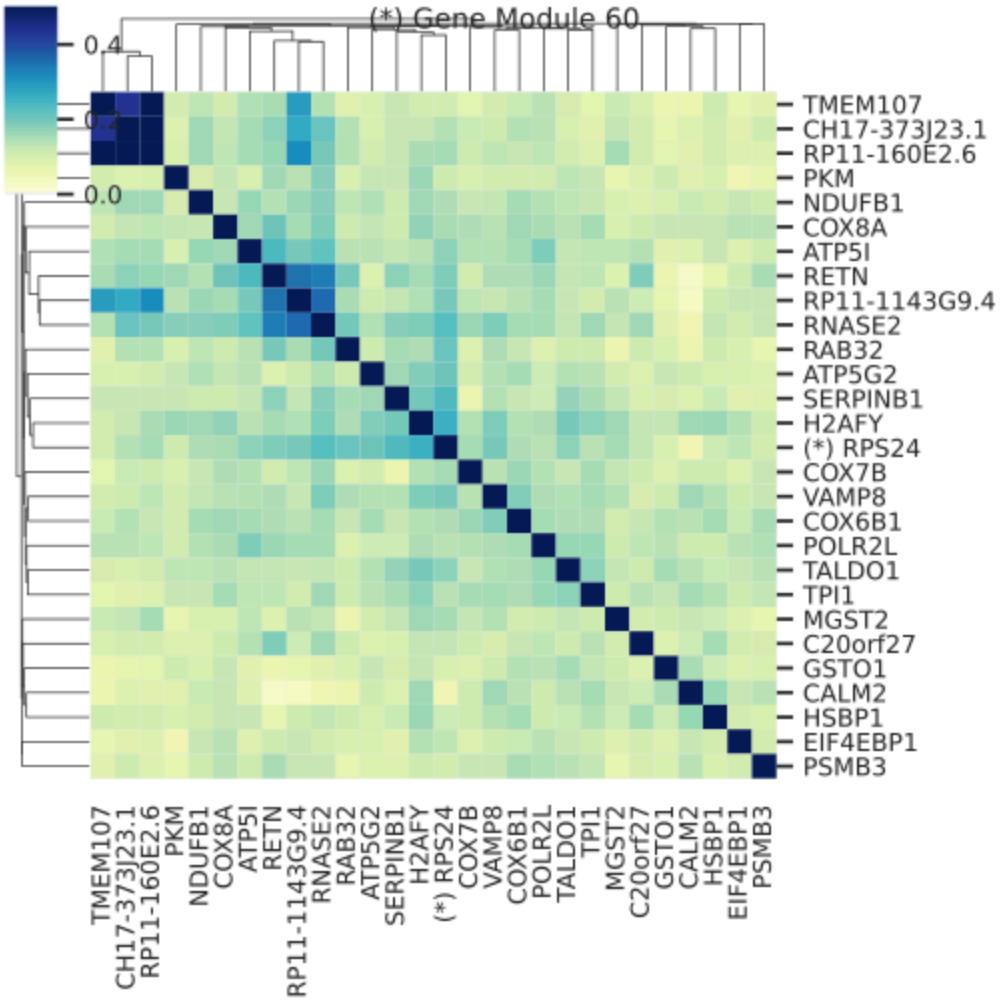
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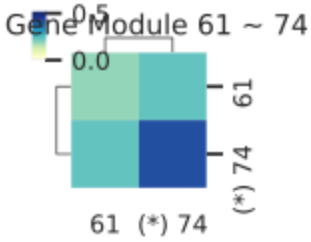


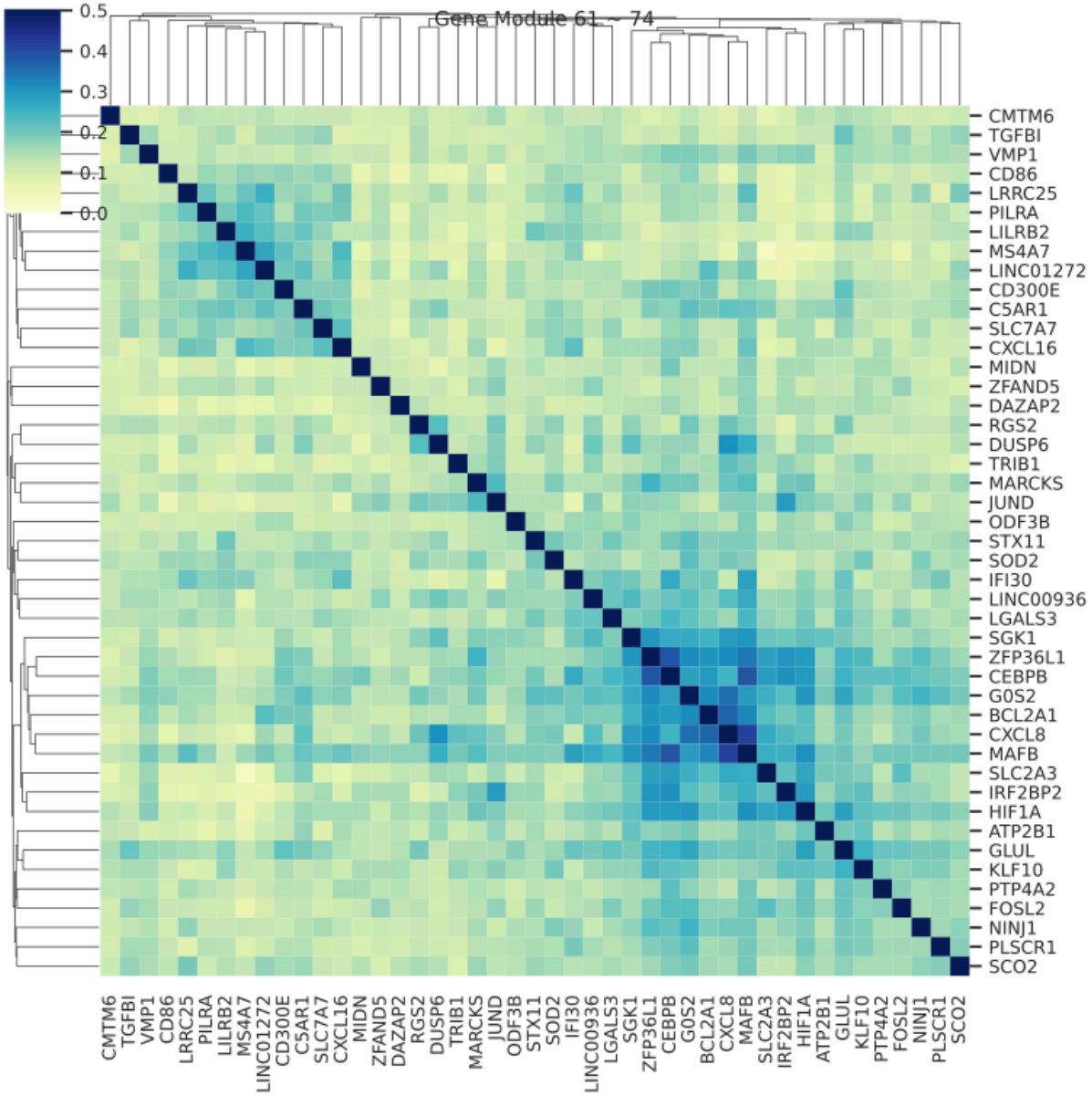
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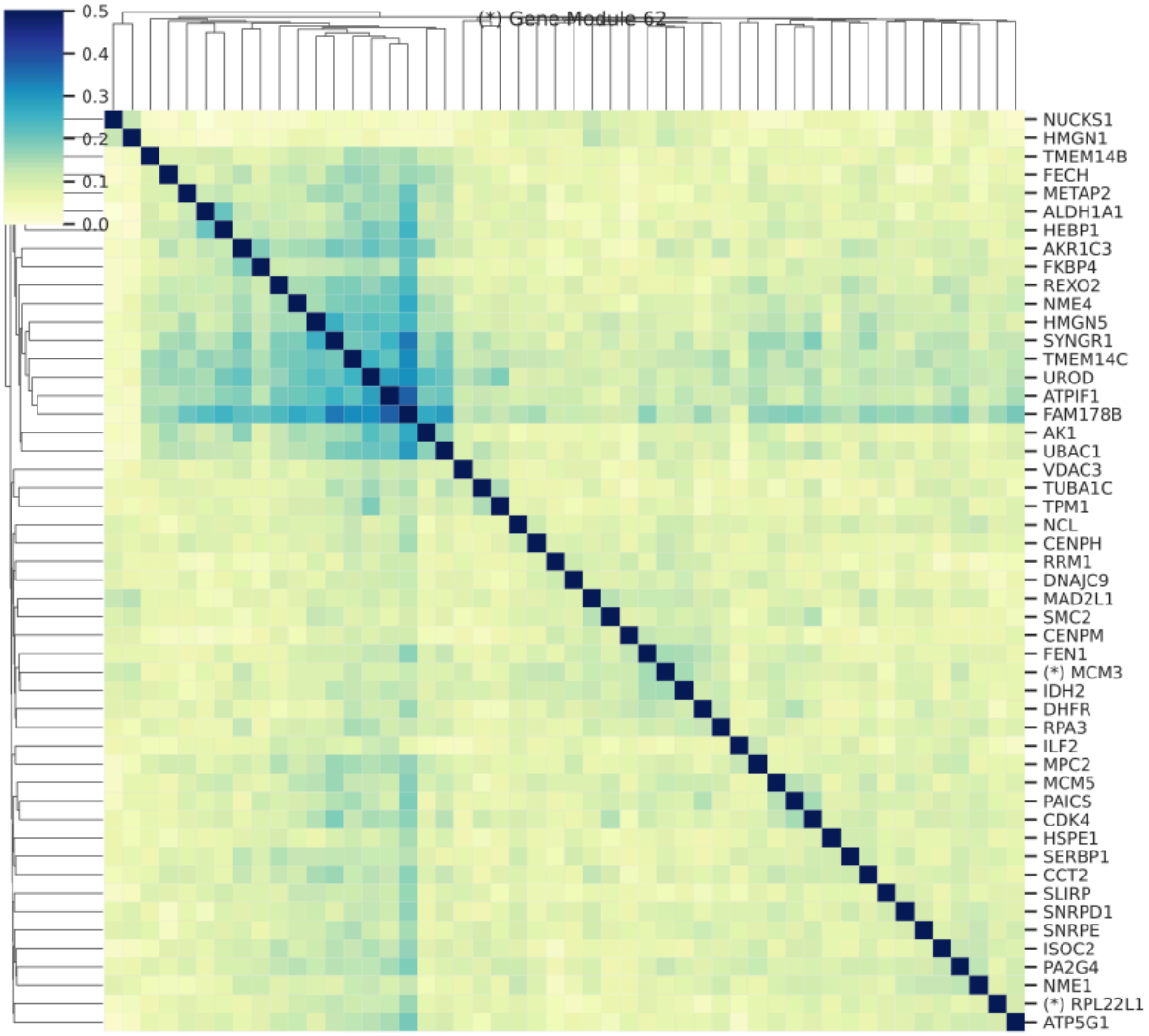


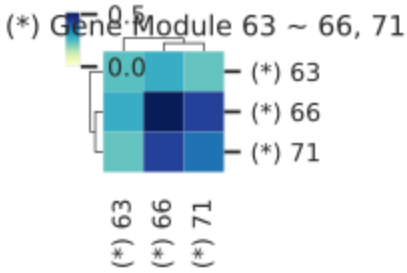

Gene Module 59 ~ 66, 74
0.5
0.4
0.3
0.2
0.1
0.0
CSF3R
FGL2
TNFSF13B
CYBB
MPEG1
CD36
APLP2
RNF130
FKBP1A
CEBPD
RAB31
PLAUR
C10orf54
NPC2
CD68
TIMP1
ASAH1
FCGRT
LGALS2
ANXA5
EMP3
TMSB10
S100A10
ANXA2
LAMTOR4
ATP6V0B
AP2S1
TCEB2
ATP5E
GNAI2
KCTD12
CTSZ
CD302
C4orf48
CAPG
MGST1
LY96
CLEC12A
CDA
RBP7
PLBD1
LILRA5
SLC11A1
TREM1
RAC1
PCBP1
RHOA
LCP1
IQGAP1

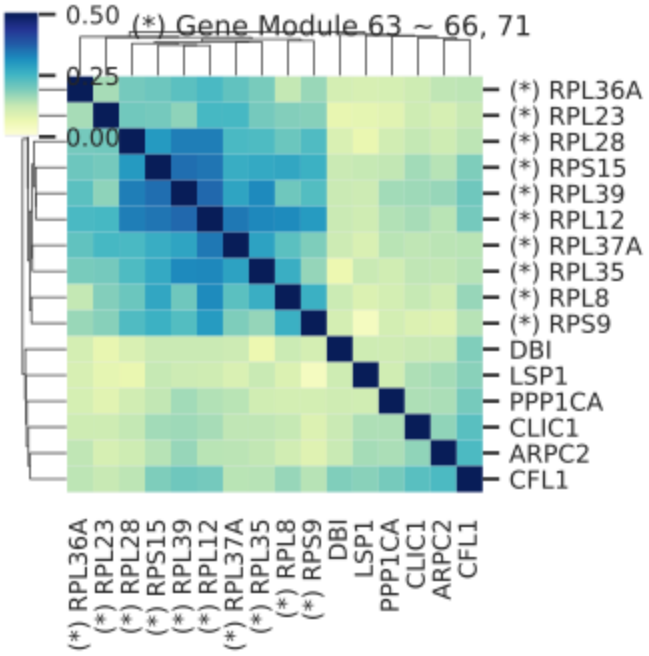

(*) Gene Module 60
0.4
0.2
0.0
TMEM107
CH17-373J23.1
RP11-160E2.6
PKM
NDUFB1
COX8A
ATP5I
RETN
RP11-1143G9.4
RNASE2
RAB32
ATP5G2
SERPINB1
H2AFY
(*) RPS24
COX7B
VAMP8
COX6B1
POLR2L
TALDO1
TPI1
MGST2
C20orf27
GSTO1
CALM2
HSBP1
EIF4EBP1
PSMB3


Gene Module 61 ~ 74
0.5
0.0
61
(*) 74
61
(*) 74

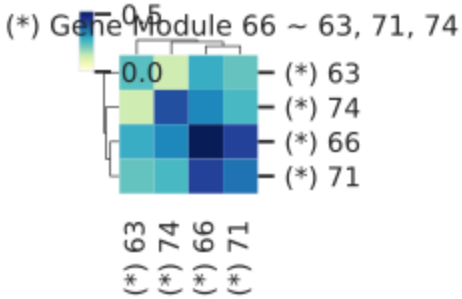

Gene Module 61 ~ 74
0.5
0.4
0.3
0.2
0.1
0.0
CMTM6
TGFBI
VMP1
CD86
LRRC25
PILRA
LILRB2
MS4A7
LINC01272
CD300E
C5AR1
SLC7A7
CXCL16
MIDN
ZFAND5
DAZAP2
RGS2
DUSP6
TRIB1
MARCKS
JUND
ODF3B
STX11
SOD2
IFI30
LINC00936
LGALS3
SGK1
ZFP36L1
CEBPB
G0S2
BCL2A1
CXCL8
MAFB
SLC2A3
IRF2BP2
HIF1A
ATP2B1
GLUL
KLF10
PTP4A2
FOSL2
NINJ1
PLSCR1
SCO2

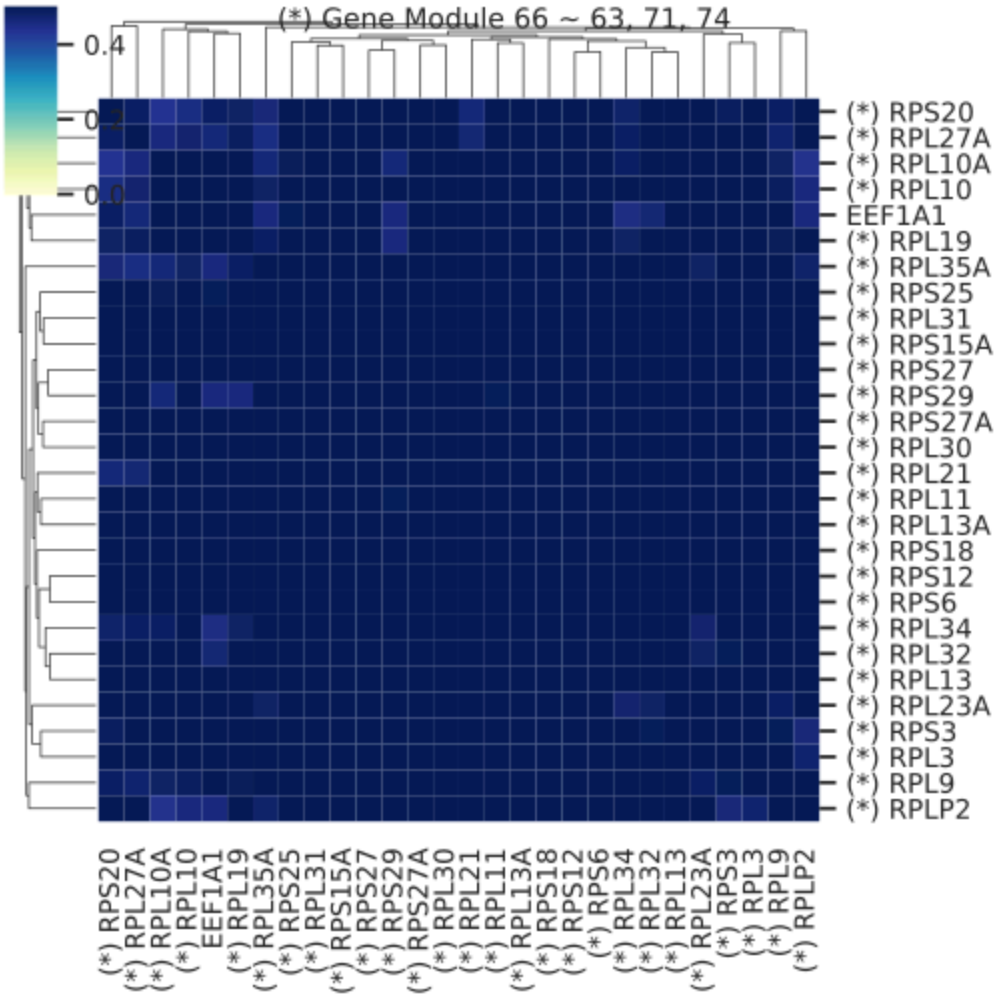

(*) Gene Module 62
0.5
0.4
0.3
0.2
0.1
0.0
NUCKS1
HMGN1
TMEM14B
FECH
METAP2
ALDH1A1
HEBP1
AKR1C3
FKBP4
REXO2
NME4
HMGN5
SYNGR1
TMEM14C
UROD
ATPIF1
FAM178B
AK1
UBAC1
VDAC3
TUBA1C
TPM1
NCL
CENPH
RRM1
DNAJC9
MAD2L1
SMC2
CENPM
FEN1
(*) MCM3
IDH2
DHFR
RPA3
ILF2
MPC2
MCM5
PAICS
CDK4
HSPE1
SERBP1
CCT2
SLIRP
SNRPD1
SNRPE
ISOC2
PA2G4
NME1
(*) RPL22L1
ATP5G1


(*) Gene Module 63 ~ 66, 71
0.5
0.0
(*) 63
(*) 66
(*) 71

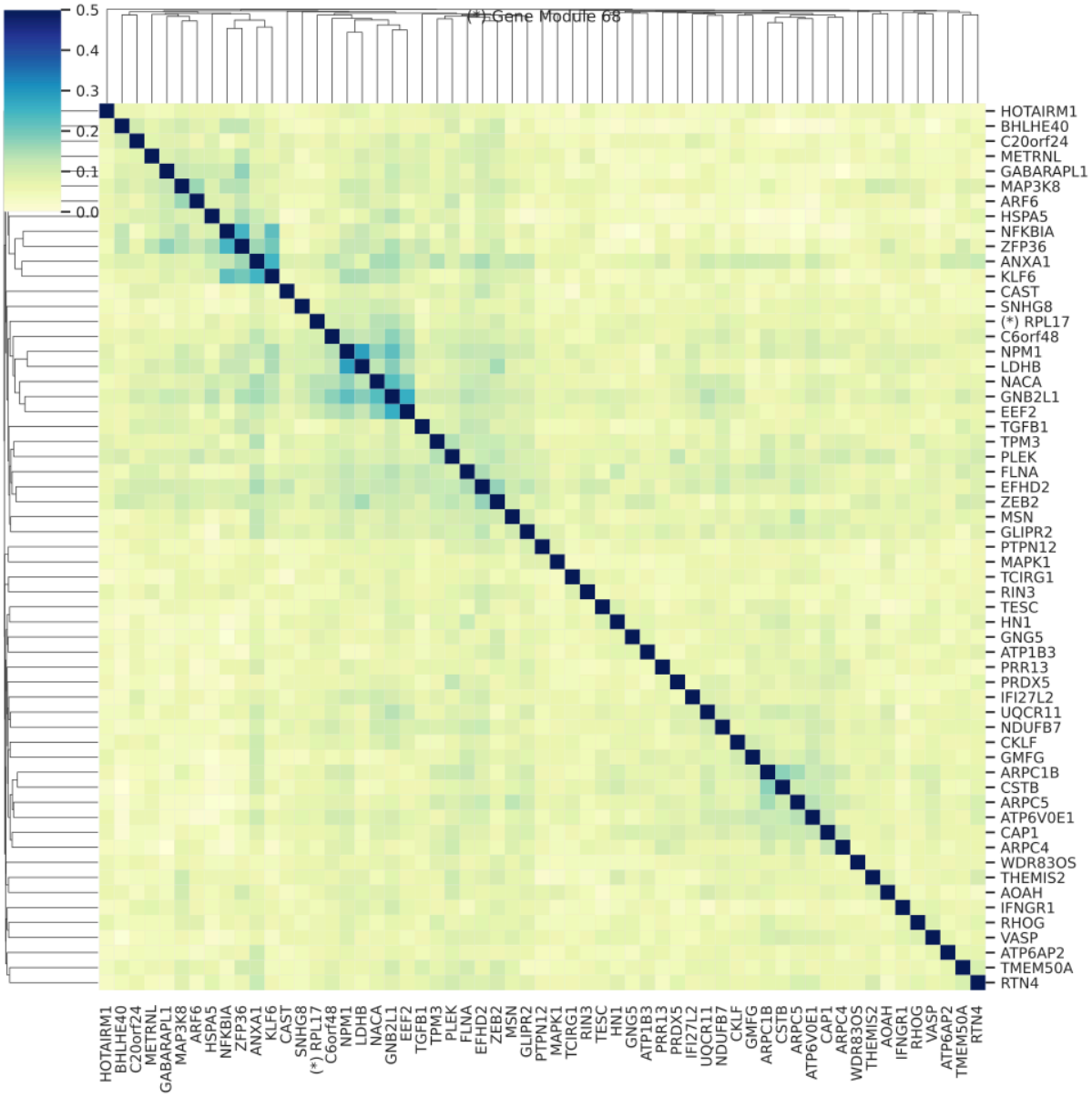


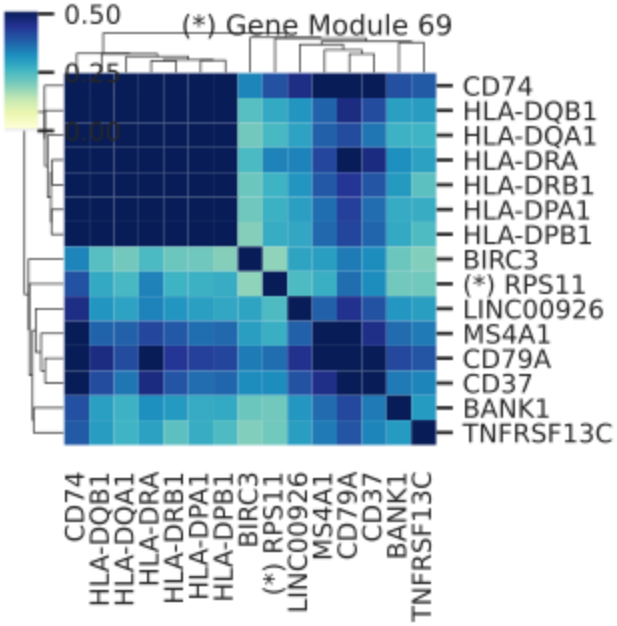
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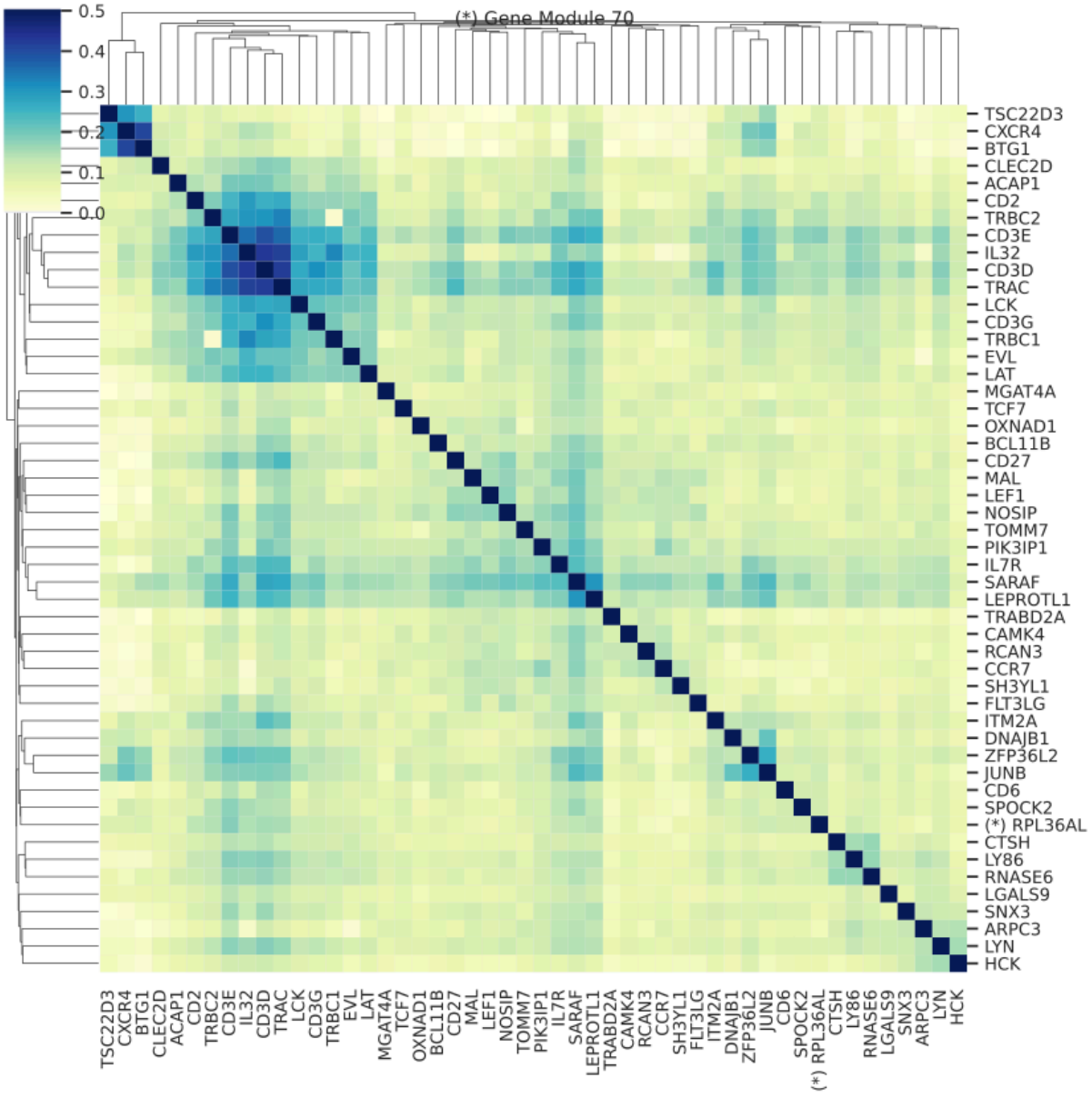


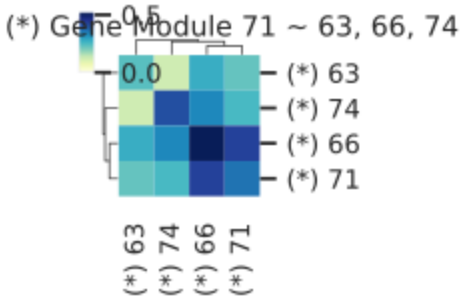


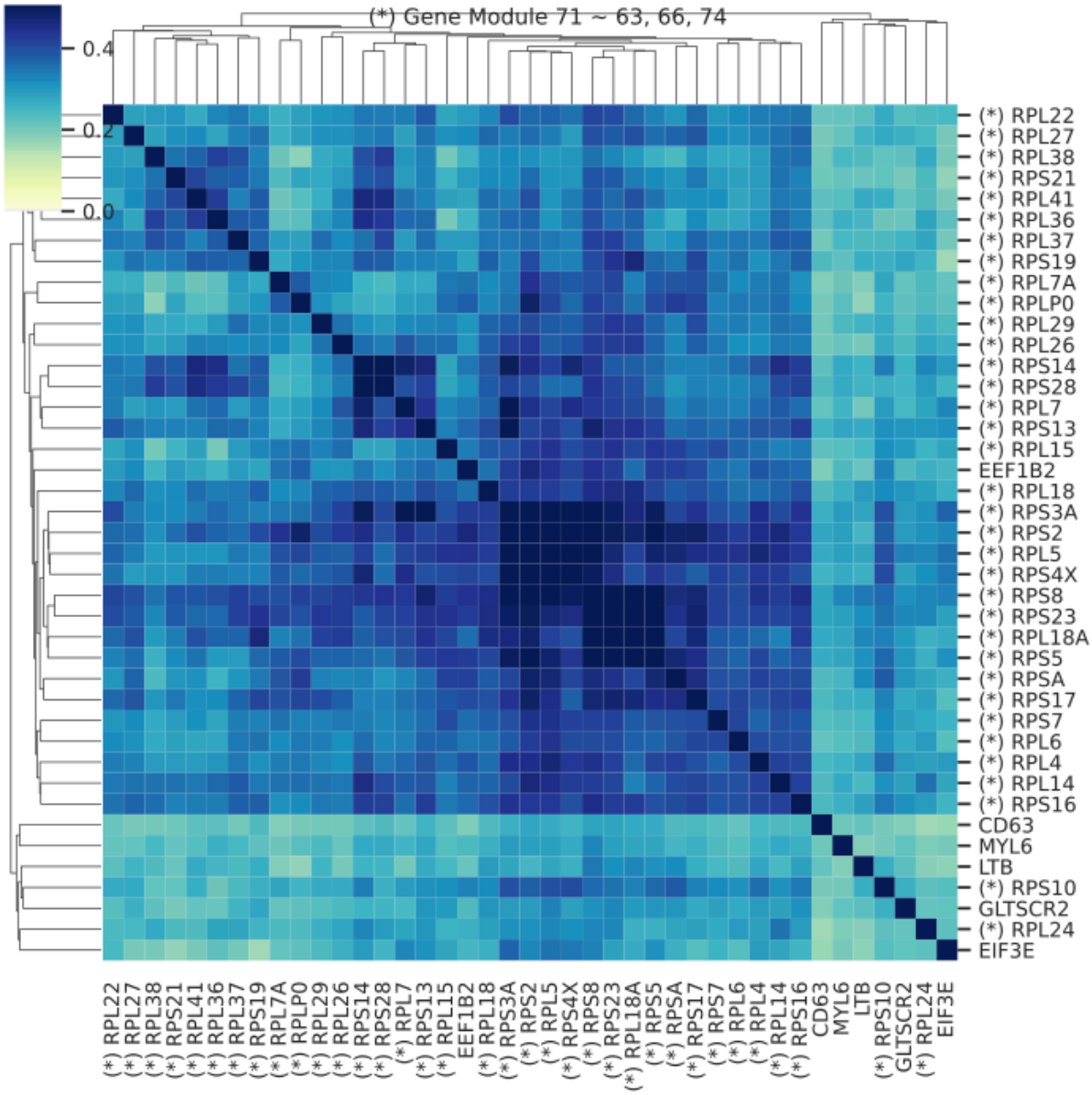
<Figure size 536.4x536.4 with 0 Axes>


0.5
0.4
0.3
0.2
0.1
0.0
(*) Gene Module 68
HOTAIRM1
BHLHE40
C20orf24
METRNL
GABARAPL1
MAP3K8
ARF6
HSPA5
NFKBIA
ZFP36
ANXA1
KLF6
CAST
SNHG8
(*) RPL17
C6orf48
NPM1
LDHB
NACA
GNB2L1
EEF2
TGFB1
TPM3
PLEK
FLNA
EFHD2
ZEB2
MSN
GLIPR2
PTPN12
MAPK1
TCIRG1
RIN3
TESC
HN1
GNG5
ATP1B3
PRR13
PRDX5
IFI27L2
UQCR11
NDUFB7
CKLF
GMFG
ARPC1B
CSTB
ARPC5
ATP6V0E1
CAP1
ARPC4
WDR83OS
THEMIS2
AOAH
IFNGR1
RHOG
VASP
ATP6AP2
TMEM50A
RTN4

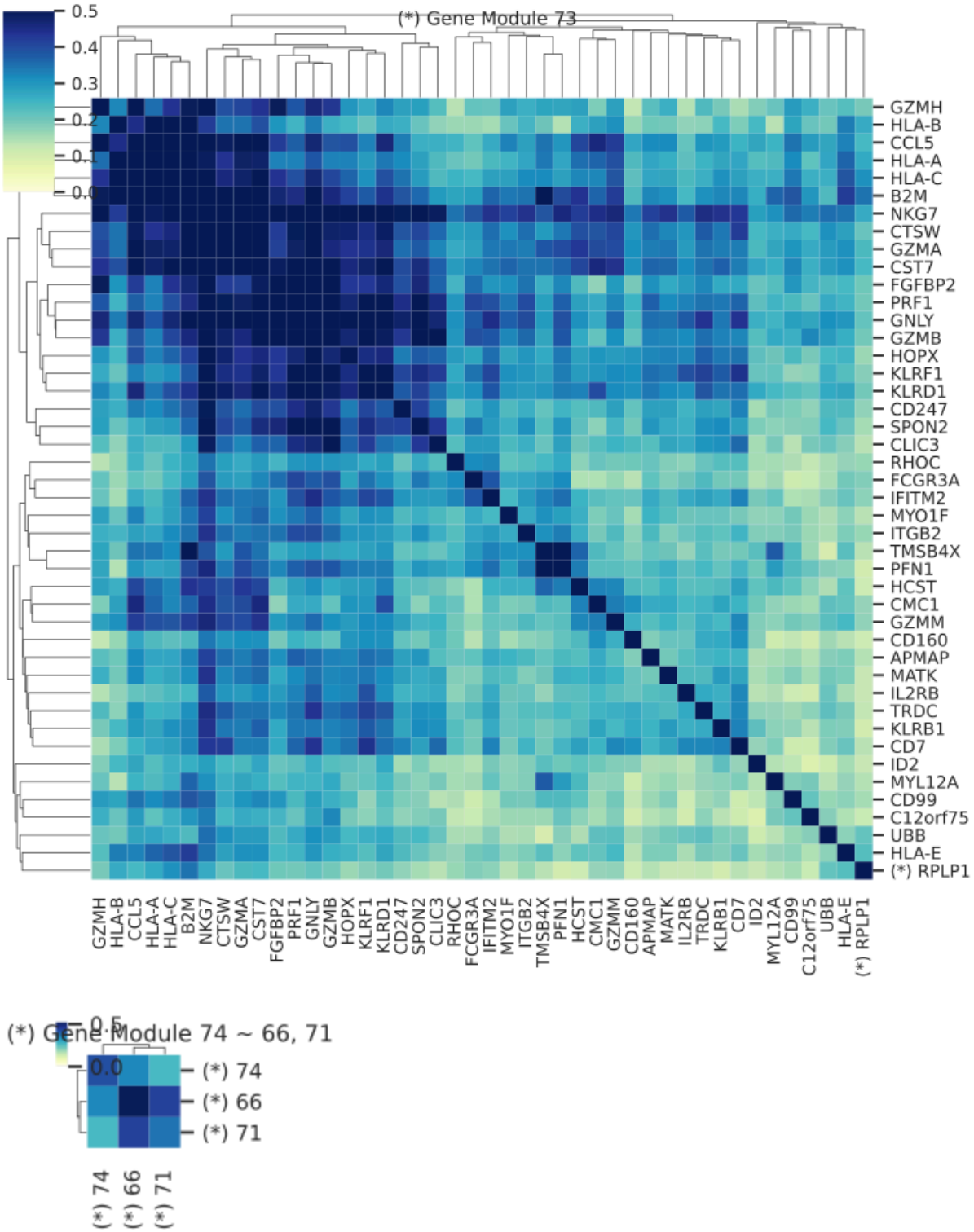

0.50
0.25
0.00
(*) Gene Module 69
CD74
HLA-DQB1
HLA-DQA1
HLA-DRA
HLA-DRB1
HLA-DPA1
HLA-DPB1
BIRC3
(*) RPS11
LINC00926
MS4A1
CD79A
CD37
BANK1
TNFRSF13C


(*) Gene Module 70
0.5
0.4
0.3
0.2
0.1
0.0
TSC22D3
CXCR4
BTG1
CLEC2D
ACAP1
CD2
TRBC2
CD3E
IL32
CD3D
TRAC
LCK
CD3G
TRBC1
EVL
LAT
MGAT4A
TCF7
OXNAD1
BCL11B
CD27
MAL
LEF1
NOSIP
TOMM7
PIK3IP1
IL7R
SARAF
LEPROTL1
TRABD2A
CAMK4
RCAN3
CCR7
SH3YL1
FLT3LG
ITM2A
DNAJB1
ZFP36L2
JUNB
CD6
SPOCK2
(*) RPL36AL
CTSH
LY86
RNASE6
LGALS9
SNX3
ARPC3
LYN
HCK

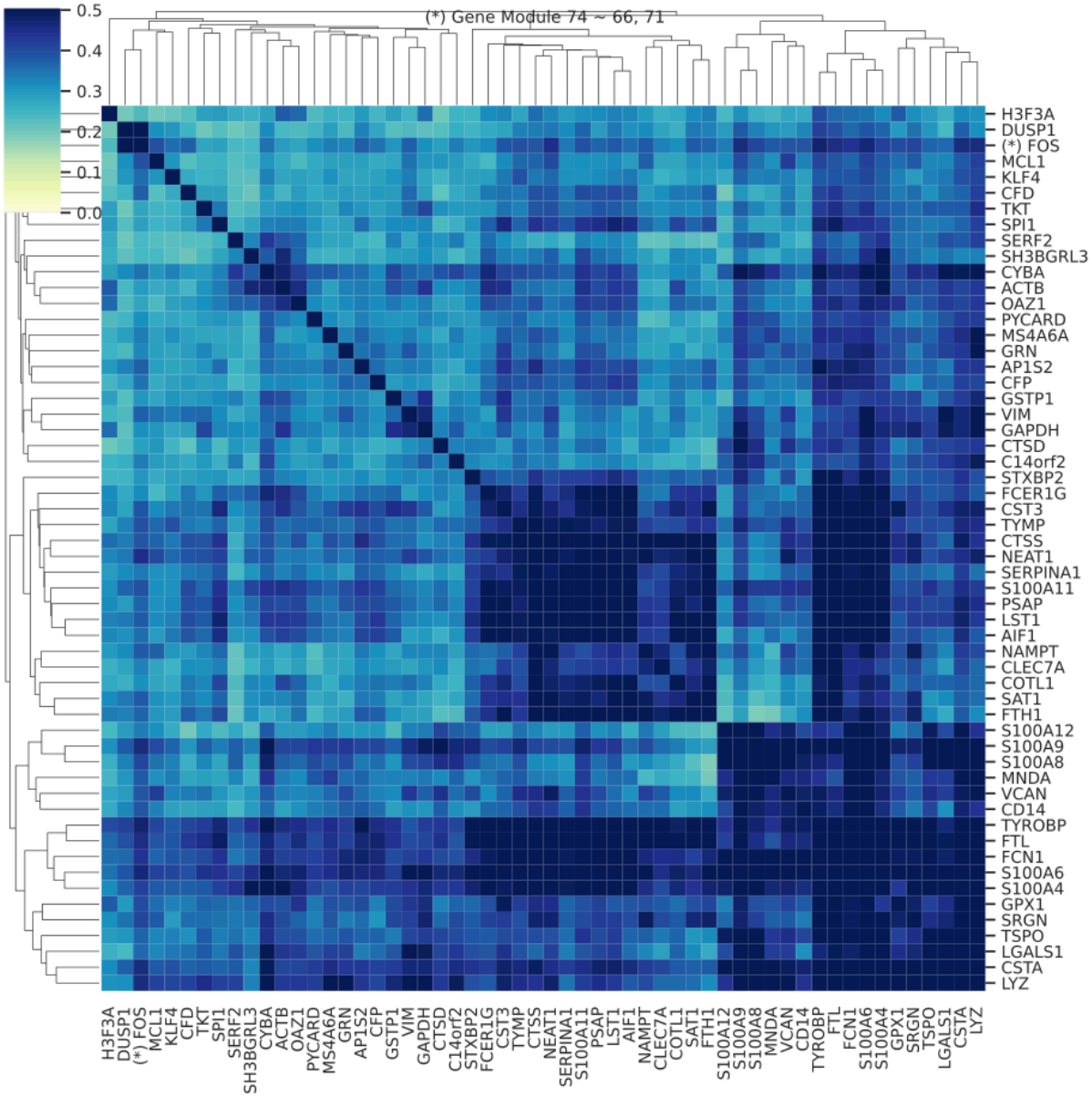

(*) Gene Module 71 ~ 63, 66, 74
0.5
0.0
(*) 63
(*) 74
(*) 66
(*) 71

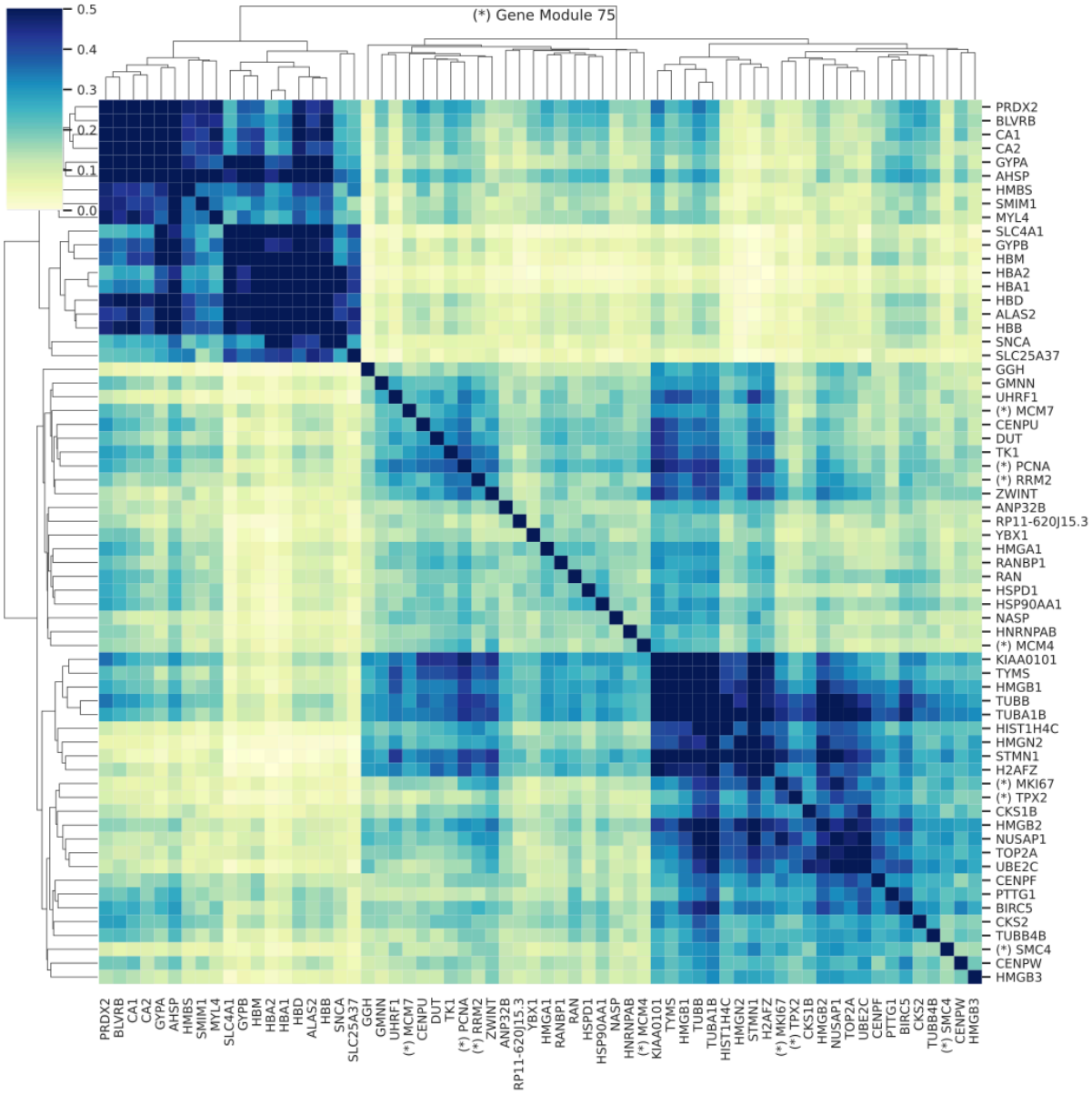


<Figure size 612x612 with 0 Axes>


(*) Gene Module 73
0.5
0.4
0.3
0.2
0.1
0.0
GZMH
HLA-B
CCL5
HLA-A
HLA-C
B2M
NKG7
CTSW
GZMA
CST7
FGFBP2
PRF1
GNLY
GZMB
HOPX
KLRF1
KLRD1
CD247
SPON2
CLIC3
RHOC
FCGR3A
IFITM2
MYO1F
ITGB2
TMSB4X
PFN1
HCST
CMC1
GZMM
CD160
APMAP
MATK
IL2RB
TRDC
KLRB1
CD7
ID2
MYL12A
CD99
C12orf75
UBB
HLA-E
(*) RPLP1

(*) Gene Module 74 ~ 66, 71
0.5
0.0
(*) 74
(*) 66
(*) 71


(*) Gene Module 74 ~ 66, 71
0.5
0.4
0.3
0.2
0.1
0.0
H3F3A
DUSP1
(*) FOS
MCL1
KLF4
CFD
TKT
SPI1
SERF2
SH3BGRL3
CYBA
ACTB
OAZ1
PYCARD
MS4A6A
GRN
AP1S2
CFP
GSTP1
VIM
GAPDH
CTSD
C14orf2
STXBP2
FCER1G
CST3
TYMP
CTSS
NEAT1
SERPINA1
S100A11
PSAP
LST1
AIF1
NAMPT
CLEC7A
COTL1
SAT1
FTH1
S100A12
S100A9
S100A8
MNDA
VCAN
CD14
TYROBP
FTL
FCN1
S100A6
S100A4
GPX1
SRGN
TSPO
LGALS1
CSTA
LYZ




```
CPU times: user 1min 32s, sys: 1min 51s, total: 3min 23s
Wall time: 2min 52s
```

Having looked at all the above (which admittedly, takes some effort), we made the following decisions (which we hope you will find to be reasonable). Again, we don't expect it to be perfect (yet), we'll revise this list in future iterations. But the better we make it now, the less iterations we'll need to make.

In [23]:
```
LATERAL_GENE_MODULES = [
    66, 71,  # Ribosomal.
    75,      # Cell-cycle, with some exceptions (erythrocytes).
]
SPECIFIC_LATERAL_GENES = ["DUSP1"]  # Stress
SPECIFIC_NON_LATERAL_GENES = [ # Excluded from 75, erythrocytes
    "AHSP", "ALAS2", "BLVRB", "CA1", "CA2", "GYPA", "GYPB", "HBA1", "HBA2",
    "HMBS", "MYL4", "PRDX2", "SLC25A37", "SLC4A1", "SMIM1", "SNCA"
]
```

In [24]:
```
%%time
for gene_module in LATERAL_GENE_MODULES:
    module_genes_mask = module_per_gene == gene_module
    update_lateral_genes(names=cells.var_names[module_genes_mask], op="add",
update_lateral_genes(names=SPECIFIC_LATERAL_GENES, op="add", show=False)
update_lateral_genes(names=SPECIFIC_NON_LATERAL_GENES, op="remove")
```

```
set hca_bm.iteration-1.cells.var[lateral_gene]: 116 true (0.4255%) out of 27
262 bools
set hca_bm.iteration-1.cells.var[lateral_gene]: 122 true (0.4475%) out of 27
262 bools
set hca_bm.iteration-1.cells.var[lateral_gene]: 179 true (0.6566%) out of 27
262 bools
set hca_bm.iteration-1.cells.var[lateral_gene]: 180 true (0.6603%) out of 27
262 bools
set hca_bm.iteration-1.cells.var[lateral_gene]: 161 true (0.5906%) out of 27
262 bools
['ANP32B', 'AURKA', 'BIRC5', 'CD63', 'CENPF', 'CENPU', 'CENPW', 'CKS1B', 'CK
S2', 'DUSP1', 'DUT', 'EEF1A1', 'EEF1B2', 'EIF3E', 'FOS', 'GGH', 'GLTSCR2',
'GMNN', 'H2AFZ', 'HIST1H4C', 'HMGA1', 'HMGB1', 'HMGB2', 'HMGB3', 'HMGN2', 'H
NRNPAB', 'HSP90AA1', 'HSP90AB1', 'HSPD1', 'KIAA0101', 'LTB', 'MCM3', 'MCM4',
'MCM7', 'MKI67', 'MYL6', 'NASP', 'NUSAP1', 'PCNA', 'PTTG1', 'RAN', 'RANBP1',
'RP11-620J15.3', 'RRM2', 'SMC4', 'STMN1', 'TK1', 'TOP2A', 'TPX2', 'TUBA1B',
'TUBB', 'TUBB4B', 'TXN', 'TYMS', 'UBE2C', 'UHRF1', 'YBX1', 'ZWINT']
and 103 RP[LS].* genes
CPU times: user 37.3 ms, sys: 224 µs, total: 37.6 ms
Wall time: 36.5 ms
```

4.1.3 Noisy genes

Another important list of genes are the genes with higher than usual variance, which will cause a lot of cells to be deviant (inconsistent with their metacells), and therefore be marked as outliers. Marking them as "noisy" allows for more variance for these genes in each metacell, reducing the number of outliers.

We currently don't provide a cell-based method for finding such genes; instead we wait until we have a metacells model and use that as a basis for decision instead.

4.1.4 Parallelization

Finally, we need to decide on how much parallelization to use. This is a purely technical decision - it does not affect the results, just the performance.

The more parallel piles we use, the faster the computation will be (up to the number of physical processors, but that is handled automatically for us).

However, having more parallel piles means using more memory. If we run out of memory, we'll need to reduce the number of parallel piles. You can track the memory usage by running `top` or `htop` during the computation.

We provide a guesstimator for the maximal number of parallel piles that will fit in memory. This is by no means perfect, but it is a starting point.

In [25]:
```
%%time
# Either use the guesstimator:
max_parallel_piles = mc.pl.guess_max_parallel_piles(cells)
# Or, if running out of memory manually override:
# max_paralle_piles = ...
print(max_parallel_piles)
mc.pl.set_max_parallel_piles(max_parallel_piles)
```

```
497
CPU times: user 8.54 s, sys: 11.1 s, total: 19.6 s
Wall time: 19.6 s
```

4.2 Computation

We are (*finally*) ready to actually group the cells into metacells. At least, for the first time. Since we don't trust the decisions we made so far too much, we don't expect these 1st iteration metacells to be worth much, either.

4.2.1 Hyper-parameters

The metacells pipeline has a lot of hyper parameters you can tweak. The defaults were chosen such that scRNA-seq data, especially 10x data, should work "well" out of the box. You should read the documentation and have a good understanding of the effect of any parameter you may want to tweak, keeping in mind the synergy between some of the parameters.

If we had to call out one hyper-parameter you might wish to tweak, it would be the target_metacell_size. This specifies the "ideal" number of cells in each metacell. The algorithm works hard to keep the actual metacell size close to this value - in particular, metacells larger than twice this size will be split, and metacells which are much smaller than this size will be merged, or dissolved (become outliers).

By default this value is set to 96. Setting it to a smaller value will create more metacells, which may allow capturing more subtle differences between cell states (e.g. along a gradient); this, however, would come at the cost of making less robust estimations of each gene's expression level in each metacell. This might be a worthwhile tradeoff if your cells are of higher quality.

In this vignette, we'll use the default across all iterations. Realistically, one may try different hyper parameters in different iterations.

One technical "hyper-paramater" you must specify is the random_seed. A non-zero value will genenrate reproducible results. A zero value will generate non-reproducible results and will

be slightly faster. We "strongly urge" you to use a non-zero as reproducible results are much easier to deal with.

4.2.2 Assigning cells to metacells

This is the core of the method. It can take a while. The dataset used in this example is far from trivial - it contains ~1/3M cells. This takes ~10 minutes to compute on a hefty (48 HT cores, 0.5TB RAM) server. You can see progress is being made by running the computation with a progress bar. This progress is rather non-linear, for example there's a pause at the start when computing rare gene modules, and it isn't smooth afterwards either. You can skip it alltogether by getting rid of the `with` statement.

In [26]:
```
%%time
with mc.ut.progress_bar():
    mc.pl.divide_and_conquer_pipeline(cells, random_seed=123456)
```

```
Detect rare gene modules...
100%|[06:34]
CPU times: user 3min 27s, sys: 3min 6s, total: 6min 33s
Wall time: 7min 27s
```

4.2.3 Collecting the metacells

The above merely computed a metacell name and index for each cell ("Outliers" and negative for outlier cells). We still need to collect all the cells of each metacell, to create a new `AnnData` where each profile is a metacell. Note that in this new metacells data, we no longer have UMIs per gene; instead, for each gene, we have an estimate of the fraction of its UMIs out of the total UMIs. Since `AnnData` can only hold a single 2D dataset, the result must be a separate object (with each "observation" being a metacell), so we copy all the per-gene annotations from the cells dataset to the result.

In [27]:
```
%%time
metacells = mc.pl.collect_metacells(cells, name="hca_bm.iteration-1.metacell
print(f"Iteration 1: {metacells.n_obs} metacells, {metacells.n_vars} genes")
```

```
set hca_bm.iteration-1.metacells.obs[grouped]: 3151 int64s
set hca_bm.iteration-1.metacells.obs[total_umis]: 3151 float64s
set hca_bm.iteration-1.metacells.layers[total_umis]: ndarray 3151 X 27262 fl
oat32s
set hca_bm.iteration-1.metacells.obs[__zeros_downsample_umis]: 3151 int64s
set hca_bm.iteration-1.metacells.layers[zeros]: ndarray 3151 X 27262 int32s
set hca_bm.iteration-1.cells.obs[metacell_name]: 297890 <U8s
set hca_bm.iteration-1.metacells.var[gene_ids]: 27262 objects
set hca_bm.iteration-1.metacells.var[bursty_lonely_gene]: 0 true (0%) out of
27262 bools
set hca_bm.iteration-1.metacells.var[properly_sampled_gene]: 27262 true (10
0%) out of 27262 bools
set hca_bm.iteration-1.metacells.var[excluded_gene]: 0 true (0%) out of 2726
2 bools
set hca_bm.iteration-1.metacells.var[full_gene_index]: 27262 int32s
set hca_bm.iteration-1.metacells.var[lateral_gene]: 161 true (0.5906%) out o
f 27262 bools
set hca_bm.iteration-1.metacells.var[lateral_genes_module]: 27262 int32s
set hca_bm.iteration-1.metacells.var[selected_gene]: 3542 true (12.99%) out
of 27262 bools
set hca_bm.iteration-1.metacells.var[rare_gene]: 72 true (0.2641%) out of 27
262 bools
set hca_bm.iteration-1.metacells.var[rare_gene_module]: 27262 int32s
set hca_bm.iteration-1.metacells.obs[metacells_rare_gene_module]: 3151 int32
s
set hca_bm.iteration-1.metacells.obs[rare_metacell]: 16 true (0.5078%) out o
f 3151 bools
set hca_bm.iteration-1.metacells.uns[outliers]: 110
set hca_bm.iteration-1.metacells.uns[metacells_algorithm]: metacells.0.9.0-d
ev.1
Iteration 1: 3151 metacells, 27262 genes
CPU times: user 8.76 s, sys: 46.5 s, total: 55.3 s
Wall time: 1min 6s
```

If the input data contains per-cell annotations of interest, then here would be a good place to convey that data to the metacells object. Since each metacell is a combination of multiple cells, this requires a decision on how exactly to do this for each per-cell annotation. We can either reduce the multiple per-cell values into a single one (such as the most frequent value or the mean value), or we can collect the fraction of cells that have each value in each metacell (for categorical data). Since `AnnData` is limited to simple 2D data, the latter is implemented as a series of per-metacell annotations (one for each possible value). This will be handled better in `DAF`.

In [28]:
```
%%time

# We'll reuse this through the iterations.
def convey_cell_annotations_to_metacells() -> None:

    # Assign a single value for each metacell based on the cells.
    mc.tl.convey_obs_to_group(
        adata=cells, gdata=metacells,
        property_name="donor_organism.organism_age", to_property_name="sex",
        method=mc.ut.most_frequent  # This is the default, for categorical d
```

```
    )
    mc.tl.convey_obs_to_group(
        adata=cells, gdata=metacells,
        property_name="donor_organism.organism_age", to_property_name="age",
        method=np.mean
    )

    # Compute the fraction of cells with each possible value in each metacel
    mc.tl.convey_obs_fractions_to_group(
        adata=cells, gdata=metacells,
        property_name="donor_organism.sex", to_property_name="sex"
    )

    # Age has just a few possible values so treat it as categorical.
    mc.tl.convey_obs_fractions_to_group(
        adata=cells, gdata=metacells,
        property_name="donor_organism.organism_age", to_property_name="age"
    )

    mc.tl.convey_obs_fractions_to_group(adata=cells, gdata=metacells, proper
    mc.tl.convey_obs_fractions_to_group(adata=cells, gdata=metacells, proper

convey_cell_annotations_to_metacells()
```

```
set hca_bm.iteration-1.metacells.obs[sex]: 3151 float32s
set hca_bm.iteration-1.metacells.obs[age]: 3151 float32s
set hca_bm.iteration-1.metacells.obs[sex_fraction_of_female]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[sex_fraction_of_male]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_26.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_29.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_32.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_36.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_39.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_50.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[age_fraction_of_52.0]: 3151 float64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM1]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM2]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM3]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM4]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM5]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM6]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM7]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[donor_fraction_of_MantonBM8]: 3151 floa
t64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_7]: 3
```

```
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_4]: 3
```

```
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_8]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_1]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_2]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_3]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_4]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_5]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_6]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_7]: 3
151 float64s
set hca_bm.iteration-1.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_8]: 3
151 float64s
CPU times: user 24.8 s, sys: 482 ms, total: 25.3 s
Wall time: 25.2 s
```

4.3 Computing for MCView

So we have our metacells data (that we only trust as much as our decisions leading to them). This is pretty opaque as of itself. `MCView` is our interactive tool for exploring the data, refining our decisions, and assigning type labels to the metacells.

However, in order to use `MCView`, we first need to compute all sort of quality control data for `MCView` to display. This again may take a while (but much less then computing the metacells above).

In [29]:
```
%%time
with mc.ut.progress_bar():
    mc.pl.compute_for_mcview(adata=cells, gdata=metacells, random_seed=12345
```

```
100%|██████████████████████████████████████████████████████████
██████████████████████████████████████████████████████[04:55]
CPU times: user 1h 18min 39s, sys: 4min 51s, total: 1h 23min 31s
Wall time: 4min 55s
```

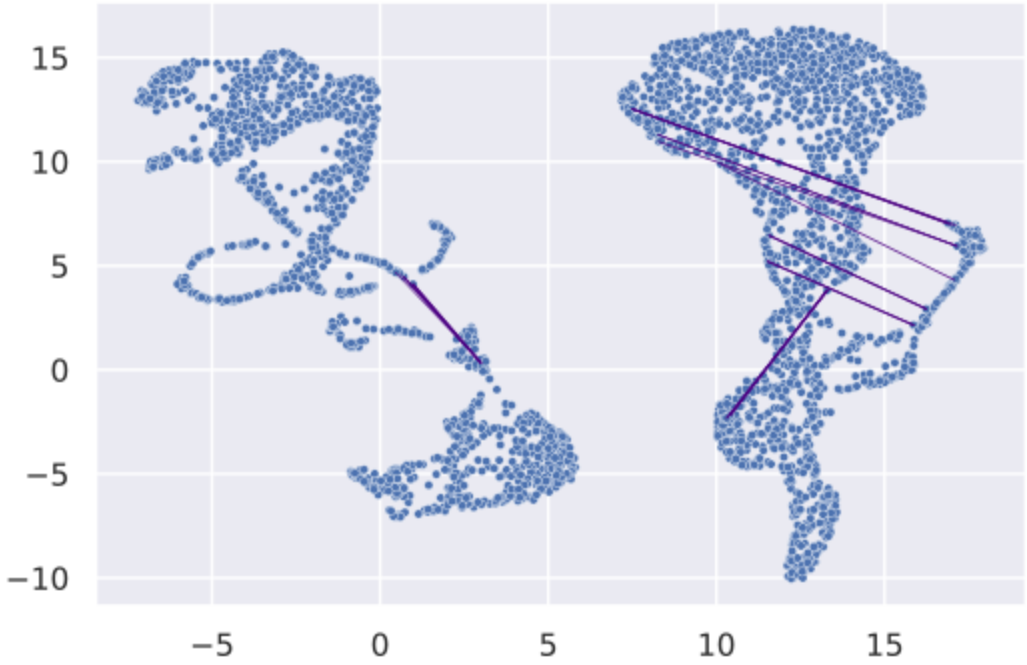
Here's a preview of the 2D UMAP view of the data (without any color annotations, as we do not have type annotations yet):

In [30]:
```
%%time

# We'll reuse this through the iterations.
def plot_umap(*, type_annotation: Optional[str]) -> None:
    if type_annotation is not None:
        type_color_csv = pd.read_csv("../captured/type_colors.csv")
        color_of_type = pd.Series(
            list(type_color_csv["color"]) + ["magenta", "magenta"],
            index=list(type_color_csv["cell_type"]) + ["Outliers", "(Missing
        )
        type_of_metacell = mc.ut.get_o_numpy(metacells, type_annotation)
        color_of_metacell = np.array(color_of_type[type_of_metacell])

    min_long_edge_size = 4
    umap_x = mc.ut.get_o_numpy(metacells, "x")
    umap_y = mc.ut.get_o_numpy(metacells, "y")
    umap_edges = sp.coo_matrix(mc.ut.get_oo_proper(metacells, "obs_outgoing_
    sb.set()
    if type_annotation is None:
        plot = sb.scatterplot(x=umap_x, y=umap_y, s=10)
    else:
        plot = sb.scatterplot(x=umap_x, y=umap_y, color=color_of_metacell, s
    for (
        source_index, target_index, weight
    ) in zip(
        umap_edges.row, umap_edges.col, umap_edges.data
    ):
        source_x = umap_x[source_index]
        target_x = umap_x[target_index]
        source_y = umap_y[source_index]
        target_y = umap_y[target_index]
        if hypot(target_x - source_x, target_y - source_y) >= min_long_edge_
            plt.plot([source_x, target_x], [source_y, target_y],
                     linewidth=weight * 2, color='indigo')
    plt.show()

plot_umap(type_annotation=None)
```




```
CPU times: user 1.28 s, sys: 5.04 s, total: 6.32 s
Wall time: 267 ms
```

4.4 Saving the data

We'll now save the results, so we can import them into `MCView`.

In [31]:
```
%%time
cells.write_h5ad("../output/iterative/iteration-1/hca_bm.cells.h5ad")
```

```
CPU times: user 584 ms, sys: 25 s, total: 25.5 s
Wall time: 53.7 s
```

In [32]:
```
%%time
metacells.write_h5ad("../output/iterative/iteration-1/hca_bm.metacells.h5ad"
```

```
CPU times: user 232 ms, sys: 689 ms, total: 921 ms
Wall time: 2.97 s
```

5. Importing into MCView

This vignette focuses on the metacells package, not `MCView`, which deserves a full vignette of its own. Still, here are some basics about how to use it.

5.1 Installing MCView

The `MCView` is written in `R` but is not a standard CRAN package. To install it, you should type (**in `R`**):

```
install.packages("remotes")
remotes::install_github("tanaylab/MCView")
```

5.2 Importing data set

Since `MCView` is written in `R`, it isn't easy to run it inside a Python notebook. Instead we've provided a small `R` script that will load the data we saved above, and import it into an `MCView` application. Here is the code of the script for reference:

```
library("MCView")

args <- commandArgs(trailingOnly=TRUE)

if (length(args) == 6) {
    prefix <- args[1]
    name <- args[2]
    title <- args[3]
    type <- args[4]
    atlas_prefix <- args[5]
    atlas_name <- args[6]
    import_dataset(
        sprintf("../mcview/%s", name),                          #
The directory to create
        sprintf("%s-%s", prefix, gsub("/", "-", name)),         #
The name of the dataset
        sprintf("../output/%s/%s.metacells.h5ad", name, prefix), #
The metacells h5ad file
        metadata_fields = "all",                                #
Ask to import all the metadata
        title = title,                                          #
A title for the GUI
        cell_type_field = type,                                 #
The name of the type field
        cell_type_colors_file = "../captured/type_colors.csv",  #
The type colors CSV file
        projection_weights_file =
sprintf("../output/%s/%s.atlas_weights.csv", name, prefix),
        atlas_project = sprintf("../mcview/%s", atlas_name),
        atlas_dataset = sprintf("%s-%s", atlas_prefix, gsub("/", "-
", atlas_name)),
    )

} else if (length(args) == 4) {
    prefix <- args[1]
    name <- args[2]
    title <- args[3]
    type <- args[4]
    import_dataset(
        sprintf("../mcview/%s", name),                          #
The directory to create
        sprintf("%s-%s", prefix, gsub("/", "-", name)),         #
The name of the dataset
        sprintf("../output/%s/%s.metacells.h5ad", name, prefix), #
The metacells h5ad file
```

```
        metadata_fields = "all",                                    #
Ask to import all the metadata
        title = title,                                              #
A title for the GUI
        cell_type_field = type,                                     #
The name of the type field
        cell_type_colors_file = "../captured/type_colors.csv"       #
The type colors CSV file
    )

} else if (length(args) == 3) {
    prefix <- args[1]
    name <- args[2]
    title <- args[3]
    import_dataset(
        sprintf("../mcview/%s", name),                              #
The directory to create
        sprintf("%s-%s", prefix, gsub("/", "-", name)),             #
The name of the dataset
        sprintf("../output/%s/%s.metacells.h5ad", name, prefix), #
The metacells h5ad file
        metadata_fields = "all",                                    #
Ask to import all the metadata
        title = title                                               #
A title for the GUI
    )

} else {
    stopifnot(FALSE)
}
```

We'll just run it as an external process using `Rscript`:

In [33]:
```
%%time
os.system("Rscript ../scripts/import_dataset.r hca_bm iterative/iteration-1
```

```
□ creating ../mcview/iterative/iteration-1
→ You can edit the app configuration at '../mcview/iterative/iteration-1/con
fig/config.yaml'
□ Importing hca_bm-iterative-iteration-1
□ Reading '../output/iterative/iteration-1/hca_bm.metacells.h5ad'
□ Processing metacell matrix
□ Processing 2d projection
□ Calculating top genes per metacell (marker genes)
□ Calculating metacell correlations of default marker genes
50%...65%...82%...95%...100%
□ Processing inner-folds matrix
□ Calculating top inner-fold genes
→ Added the Inner-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-1/config/
config.yaml'
□ Processing inner-stdev matrix
□ Calculating top inner-stdev genes
→ Added the Stdev-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-1/config/
config.yaml'
□ Clustering in order to get initial annotation.
□ using 172 genes
□ clustering k = 64
□ number of features = 172
□ Generating cell type colors using chameleon package.
□ Coloring using pre-calculated 3D umap
□ Clustering in order to get gene modules. k = 31
□ Number of genes considered = 1000
□ Loading previously calculated 30 correlated and anti-correlated genes for
each gene
✔ hca_bm-iterative-iteration-1 dataset imported succesfully to '../mcview/it
erative/iteration-1' project
• You can now run the app using: run_app("../mcview/iterative/iteration-1")
• or create a bundle using: create_bundle("../mcview/iterative/iteration-1",
name = "name_of_bundle")
CPU times: user 20.6 ms, sys: 915 ms, total: 935 ms
Wall time: 1min 47s
```

Out[33]: 0

6. Compute the 2nd iteration metacells

Having looked at our data in `MCView`, we are ready to revise our original decisions and compute higher quality metacells. In general, we will have to iterate several times, recomputing the metacells in each iteration, until the result is "good enough". We'll rename the data to reflect this. Note that the cells contain the old assignment to metacells; this is harmless as it will be replaced by the new metacells we'll compute.

In [34]:
```
%%time
mc.ut.set_name(cells, "hca_bm.iteration-2.cells")
print(f"Iteration 2: {cells.n_obs} cells, {cells.n_vars} genes")
```

```
Iteration 2: 297890 cells, 27262 genes
CPU times: user 102 µs, sys: 17 µs, total: 119 µs
Wall time: 113 µs
```

6.1 Decisions

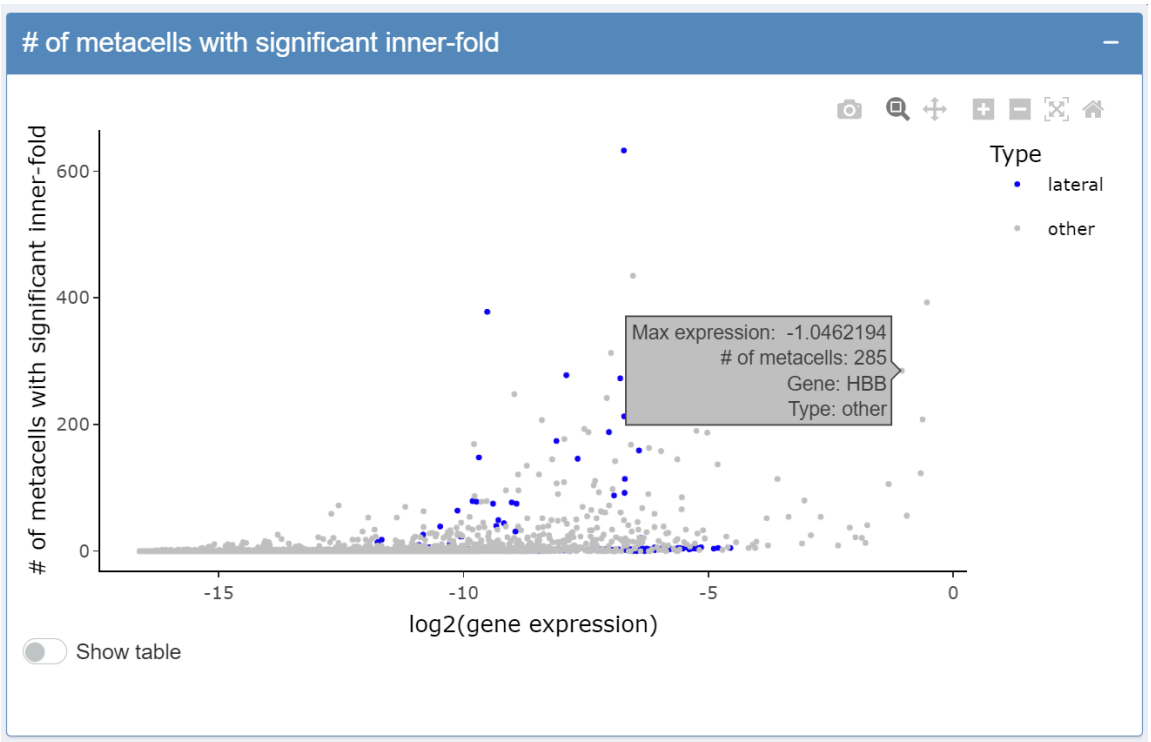
The revised decisions below are based on the analysis we performed in `MCView`. When anlyzing the first iteration results in `MCView`, we didn't bother to even try to add type annotations to the data, which makes this simpler. In the following iterations we'll add such annotations and convey them between the iterations.

6.1.2 Adding noisy genes

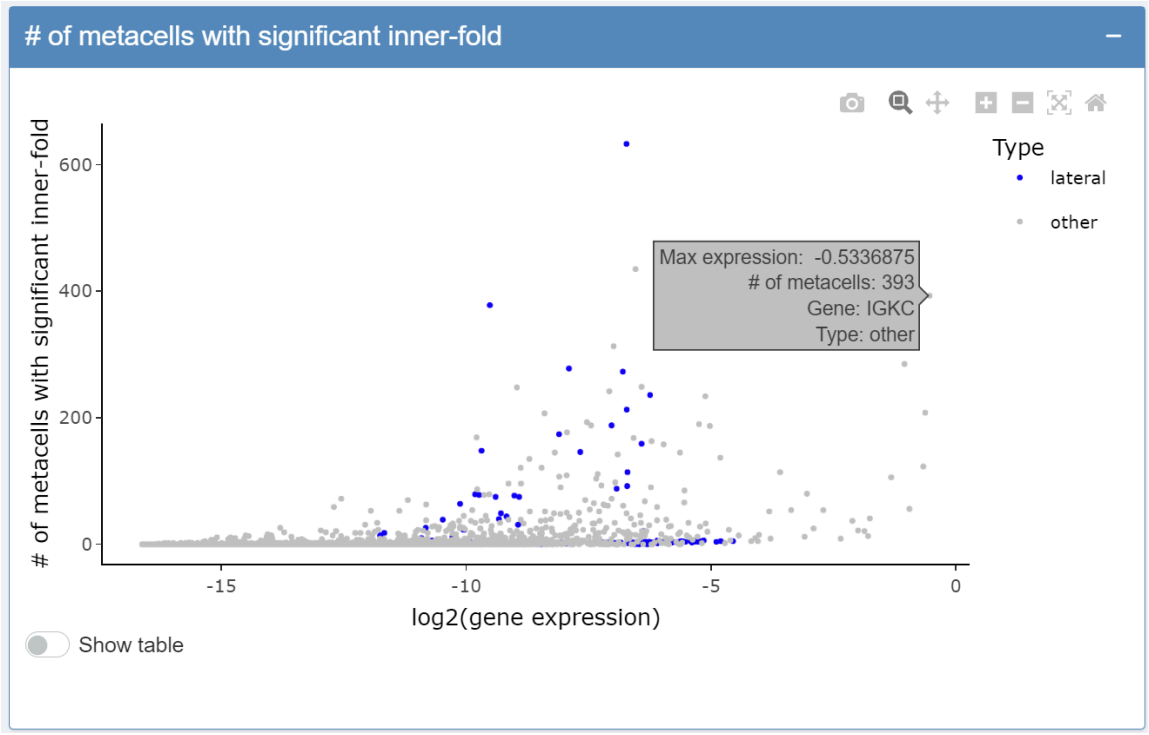
Looking in `MCView`, we have discovered genes which we should mark as noisy, because they have very high variance for cells with the "same" state (that is, within a metacell), and we think this makes biological sense.

The main tool we use for detecting noisy genes is the "# of metacells with significant inner-fold" QC graph. Assuming the algorithm works well, genes which have a high inner fold in many metacells must be due to them being extra noisy. These are also worth considering as lateral genes, but sometimes they are important for distinguishing between different cell types.

For example, `HBB` is noisy, but is a clear erythrocytes marker, so it should *not* be lateral:



While `IGKC` is also noisy, but is an immunoglobulin, so it *should* be lateral:



In [35]:
```
NOISY_GENE_NAMES = [
    "CCL3", "CCL4", "CCL5", "CXCL8", "DUSP1", "FOS", "G0S2", "HBB", "HIST1H4
    "IGLC2", "JUN", "JUNB", "KLRB1", "MT2A", "RP11", "RPS26", "RPS4Y1", "TRE
]
```

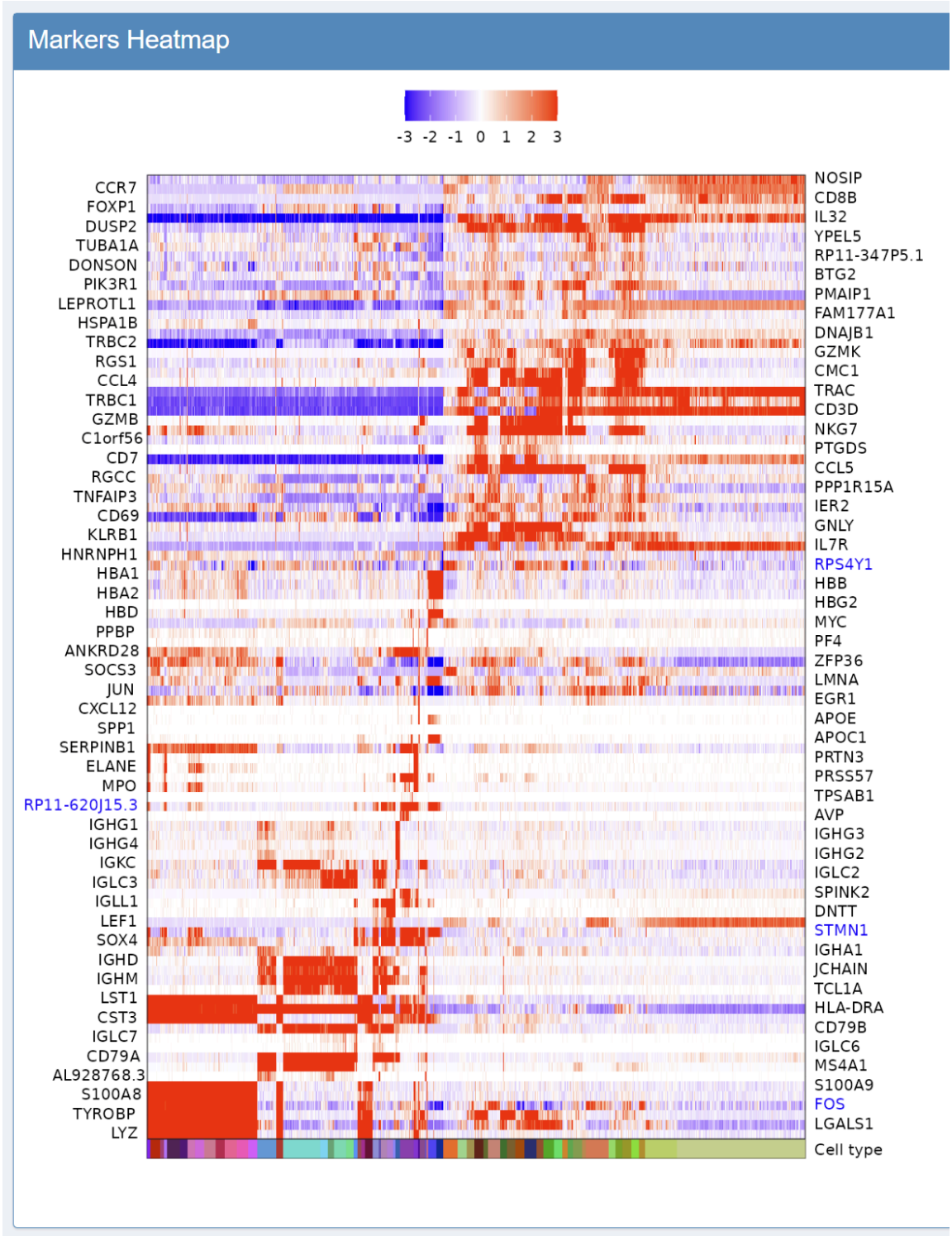
In [36]:
```
%%time
mc.pl.mark_noisy_genes(cells, noisy_gene_names=NOISY_GENE_NAMES)
```

```
set hca_bm.iteration-2.cells.var[noisy_gene]: 21 true (0.07703%) out of 2726
2 bools
CPU times: user 9.07 ms, sys: 12 µs, total: 9.08 ms
Wall time: 8.68 ms
```

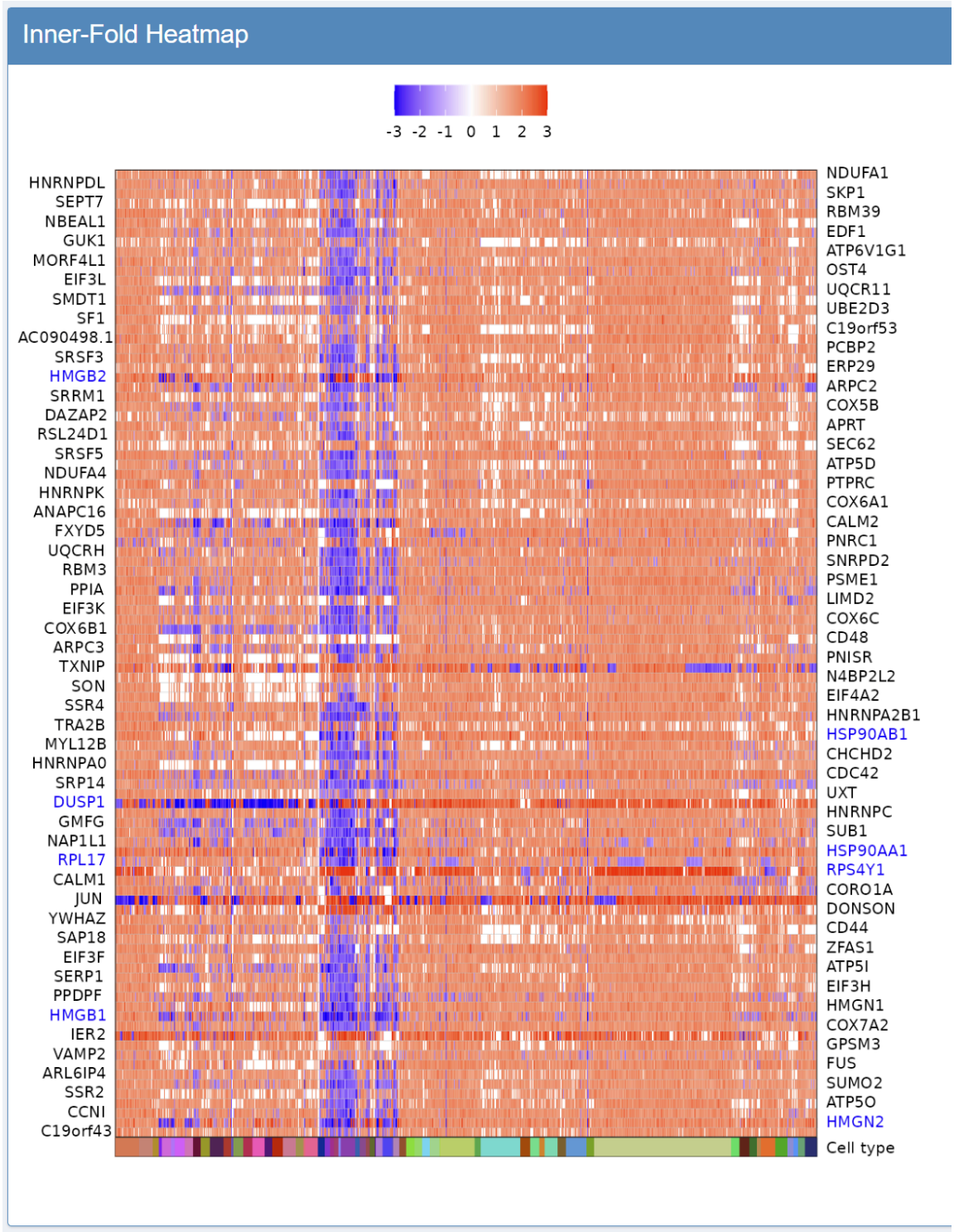
6.1.3 Adding lateral genes

Similarly, even given a cursory analysis of the data (without doing any manual type annotations), we already discovered more genes we need to mark as lateral (in additions to the lateral genes we noticed while looking for noisy genes).

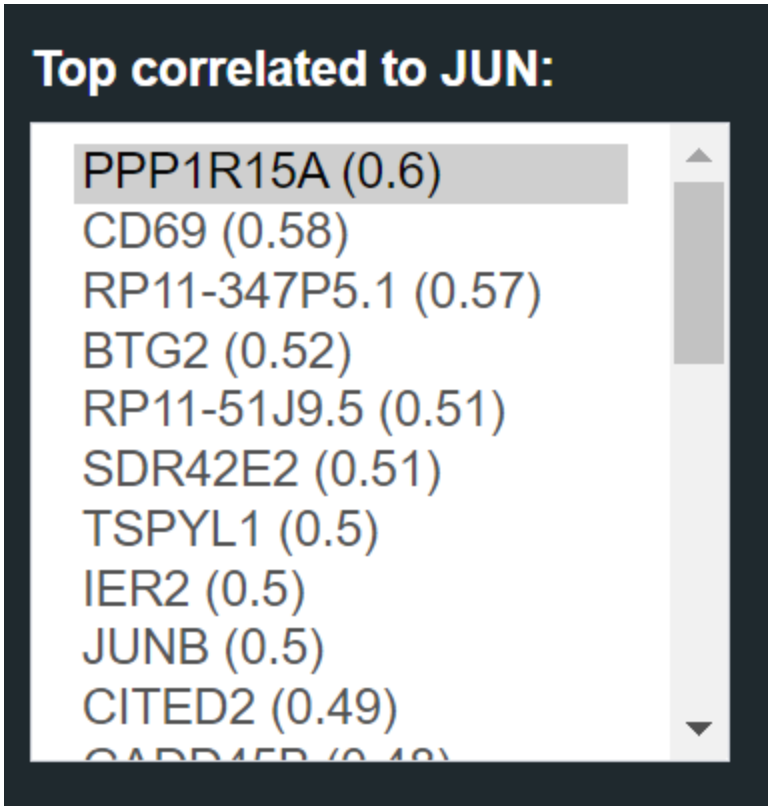
The main tool for identifying additional lateral genes is the marker heatmap. In general, we should have a pretty good idea about the function of each gene in this heatmap. When looking at the one we got, we noticed several genes which do not differentiate between the cell types we are interested in isolating, for example, `JUN` and `TUBA1A` (cell death).



Another tool is looking at the inner fold heatmap. This shows genes that the metacells algorithm did not manage to homogenize well, so it is a good place to look for residual lateral behavior genes. For example, `DONSON` is related to cell cycle.



Finally, it is worth looking at the genes which are highly correlated with known lateral genes (including the ones we found above). For example, `JUNB` is correlated with `JUN`.



In [37]:
```
ADD_LATERAL_GENE_NAMES = [
    "B2M", "BTG2", "CALM1", "COX4I1", "DNAJB1", "DONSON", "FTH1", "G0S2", "G
    "HLA-B", "HLA-C", "HLA-DMB", "HLA-DPA1", "HLA-DPB1", "HLA-DQB1", "HLA-DR
    "HSPA1A", "HSPA1B", "HSPA6", "HSPH1", "IER2", "IGHA1", "IGHA2", "IGHD",
    "IGLC2", "IGLC3", "IGLC6", "IGLL1", "JUN", "JUNB", "LEPROTL1", "LGALS1",
    "PDLIM1", "PLK3", "PPP1R15A", "PTMA", "RGCC", "SRSF7", "TMSB4X", "TUBA1A
    "YPEL5", "ZFP36"
]
```

In [38]:
```
%%time
update_lateral_genes(names=ADD_LATERAL_GENE_NAMES, op="add")
```

```
set hca_bm.iteration-2.cells.var[lateral_gene]: 214 true (0.785%) out of 272
62 bools
```

```
['ANP32B', 'AURKA', 'B2M', 'BIRC5', 'BTG2', 'CALM1', 'CD63', 'CENPF', 'CENP
U', 'CENPW', 'CKS1B', 'CKS2', 'COX4I1', 'DNAJB1', 'DONSON', 'DUSP1', 'DUT',
'EEF1A1', 'EEF1B2', 'EIF3E', 'FOS', 'FTH1', 'G0S2', 'GGH', 'GLTSCR2', 'GMN
N', 'GNB2L1', 'H2AFZ', 'H3F3B', 'HIST1H4C', 'HLA-A', 'HLA-B', 'HLA-C', 'HLA-
DMB', 'HLA-DPA1', 'HLA-DPB1', 'HLA-DQB1', 'HLA-DRA', 'HLA-E', 'HLA-F', 'HMGA
1', 'HMGB1', 'HMGB2', 'HMGB3', 'HMGN2', 'HNRNPAB', 'HSP90AA1', 'HSP90AB1',
'HSPA1A', 'HSPA1B', 'HSPA6', 'HSPD1', 'HSPH1', 'IER2', 'IGHA1', 'IGHA2', 'IG
HD', 'IGHG3', 'IGHM', 'IGKC', 'IGLC2', 'IGLC3', 'IGLC6', 'IGLL1', 'JUN', 'JU
NB', 'KIAA0101', 'LEPROTL1', 'LGALS1', 'LTB', 'MCM3', 'MCM4', 'MCM7', 'MKI6
7', 'MT2A', 'MYL12A', 'MYL6', 'NASP', 'NUSAP1', 'PCNA', 'PDLIM1', 'PLK3', 'P
PP1R15A', 'PTMA', 'PTTG1', 'RAN', 'RANBP1', 'RGCC', 'RP11-620J15.3', 'RRM2',
'SMC4', 'SRSF7', 'STMN1', 'TK1', 'TMSB4X', 'TOP2A', 'TPX2', 'TUBA1A', 'TUBA1
B', 'TUBB', 'TUBB4B', 'TXN', 'TYMS', 'UBA52', 'UBC', 'UBE2C', 'UHRF1', 'YBX
1', 'YPEL5', 'ZFP36', 'ZWINT']
and 103 RP[LS].* genes
CPU times: user 7.91 ms, sys: 0 ns, total: 7.91 ms
Wall time: 7.66 ms
```

6.1.4 Parallelization

Since this will be essentially the same data set as before, normally we'll just reuse the same amount of parallelization as in the first iteration for the rest of the process.

6.2 Computation

The steps here just repeat what we did in the first iteration.

In [39]:
```
%%time

def compute_next_iteration(next_iteration_index: int) -> None:

    print("# DIVIDE AND CONQUER...")
    global metacells
    metacells = None # So can be gc-ed
    mc.pl.divide_and_conquer_pipeline(cells, random_seed=123456)

    print("# COLLECT METACELLS...")
    metacells = mc.pl.collect_metacells(
        cells, name=f"hca_bm.iteration-{next_iteration_index}.metacells", ra
    )
    print(f"Iteration {next_iteration_index}: {metacells.n_obs} metacells, {

    print("# CONVEY CELL ANNOTATIONS...")
    convey_cell_annotations_to_metacells()

def finalize_next_iteration(next_iteration_index: int, *, with_types: bool)
    print("# COMPUTE FOR MCVIEW...")
    mc.pl.compute_for_mcview(adata=cells, gdata=metacells, random_seed=12345

    print("# PLOT UMAP...")
    if with_types:
        type_annotation = f"type.iteration-{next_iteration_index}.auto"
    else:
        type_annotation = None
```

```
    plot_umap(type_annotation=type_annotation)

    print("# SAVE CELLS...")
    cells.write_h5ad(f"../output/iterative/iteration-{next_iteration_index}/

    print("# SAVE METACELLS...")
    metacells.write_h5ad(f"../output/iterative/iteration-{next_iteration_inc

    print("# IMPORT TO MCVIEW...")
    os.system(
        f"Rscript ../scripts/import_dataset.r hca_bm iterative/iteration-{ne
        f"'HCABM IT|{next_iteration_index}'"
    )

def next_iteration_without_types(next_iteration_index: int) -> None:
    compute_next_iteration(next_iteration_index)
    finalize_next_iteration(next_iteration_index, with_types=False)

next_iteration_without_types(2)
```

```
set hca_bm.iteration-2.cells.var[selected_gene]: * -> False
# DIVIDE AND CONQUER...
set hca_bm.iteration-2.cells.var[rare_gene]: 72 true (0.2641%) out of 27262
bools
set hca_bm.iteration-2.cells.var[rare_gene_module]: 27190 outliers (99.74%)
and 72 grouped (0.2641%) out of 27262 int32 elements with 7 groups with mean
size 10.29
set hca_bm.iteration-2.cells.obs[cells_rare_gene_module]: 296459 outliers (9
9.52%) and 1431 grouped (0.4804%) out of 297890 int32 elements with 7 groups
with mean size 204.4
set hca_bm.iteration-2.cells.obs[rare_cell]: 1431 true (0.4804%) out of 2978
90 bools
set hca_bm.iteration-2.cells.var[selected_gene]: 3550 true (13.02%) out of 2
7262 bools
set hca_bm.iteration-2.cells.obs[metacell]: 297890 int32s
set hca_bm.iteration-2.cells.obs[dissolved]: 0 true (0%) out of 297890 bools
set hca_bm.iteration-2.cells.obs[metacell_level]: 297890 int32s
# COLLECT METACELLS...
```

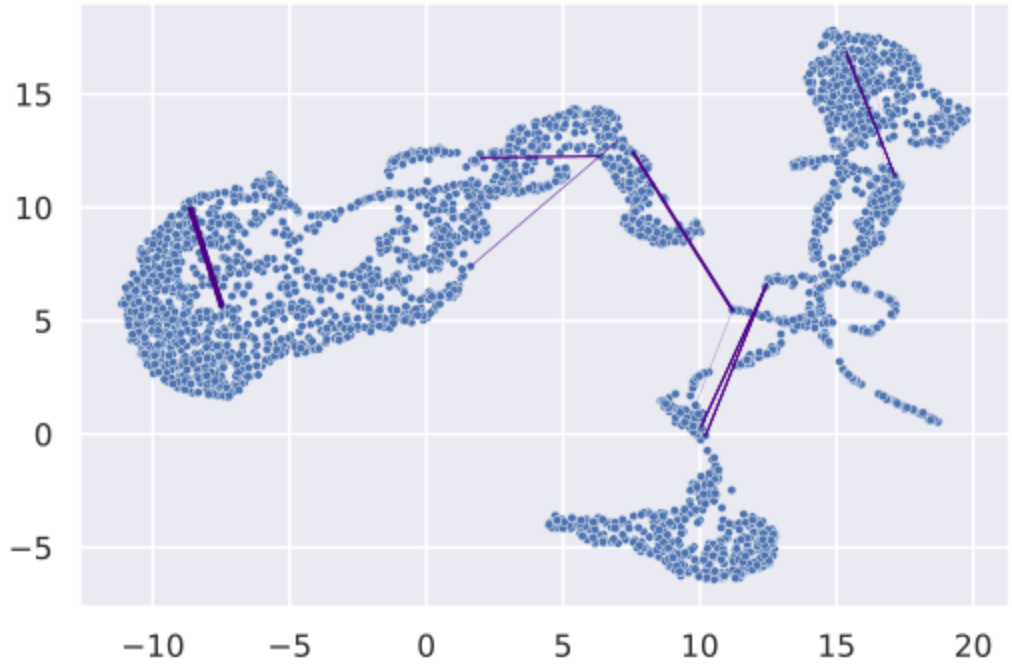
```
set hca_bm.iteration-2.metacells.obs[grouped]: 3163 int64s
set hca_bm.iteration-2.metacells.obs[total_umis]: 3163 float64s
set hca_bm.iteration-2.metacells.layers[total_umis]: ndarray 3163 X 27262 fl
oat32s
set hca_bm.iteration-2.metacells.obs[__zeros_downsample_umis]: 3163 int64s
set hca_bm.iteration-2.metacells.layers[zeros]: ndarray 3163 X 27262 int32s
set hca_bm.iteration-2.cells.obs[metacell_name]: 297890 <U8s
set hca_bm.iteration-2.metacells.var[gene_ids]: 27262 objects
set hca_bm.iteration-2.metacells.var[bursty_lonely_gene]: 0 true (0%) out of
27262 bools
set hca_bm.iteration-2.metacells.var[properly_sampled_gene]: 27262 true (10
0%) out of 27262 bools
set hca_bm.iteration-2.metacells.var[excluded_gene]: 0 true (0%) out of 2726
2 bools
set hca_bm.iteration-2.metacells.var[full_gene_index]: 27262 int32s
set hca_bm.iteration-2.metacells.var[lateral_gene]: 214 true (0.785%) out of
27262 bools
set hca_bm.iteration-2.metacells.var[lateral_genes_module]: 27262 int32s
set hca_bm.iteration-2.metacells.var[selected_gene]: 3550 true (13.02%) out
of 27262 bools
set hca_bm.iteration-2.metacells.var[rare_gene]: 72 true (0.2641%) out of 27
262 bools
set hca_bm.iteration-2.metacells.var[rare_gene_module]: 27262 int32s
set hca_bm.iteration-2.metacells.var[noisy_gene]: 21 true (0.07703%) out of
27262 bools
set hca_bm.iteration-2.metacells.obs[metacells_rare_gene_module]: 3163 int32
s
set hca_bm.iteration-2.metacells.obs[rare_metacell]: 16 true (0.5058%) out o
f 3163 bools
set hca_bm.iteration-2.metacells.uns[outliers]: 159
set hca_bm.iteration-2.metacells.uns[metacells_algorithm]: metacells.0.9.0-d
ev.1
Iteration 2: 3163 metacells, 27262 genes
# CONVEY CELL ANNOTATIONS...
```

```
set hca_bm.iteration-2.metacells.obs[sex]: 3163 float32s
set hca_bm.iteration-2.metacells.obs[age]: 3163 float32s
set hca_bm.iteration-2.metacells.obs[sex_fraction_of_female]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[sex_fraction_of_male]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_26.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_29.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_32.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_36.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_39.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_50.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[age_fraction_of_52.0]: 3163 float64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM1]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM2]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM3]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM4]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM5]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM6]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM7]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[donor_fraction_of_MantonBM8]: 3163 floa
t64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_7]: 3
```

```
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_4]: 3
```

```
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_1]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_2]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_3]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_4]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_5]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_6]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_7]: 3
163 float64s
set hca_bm.iteration-2.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_8]: 3
163 float64s
set hca_bm.iteration-2.metacells.uns[mcview_format]: 1.0
set hca_bm.iteration-2.metacells.uns[outliers]: 159
# COMPUTE FOR MCVIEW...
```

```
set hca_bm.iteration-2.metacells.var[marker_gene]: 3377 true (12.39%) out of
27262 bools
set hca_bm.iteration-2.metacells.obsp[obs_balanced_ranks]: 38585 nonzero (0.
3857%) out of 10004569 elements
set hca_bm.iteration-2.metacells.obsp[obs_pruned_ranks]: 12639 nonzero (0.12
63%) out of 10004569 elements
set hca_bm.iteration-2.metacells.obsp[obs_outgoing_weights]: 12639 nonzero
(0.1263%) out of 10004569 elements
set hca_bm.iteration-2.metacells.obsp[umap_distances]: csr_matrix 3163 X 316
3 float32s (10001406 > 0, 99.97%)
set hca_bm.iteration-2.metacells.obs[u]: 3163 float32s
set hca_bm.iteration-2.metacells.obs[v]: 3163 float32s
set hca_bm.iteration-2.metacells.obs[w]: 3163 float32s
set hca_bm.iteration-2.metacells.obsp[obs_balanced_ranks]: 38585 nonzero (0.
3857%) out of 10004569 elements
set hca_bm.iteration-2.metacells.obsp[obs_pruned_ranks]: 12639 nonzero (0.12
63%) out of 10004569 elements
set hca_bm.iteration-2.metacells.obsp[obs_outgoing_weights]: 12639 nonzero
(0.1263%) out of 10004569 elements
set hca_bm.iteration-2.metacells.obsp[umap_distances]: csr_matrix 3163 X 316
3 float32s (10001406 > 0, 99.97%)
set hca_bm.iteration-2.metacells.obs[x]: 3163 float32s
set hca_bm.iteration-2.metacells.obs[y]: 3163 float32s
set hca_bm.iteration-2.cells.obs[most_similar]: 297890 int32s
set hca_bm.iteration-2.cells.layers[deviant_fold]: csr_matrix 297890 X 27262
float32s (176746871 > 0, 2.176%)
set hca_bm.iteration-2.metacells.layers[inner_fold]: csr_matrix 3163 X 27262
float32s (1800512 > 0, 2.088%)
set hca_bm.iteration-2.metacells.var[significant_inner_folds_count]: 27262 i
nt64s with mean 0.8005
set hca_bm.iteration-2.metacells.layers[inner_stdev_log]: csr_matrix 3163 X
27262 float32s (1757191 > 0, 2.038%)
set hca_bm.iteration-2.metacells.varp[var_similarity]: csr_matrix 27262 X 27
262 float32s (1316501 > 0, 0.1771%)
# PLOT UMAP...
```



```
# SAVE CELLS...
# SAVE METACELLS...
# IMPORT TO MCVIEW...
□ creating ../mcview/iterative/iteration-2
→ You can edit the app configuration at '../mcview/iterative/iteration-2/con
fig/config.yaml'
□ Importing hca_bm-iterative-iteration-2
□ Reading '../output/iterative/iteration-2/hca_bm.metacells.h5ad'
□ Processing metacell matrix
□ Processing 2d projection
□ Calculating top genes per metacell (marker genes)
□ Calculating metacell correlations of default marker genes
□ Processing inner-folds matrix
□ Calculating top inner-fold genes
→ Added the Inner-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-2/config/
config.yaml'
□ Processing inner-stdev matrix
□ Calculating top inner-stdev genes
→ Added the Stdev-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-2/config/
config.yaml'
□ Clustering in order to get initial annotation.
□ using 166 genes
□ clustering k = 64
□ number of features = 166
□ Generating cell type colors using chameleon package.
□ Coloring using pre-calculated 3D umap
□ Clustering in order to get gene modules. k = 31
□ Number of genes considered = 1000
□ Loading previously calculated 30 correlated and anti-correlated genes for
each gene
✔ hca_bm-iterative-iteration-2 dataset imported succesfully to '../mcview/it
erative/iteration-2' project
• You can now run the app using: run_app("../mcview/iterative/iteration-2")
• or create a bundle using: create_bundle("../mcview/iterative/iteration-2",
name = "name_of_bundle")
CPU times: user 1h 33min, sys: 8min 14s, total: 1h 41min 15s
Wall time: 16min 26s
```

7. Annotating types in MCView

This time, we expect the metacells to have sufficient quality so it is worthwhile to provide manual type annotations for them. When we'll iterate further, we'll use the previous iteration annotations as the basis for the next one, that is, the annotations are one more decision we refine during the iterations.

7.1 Type annotation decisions

The basic procedure we use for annotating metacell types is to look at the markers heatmap to identify interesting genes, and then using the gene-gene plots to identify groups of

metacells that present the right combination of markers to justify some type label. Here's an example gene-gene plot showing a gradient between HSC (Hematopoietic Stem Cell) and LMPP (Lymphoid-primed Multipotent Progenitors):



If some metacells express a combination of markers for two unrelated types, we will consider labeling them as doublets. All these decisions require prior biological knowledge of the genes and the system. Here's an example of a doublet metacell containing a mixture of Ery (erythrocytes) and T-cells:



To kickstart the process, `MCView` will automatically cluster the metacells innto some groups, and give a color to each. This makes the markers heatmap at least somewhat informative even before manual type annotation has begun. Note these clusters are just as a very crude starting point. Sometimes a cluster will map well to a single cell type, but most likely, a cell type would contain metacells from several clusters, or a cluster will need to be split between two cell types.

7.2 Updating types in MCView

The `MCView` application is stateless, that is, any type annotations created in it are not persistent. If you close your browser's window, any annotations you made will be lost, unless you first export the types as a CSV file. You can then start a new instance of the application in your browser and import this file back to continue where you left off. We saved a copy of this exported file in `iterative.iteration-2.types.csv` which we'll use below.

You can also export a small convenience file which just contains the mapping between types and colors. In this vignette we'll assume this mapping does not change between the applications. We saved a copy of this file in `captured/type_colors.csv` - in fact, you

will notice that we use this in the `scripts/import_dataset.r` script above when we create applications with type annotations already embedded in them.

These `MCView` mechanisms make it easy for a web server to deal with multiple simultaneous users which may modifying things at the same time. For the web server, the application files are read-only, which is exactly what you want when publishing data for users to see. This even allows the users to work on alternative type annotations without impacting each another.

However, this makes it more difficult for a single user which actively edits the data. The way around it is to do embed the exported file into the application's directory, that is, modify the application's disk files. From that point on, all users that will open the application will see the new type annotations.

We'll do this now, using the following simple script:

```
library("MCView")

args <- commandArgs(trailingOnly=TRUE)
stopifnot(length(args) == 2)
name <- args[1]
types <- args[2]

update_metacell_types(
    sprintf("../mcview/%s", name),                  # The directory
of the app
    sprintf("../hca_bm-%s", gsub("/", "-", name)),  # The name of
the dataset
    types                                           # The types CSV
files
)
```

In [40]:
```
%%time
os.system(
    "Rscript ../scripts/update_types.r iterative/iteration-2 "
    "../captured/iterative.iteration-2.types.csv"
)
```

```
CPU times: user 930 µs, sys: 951 ms, total: 952 ms
Wall time: 4.73 s
✔ Succesfully changed metacell cell type assignments
 File has a field named 'color', updating also cell type colors.
✔ Succesfully changed cell type color assignments
```

Out[40]: 0

8. Compute the 3rd iteration metacells

We'll again rename the data.

In [41]:
```
%%time
mc.ut.set_name(cells, "hca_bm.iteration-3.cells")
print(f"Iteration 3: {cells.n_obs} cells, {cells.n_vars} genes")
```

```
Iteration 3: 297890 cells, 27262 genes
CPU times: user 237 µs, sys: 31 µs, total: 268 µs
Wall time: 243 µs
```

8.1 Capturing previous type annotations

We are about to compute a new set of metacells, so these types we assigned to the old set of metacells can't be directly applied to them. Instead, we will apply the type annotations to the cells, that is, give each cell the type of its (old) metacell; then, after computing the new metacells, we'll give each one the most frequent type of the cells it contains. This is far from perfect, so we'll need to go to `MCView` again and verify and modify these types, but it is an improvement over starting the annotation from scratch.

A tricky point is that some cells are outliers (do not belong in any metacell), so we have to explicitly allow for that when applying the types to the cells.

In [42]:
```
%%time

# We'll reuse this through the iterations.
def capture_type_annotations_from_iteration(previous_iteration_index: int) -
    metacell_types_csv = \
        pd.read_csv(f"../captured/iterative.iteration-{previous_iteration_ir
    type_of_metacell = pd.Series(
        list(metacell_types_csv["cell_type"]) + ["Outliers"],
        index=list(metacell_types_csv["metacell"]) + ["Outliers"]
    )

    previous_metacell_of_cell = cells.obs["metacell_name"]
    type_of_cell = np.array(type_of_metacell[previous_metacell_of_cell])
    mc.ut.set_o_data(cells, f"type.iteration-{previous_iteration_index}.manu
    return type_of_cell

type_of_cell = capture_type_annotations_from_iteration(2)
```

```
set hca_bm.iteration-3.cells.obs[type.iteration-2.manual]: 297890 objects
CPU times: user 48 ms, sys: 3 ms, total: 51 ms
Wall time: 50.6 ms
```

8.2 Decisions

The revised decisions below are based on the analysis we performed in `MCView`.

8.2.1 Removing doublets

Ideally, all doublets should have been removed from even the "full" data we started with. However, in the real world, we often find a few metacells which we think are doublets. Our

best course of action is to remove them from the data, so they will be excluded in the next iteration.

In [43]:
```
%%time
doublet_cells_mask = type_of_cell == "doublets"
print(f"{np.sum(doublet_cells_mask)} cells will be excluded as doublets")
```

```
815 cells will be excluded as doublets
CPU times: user 3.35 ms, sys: 46 µs, total: 3.4 ms
Wall time: 3.16 ms
```

We'll just slice the data to remove the excluded doublet cells.

In [44]:
```
%%time
cells = mc.ut.slice(cells, obs=~doublet_cells_mask, name="hca_bm.cells.3")
print(f"Iteration 3 (~doublets): {cells.n_obs} cells, {cells.n_vars} genes")
```

```
Iteration 3 (~doublets): 297075 cells, 27262 genes
CPU times: user 26.6 s, sys: 24.1 s, total: 50.7 s
Wall time: 50.7 s
```

8.2.2 Adding lateral genes

Even though we went through two iterations, we still discovered more lateral genes, using the same techniques as described above. Such is the nature of the beast.

In [45]:
```
ADD_LATERAL_GENE_NAMES = [
    "APOE", "Acsm3", "CD69", "Cdk4", "EMP3", "FOSB", "Fkbp4", "HIST1H1C", "H
    "HLA-DMA", "HLA-DQA1", "HLA-DRB1", "HLA_DOA", "HSPE1", "ID2", "IGHG1", "
    "IGKV1-12", "IGKV1-39", "IGKV1-5", "IGKV3", "IGKV3-15", "IGKV4-1", "IGLC
    "LINC01206", "PA2G4", "RP11-160E2.6", "RP11-53B5.1", "RP1143G9.4", "RP5-
    "RP5-1171I10.5", "SRGN", "VSIR"
]
```

In [46]:
```
%%time
update_lateral_genes(names=ADD_LATERAL_GENE_NAMES, op="add")
```

```
set hca_bm.cells.3.var[lateral_gene]: 246 true (0.9024%) out of 27262 bools
```

```
['ACSM3', 'ANP32B', 'APOE', 'AURKA', 'B2M', 'BIRC5', 'BTG2', 'CALM1', 'CD6
3', 'CD69', 'CDK4', 'CENPF', 'CENPU', 'CENPW', 'CKS1B', 'CKS2', 'COX4I1', 'D
NAJB1', 'DONSON', 'DUSP1', 'DUT', 'EEF1A1', 'EEF1B2', 'EIF3E', 'EMP3', 'FKBP
4', 'FOS', 'FOSB', 'FTH1', 'G0S2', 'GGH', 'GLTSCR2', 'GMNN', 'GNB2L1', 'H2AF
Z', 'H3F3B', 'HIST1H1C', 'HIST1H2AC', 'HIST1H2BG', 'HIST1H4C', 'HLA-A', 'HLA
-B', 'HLA-C', 'HLA-DMA', 'HLA-DMB', 'HLA-DPA1', 'HLA-DPB1', 'HLA-DQA1', 'HLA
-DQB1', 'HLA-DRA', 'HLA-DRB1', 'HLA-E', 'HLA-F', 'HMGA1', 'HMGB1', 'HMGB2',
'HMGB3', 'HMGN2', 'HNRNPAB', 'HSP90AA1', 'HSP90AB1', 'HSPA1A', 'HSPA1B', 'HS
PA6', 'HSPD1', 'HSPE1', 'HSPH1', 'ID2', 'IER2', 'IGHA1', 'IGHA2', 'IGHD', 'I
GHG1', 'IGHG2', 'IGHG3', 'IGHG4', 'IGHM', 'IGKC', 'IGKV1-12', 'IGKV1-39', 'I
GKV1-5', 'IGKV3-15', 'IGKV4-1', 'IGLC2', 'IGLC3', 'IGLC6', 'IGLC7', 'IGLL1',
'IGLV2-34', 'JUN', 'JUNB', 'KIAA0101', 'LEPROTL1', 'LGALS1', 'LINC01206', 'L
TB', 'MCM3', 'MCM4', 'MCM7', 'MKI67', 'MT2A', 'MYL12A', 'MYL6', 'NASP', 'NUS
AP1', 'PA2G4', 'PCNA', 'PDLIM1', 'PLK3', 'PPP1R15A', 'PTMA', 'PTTG1', 'RAN',
'RANBP1', 'RGCC', 'RP11-160E2.6', 'RP11-53B5.1', 'RP11-620J15.3', 'RP5-1025A
1.3', 'RP5-1171I10.5', 'RRM2', 'SMC4', 'SRGN', 'SRSF7', 'STMN1', 'TK1', 'TMS
B4X', 'TOP2A', 'TPX2', 'TUBA1A', 'TUBA1B', 'TUBB', 'TUBB4B', 'TXN', 'TYMS',
'UBA52', 'UBC', 'UBE2C', 'UHRF1', 'YBX1', 'YPEL5', 'ZFP36', 'ZWINT']
and 103 RP[LS].* genes
CPU times: user 6.36 ms, sys: 4.78 ms, total: 11.1 ms
Wall time: 10.6 ms
```

8.2.3 Adding noisy genes

This time, we haven't found any new genes to mark as noisy. These genes are much easier to detect so they typically stabilize early in the iterations.

8.3 Computation

The steps here just again repeat what we did in the previous iterations. However, this time we can also now automatically assign a type for each new metacell, by picking the most frequent type of the cells in it. We do not expect this to be perfect, but to serve as a more convenient starting point when reviewing the data in `MCView`.

In [47]:
```
%%time

def next_iteration_with_types(next_iteration_index: int) -> None:
    compute_next_iteration(next_iteration_index)

    print("# APPLY PREVIOUS ITERATION TYPES")  # TRICKY: Uses *NEW* metacell
    mc.tl.convey_obs_to_group(
        adata=cells, gdata=metacells,
        property_name=f"type.iteration-{next_iteration_index - 1}.manual",
        to_property_name=f"type.iteration-{next_iteration_index}.auto",
    )

    finalize_next_iteration(next_iteration_index, with_types=True)

next_iteration_with_types(3)
```

```
set hca_bm.cells.3.var[selected_gene]: * -> False
# DIVIDE AND CONQUER...
```

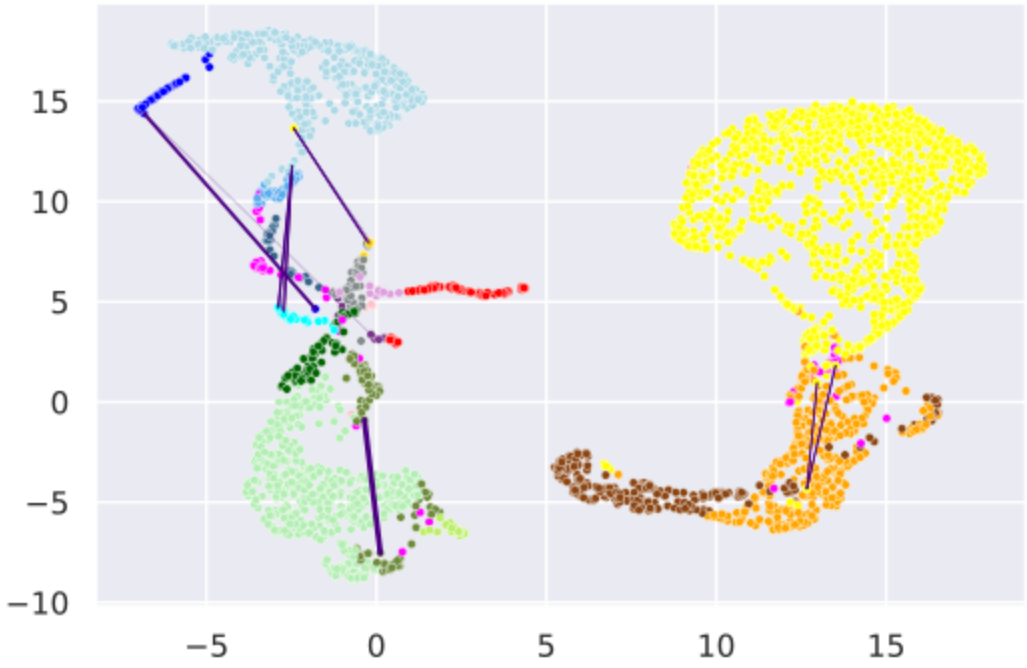
```
set hca_bm.cells.3.var[rare_gene]: 72 true (0.2641%) out of 27262 bools
set hca_bm.cells.3.var[rare_gene_module]: 27190 outliers (99.74%) and 72 gro
uped (0.2641%) out of 27262 int32 elements with 7 groups with mean size 10.2
9
set hca_bm.cells.3.obs[cells_rare_gene_module]: 295644 outliers (99.52%) and
1431 grouped (0.4817%) out of 297075 int32 elements with 7 groups with mean
size 204.4
set hca_bm.cells.3.obs[rare_cell]: 1431 true (0.4817%) out of 297075 bools
set hca_bm.cells.3.var[selected_gene]: 3592 true (13.18%) out of 27262 bools
set hca_bm.cells.3.obs[metacell]: 297075 int32s
set hca_bm.cells.3.obs[dissolved]: 0 true (0%) out of 297075 bools
set hca_bm.cells.3.obs[metacell_level]: 297075 int32s
# COLLECT METACELLS...
set hca_bm.iteration-3.metacells.obs[grouped]: 3157 int64s
set hca_bm.iteration-3.metacells.obs[total_umis]: 3157 float64s
set hca_bm.iteration-3.metacells.layers[total_umis]: ndarray 3157 X 27262 fl
oat32s
set hca_bm.iteration-3.metacells.obs[__zeros_downsample_umis]: 3157 int64s
set hca_bm.iteration-3.metacells.layers[zeros]: ndarray 3157 X 27262 int32s
set hca_bm.cells.3.obs[metacell_name]: 297075 <U8s
set hca_bm.iteration-3.metacells.var[gene_ids]: 27262 objects
set hca_bm.iteration-3.metacells.var[bursty_lonely_gene]: 0 true (0%) out of
27262 bools
set hca_bm.iteration-3.metacells.var[properly_sampled_gene]: 27262 true (10
0%) out of 27262 bools
set hca_bm.iteration-3.metacells.var[excluded_gene]: 0 true (0%) out of 2726
2 bools
set hca_bm.iteration-3.metacells.var[full_gene_index]: 27262 int32s
set hca_bm.iteration-3.metacells.var[lateral_gene]: 246 true (0.9024%) out o
f 27262 bools
set hca_bm.iteration-3.metacells.var[lateral_genes_module]: 27262 int32s
set hca_bm.iteration-3.metacells.var[selected_gene]: 3592 true (13.18%) out
of 27262 bools
set hca_bm.iteration-3.metacells.var[rare_gene]: 72 true (0.2641%) out of 27
262 bools
set hca_bm.iteration-3.metacells.var[rare_gene_module]: 27262 int32s
set hca_bm.iteration-3.metacells.var[noisy_gene]: 21 true (0.07703%) out of
27262 bools
set hca_bm.iteration-3.metacells.obs[metacells_rare_gene_module]: 3157 int32
s
set hca_bm.iteration-3.metacells.obs[rare_metacell]: 17 true (0.5385%) out o
f 3157 bools
set hca_bm.iteration-3.metacells.uns[outliers]: 154
set hca_bm.iteration-3.metacells.uns[metacells_algorithm]: metacells.0.9.0-d
ev.1
Iteration 3: 3157 metacells, 27262 genes
# CONVEY CELL ANNOTATIONS...
```

```
set hca_bm.iteration-3.metacells.obs[sex]: 3157 float32s
set hca_bm.iteration-3.metacells.obs[age]: 3157 float32s
set hca_bm.iteration-3.metacells.obs[sex_fraction_of_female]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[sex_fraction_of_male]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_26.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_29.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_32.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_36.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_39.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_50.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[age_fraction_of_52.0]: 3157 float64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM1]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM2]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM3]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM4]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM5]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM6]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM7]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[donor_fraction_of_MantonBM8]: 3157 floa
t64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_7]: 3
```

```
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_4]: 3
```

```
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_8]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_1]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_2]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_3]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_4]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_5]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_6]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_7]: 3
157 float64s
set hca_bm.iteration-3.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_8]: 3
157 float64s
# APPLY PREVIOUS ITERATION TYPES
set hca_bm.iteration-3.metacells.obs[type.iteration-3.auto]: 3157 <U9s
set hca_bm.iteration-3.metacells.uns[mcview_format]: 1.0
set hca_bm.iteration-3.metacells.uns[outliers]: 154
# COMPUTE FOR MCVIEW...
```

```
set hca_bm.iteration-3.metacells.var[marker_gene]: 3463 true (12.7%) out of
27262 bools
set hca_bm.iteration-3.metacells.obsp[obs_balanced_ranks]: 38527 nonzero (0.
3866%) out of 9966649 elements
set hca_bm.iteration-3.metacells.obsp[obs_pruned_ranks]: 12601 nonzero (0.12
64%) out of 9966649 elements
set hca_bm.iteration-3.metacells.obsp[obs_outgoing_weights]: 12601 nonzero
(0.1264%) out of 9966649 elements
set hca_bm.iteration-3.metacells.obsp[umap_distances]: csr_matrix 3157 X 315
7 float32s (9963492 > 0, 99.97%)
set hca_bm.iteration-3.metacells.obs[u]: 3157 float32s
set hca_bm.iteration-3.metacells.obs[v]: 3157 float32s
set hca_bm.iteration-3.metacells.obs[w]: 3157 float32s
set hca_bm.iteration-3.metacells.obsp[obs_balanced_ranks]: 38527 nonzero (0.
3866%) out of 9966649 elements
set hca_bm.iteration-3.metacells.obsp[obs_pruned_ranks]: 12601 nonzero (0.12
64%) out of 9966649 elements
set hca_bm.iteration-3.metacells.obsp[obs_outgoing_weights]: 12601 nonzero
(0.1264%) out of 9966649 elements
set hca_bm.iteration-3.metacells.obsp[umap_distances]: csr_matrix 3157 X 315
7 float32s (9963492 > 0, 99.97%)
set hca_bm.iteration-3.metacells.obs[x]: 3157 float32s
set hca_bm.iteration-3.metacells.obs[y]: 3157 float32s
set hca_bm.cells.3.obs[most_similar]: 297075 int32s
set hca_bm.cells.3.layers[deviant_fold]: csr_matrix 297075 X 27262 float32s
(175490330 > 0, 2.167%)
set hca_bm.iteration-3.metacells.layers[inner_fold]: csr_matrix 3157 X 27262
float32s (1794298 > 0, 2.085%)
set hca_bm.iteration-3.metacells.var[significant_inner_folds_count]: 27262 i
nt64s with mean 0.8215
set hca_bm.iteration-3.metacells.layers[inner_stdev_log]: csr_matrix 3157 X
27262 float32s (1751231 > 0, 2.035%)
set hca_bm.iteration-3.metacells.varp[var_similarity]: csr_matrix 27262 X 27
262 float32s (1317236 > 0, 0.1772%)
# PLOT UMAP...
```



```
# SAVE CELLS...
# SAVE METACELLS...
# IMPORT TO MCVIEW...
```

```
 creating ../mcview/iterative/iteration-3
→ You can edit the app configuration at '../mcview/iterative/iteration-3/con
fig/config.yaml'
 Importing hca_bm-iterative-iteration-3
 Reading '../output/iterative/iteration-3/hca_bm.metacells.h5ad'
 Processing metacell matrix
 Processing 2d projection
 Calculating top genes per metacell (marker genes)
 Calculating metacell correlations of default marker genes
 Processing inner-folds matrix
 Calculating top inner-fold genes
→ Added the Inner-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-3/config/
config.yaml'
 Processing inner-stdev matrix
 Calculating top inner-stdev genes
→ Added the Stdev-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-3/config/
config.yaml'
 Clustering in order to get initial annotation.
 using 164 genes
 clustering k = 64
 number of features = 164
 Generating cell type colors using chameleon package.
 Coloring using pre-calculated 3D umap
 Clustering in order to get gene modules. k = 31
 Number of genes considered = 1000
 Loading previously calculated 30 correlated and anti-correlated genes for
each gene
✔ hca_bm-iterative-iteration-3 dataset imported succesfully to '../mcview/it
erative/iteration-3' project
• You can now run the app using: run_app("../mcview/iterative/iteration-3")
• or create a bundle using: create_bundle("../mcview/iterative/iteration-3",
name = "name_of_bundle")
CPU times: user 1h 45min, sys: 9min 15s, total: 1h 54min 15s
Wall time: 17min 4s
```

9. Computing 4th iteration metacells

We'll do another iteration, similar to the previous one. Depending on the intended usage, it may take many iterations until the results are "good enough".

In [48]:
```
%%time
mc.ut.set_name(cells, "hca_bm.iteration-4.cells")
print(f"Iteration 4: {cells.n_obs} cells, {cells.n_vars} genes")
type_of_cell = capture_type_annotations_from_iteration(3)
```

```
set hca_bm.iteration-4.cells.obs[type.iteration-3.manual]: 297075 objects
```

```
Iteration 4: 297075 cells, 27262 genes
CPU times: user 49.1 ms, sys: 9.02 ms, total: 58.1 ms
Wall time: 76.2 ms
```

9.1 Decisions

The revised decisions below are based on the analysis we performed in `MCView`, similar to what we did in the previous iterations.

9.1.1 Removing excluded genes

This time we noticed `NEAT1` is causing problems, and upon consideration, decided it is not sufficient to mark it as lateral. Instead we want to exclude it from the data, so it will not impact the denominator (total UMIs in each cell), as it is a very high expressing gene.

In general we care about the fraction of the UMIs of each gene out of the total for each cell. A very highly expressed gene may therefore cause other genes to appear to be unner-expressed. This is part of the reason why, for example, we excluded the `MT-.*` mytochondrial genes from the analysis. One should be careful though because this is a very blunt tool to apply to the data.

In [49]:
```
EXCLUDED_GENE_NAMES = [ "NEAT1" ]
```

We'll just compute a mask of the additional (well, single) excluded gene and apply it below.

In [50]:
```
%%time
excluded_genes_mask = np.array(mc.tl.find_named_genes(cells, names=EXCLUDED_
print(f"{np.sum(excluded_genes_mask)} genes will be excluded")
```

```
1 genes will be excluded
CPU times: user 6.33 ms, sys: 927 µs, total: 7.26 ms
Wall time: 7.03 ms
```

9.1.2 Removing doublets

We discovered a few more doublets. Blood data is notorious for containing doublets because blood cells are "sticky".

In [51]:
```
%%time
doublet_cells_mask = type_of_cell == "doublets"
print(f"{np.sum(doublet_cells_mask)} cells will be excluded as doublets")
```

```
382 cells will be excluded as doublets
CPU times: user 3.45 ms, sys: 0 ns, total: 3.45 ms
Wall time: 3.19 ms
```

We'll again simply slice the set of cells and genes we use next:

In [52]:
```
%%time
cells = mc.ut.slice(cells, obs=~doublet_cells_mask, vars=~excluded_genes_mas
```

```
print(f"Iteration 4 (~doubles, ~genes): {cells.n_obs} cells, {cells.n_vars}
```

```
Iteration 4 (~doubles, ~genes): 296693 cells, 27261 genes
CPU times: user 28.7 s, sys: 34.1 s, total: 1min 2s
Wall time: 1min 2s
```

9.1.3 Adding lateral genes

Even though we went through three iterations so far, we still discovered more lateral genes. This only *appears* to be an endless process, though; it *does* converge. Eventually.

In [53]:
```
ADD_LATERAL_GENE_NAMES = [
    "CH17-373J23.1", "CXCR4", "GPR183", "HBM", "IGLL5", "NFKBIA", "TSC22D3",
    "RGS1", "RGS2", "RGS3",
]
```

In [54]:
```
%%time
update_lateral_genes(names=ADD_LATERAL_GENE_NAMES, op="add")
```

```
set hca_bm.cells.4.var[lateral_gene]: 257 true (0.9427%) out of 27261 bools
['ACSM3', 'ANP32B', 'APOE', 'AURKA', 'B2M', 'BIRC5', 'BTG2', 'CALM1', 'CD6
3', 'CD69', 'CDK4', 'CENPF', 'CENPU', 'CENPW', 'CH17-373J23.1', 'CKS1B', 'CK
S2', 'COX4I1', 'CXCR4', 'DNAJB1', 'DONSON', 'DUSP1', 'DUT', 'EEF1A1', 'EEF1B
2', 'EIF3E', 'EMP3', 'FKBP4', 'FOS', 'FOSB', 'FTH1', 'G0S2', 'GGH', 'GLTSCR
2', 'GMNN', 'GNB2L1', 'GPR183', 'H2AFZ', 'H3F3B', 'HBM', 'HIST1H1C', 'HIST1H
2AC', 'HIST1H2BG', 'HIST1H4C', 'HLA-A', 'HLA-B', 'HLA-C', 'HLA-DMA', 'HLA-DM
B', 'HLA-DPA1', 'HLA-DPB1', 'HLA-DQA1', 'HLA-DQB1', 'HLA-DRA', 'HLA-DRB1',
'HLA-E', 'HLA-F', 'HMGA1', 'HMGB1', 'HMGB2', 'HMGB3', 'HMGN2', 'HNRNPAB', 'H
SP90AA1', 'HSP90AB1', 'HSPA1A', 'HSPA1B', 'HSPA6', 'HSPD1', 'HSPE1', 'HSPH
1', 'ID2', 'IER2', 'IGHA1', 'IGHA2', 'IGHD', 'IGHG1', 'IGHG2', 'IGHG3', 'IGH
G4', 'IGHM', 'IGKC', 'IGKV1-12', 'IGKV1-39', 'IGKV1-5', 'IGKV3-15', 'IGKV4-
1', 'IGLC2', 'IGLC3', 'IGLC6', 'IGLC7', 'IGLL1', 'IGLL5', 'IGLV2-34', 'JUN',
'JUNB', 'KIAA0101', 'LEPROTL1', 'LGALS1', 'LINC01206', 'LTB', 'MCM3', 'MCM
4', 'MCM7', 'MKI67', 'MT2A', 'MYL12A', 'MYL6', 'NASP', 'NFKBIA', 'NUSAP1',
'PA2G4', 'PCNA', 'PDLIM1', 'PLK3', 'PPP1R15A', 'PTMA', 'PTTG1', 'RAN', 'RANB
P1', 'RGCC', 'RGS1', 'RGS2', 'RGS3', 'RP11-1143G9.4', 'RP11-160E2.6', 'RP11-
53B5.1', 'RP11-620J15.3', 'RP5-1025A1.3', 'RP5-1171I10.5', 'RRM2', 'SMC4',
'SRGN', 'SRSF7', 'STMN1', 'TK1', 'TMSB4X', 'TOP2A', 'TPX2', 'TSC22D3', 'TUBA
1A', 'TUBA1B', 'TUBB', 'TUBB4B', 'TXN', 'TYMS', 'UBA52', 'UBC', 'UBE2C', 'UH
RF1', 'YBX1', 'YPEL5', 'ZFP36', 'ZWINT']
and 103 RP[LS].* genes
CPU times: user 6.73 ms, sys: 5.09 ms, total: 11.8 ms
Wall time: 11.3 ms
```

9.2 Computation

The steps here just again repeat what we did in the previous iterations.

In [55]:
```
%%time
next_iteration_with_types(4)
```

```
set hca_bm.cells.4.var[selected_gene]: * -> False
# DIVIDE AND CONQUER...
```

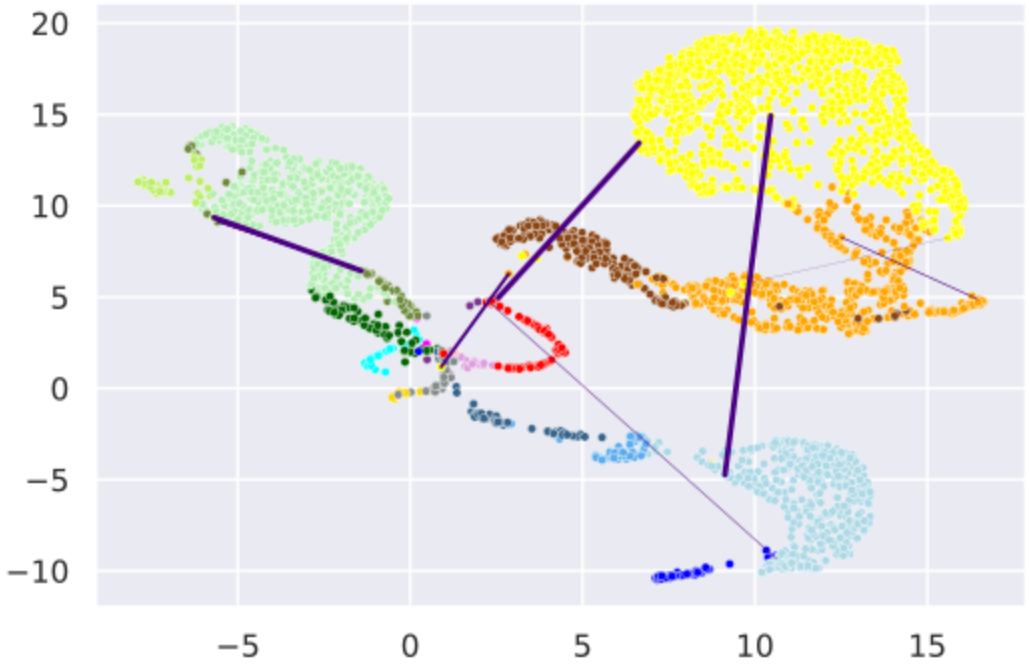
```
set hca_bm.cells.4.var[rare_gene]: 73 true (0.2678%) out of 27261 bools
set hca_bm.cells.4.var[rare_gene_module]: 27188 outliers (99.73%) and 73 gro
uped (0.2678%) out of 27261 int32 elements with 7 groups with mean size 10.4
3
set hca_bm.cells.4.obs[cells_rare_gene_module]: 295318 outliers (99.54%) and
1375 grouped (0.4634%) out of 296693 int32 elements with 7 groups with mean
size 196.4
set hca_bm.cells.4.obs[rare_cell]: 1375 true (0.4634%) out of 296693 bools
set hca_bm.cells.4.var[selected_gene]: 3547 true (13.01%) out of 27261 bools
set hca_bm.cells.4.obs[metacell]: 296693 int32s
set hca_bm.cells.4.obs[dissolved]: 0 true (0%) out of 296693 bools
set hca_bm.cells.4.obs[metacell_level]: 296693 int32s
# COLLECT METACELLS...
set hca_bm.iteration-4.metacells.obs[grouped]: 3149 int64s
set hca_bm.iteration-4.metacells.obs[total_umis]: 3149 float64s
set hca_bm.iteration-4.metacells.layers[total_umis]: ndarray 3149 X 27261 fl
oat32s
set hca_bm.iteration-4.metacells.obs[__zeros_downsample_umis]: 3149 int64s
set hca_bm.iteration-4.metacells.layers[zeros]: ndarray 3149 X 27261 int32s
set hca_bm.cells.4.obs[metacell_name]: 296693 <U8s
set hca_bm.iteration-4.metacells.var[gene_ids]: 27261 objects
set hca_bm.iteration-4.metacells.var[bursty_lonely_gene]: 0 true (0%) out of
27261 bools
set hca_bm.iteration-4.metacells.var[properly_sampled_gene]: 27261 true (10
0%) out of 27261 bools
set hca_bm.iteration-4.metacells.var[excluded_gene]: 0 true (0%) out of 2726
1 bools
set hca_bm.iteration-4.metacells.var[full_gene_index]: 27261 int32s
set hca_bm.iteration-4.metacells.var[lateral_gene]: 257 true (0.9427%) out o
f 27261 bools
set hca_bm.iteration-4.metacells.var[lateral_genes_module]: 27261 int32s
set hca_bm.iteration-4.metacells.var[selected_gene]: 3547 true (13.01%) out
of 27261 bools
set hca_bm.iteration-4.metacells.var[rare_gene]: 73 true (0.2678%) out of 27
261 bools
set hca_bm.iteration-4.metacells.var[rare_gene_module]: 27261 int32s
set hca_bm.iteration-4.metacells.var[noisy_gene]: 21 true (0.07703%) out of
27261 bools
set hca_bm.iteration-4.metacells.obs[metacells_rare_gene_module]: 3149 int32
s
set hca_bm.iteration-4.metacells.obs[rare_metacell]: 17 true (0.5399%) out o
f 3149 bools
set hca_bm.iteration-4.metacells.uns[outliers]: 140
set hca_bm.iteration-4.metacells.uns[metacells_algorithm]: metacells.0.9.0-d
ev.1
Iteration 4: 3149 metacells, 27261 genes
# CONVEY CELL ANNOTATIONS...
```

```
set hca_bm.iteration-4.metacells.obs[sex]: 3149 float32s
set hca_bm.iteration-4.metacells.obs[age]: 3149 float32s
set hca_bm.iteration-4.metacells.obs[sex_fraction_of_female]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[sex_fraction_of_male]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_26.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_29.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_32.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_36.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_39.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_50.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[age_fraction_of_52.0]: 3149 float64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM1]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM2]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM3]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM4]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM5]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM6]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM7]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[donor_fraction_of_MantonBM8]: 3149 floa
t64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM1_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_7]: 3
```

```
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM2_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM3_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM4_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM5_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_4]: 3
```

```
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM6_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM7_HiSeq_8]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_1]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_2]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_3]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_4]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_5]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_6]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_7]: 3
149 float64s
set hca_bm.iteration-4.metacells.obs[batch_fraction_of_MantonBM8_HiSeq_8]: 3
149 float64s
# APPLY PREVIOUS ITERATION TYPES
set hca_bm.iteration-4.metacells.obs[type.iteration-4.auto]: 3149 <U9s
set hca_bm.iteration-4.metacells.uns[mcview_format]: 1.0
set hca_bm.iteration-4.metacells.uns[outliers]: 140
# COMPUTE FOR MCVIEW...
```

```
set hca_bm.iteration-4.metacells.var[marker_gene]: 3334 true (12.23%) out of
27261 bools
set hca_bm.iteration-4.metacells.obsp[obs_balanced_ranks]: 38473 nonzero (0.
388%) out of 9916201 elements
set hca_bm.iteration-4.metacells.obsp[obs_pruned_ranks]: 12579 nonzero (0.12
69%) out of 9916201 elements
set hca_bm.iteration-4.metacells.obsp[obs_outgoing_weights]: 12579 nonzero
(0.1269%) out of 9916201 elements
set hca_bm.iteration-4.metacells.obsp[umap_distances]: csr_matrix 3149 X 314
9 float32s (9913050 > 0, 99.97%)
set hca_bm.iteration-4.metacells.obs[u]: 3149 float32s
set hca_bm.iteration-4.metacells.obs[v]: 3149 float32s
set hca_bm.iteration-4.metacells.obs[w]: 3149 float32s
set hca_bm.iteration-4.metacells.obsp[obs_balanced_ranks]: 38473 nonzero (0.
388%) out of 9916201 elements
set hca_bm.iteration-4.metacells.obsp[obs_pruned_ranks]: 12579 nonzero (0.12
69%) out of 9916201 elements
set hca_bm.iteration-4.metacells.obsp[obs_outgoing_weights]: 12579 nonzero
(0.1269%) out of 9916201 elements
set hca_bm.iteration-4.metacells.obsp[umap_distances]: csr_matrix 3149 X 314
9 float32s (9913050 > 0, 99.97%)
set hca_bm.iteration-4.metacells.obs[x]: 3149 float32s
set hca_bm.iteration-4.metacells.obs[y]: 3149 float32s
set hca_bm.cells.4.obs[most_similar]: 296693 int32s
set hca_bm.cells.4.layers[deviant_fold]: csr_matrix 296693 X 27261 float32s
(174380391 > 0, 2.156%)
set hca_bm.iteration-4.metacells.layers[inner_fold]: csr_matrix 3149 X 27261
float32s (1789179 > 0, 2.084%)
set hca_bm.iteration-4.metacells.var[significant_inner_folds_count]: 27261 i
nt64s with mean 0.87
set hca_bm.iteration-4.metacells.layers[inner_stdev_log]: csr_matrix 3149 X
27261 float32s (1745855 > 0, 2.034%)
set hca_bm.iteration-4.metacells.varp[var_similarity]: csr_matrix 27261 X 27
261 float32s (1315226 > 0, 0.177%)
# PLOT UMAP...
```



```
# SAVE CELLS...
# SAVE METACELLS...
# IMPORT TO MCVIEW...
 creating ../mcview/iterative/iteration-4
→ You can edit the app configuration at '../mcview/iterative/iteration-4/con
fig/config.yaml'
 Importing hca_bm-iterative-iteration-4
 Reading '../output/iterative/iteration-4/hca_bm.metacells.h5ad'
 Processing metacell matrix
 Processing 2d projection
 Calculating top genes per metacell (marker genes)
 Calculating metacell correlations of default marker genes
 Processing inner-folds matrix
 Calculating top inner-fold genes
→ Added the Inner-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-4/config/
config.yaml'
 Processing inner-stdev matrix
 Calculating top inner-stdev genes
→ Added the Stdev-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/iteration-4/config/
config.yaml'
 Clustering in order to get initial annotation.
 using 163 genes
 clustering k = 64
 number of features = 163
 Generating cell type colors using chameleon package.
 Coloring using pre-calculated 3D umap
 Clustering in order to get gene modules. k = 31
 Number of genes considered = 1000
 Loading previously calculated 30 correlated and anti-correlated genes for
each gene
✔ hca_bm-iterative-iteration-4 dataset imported succesfully to '../mcview/it
erative/iteration-4' project
• You can now run the app using: run_app("../mcview/iterative/iteration-4")
• or create a bundle using: create_bundle("../mcview/iterative/iteration-4",
name = "name_of_bundle")
CPU times: user 1h 42min 12s, sys: 8min 40s, total: 1h 50min 52s
Wall time: 16min 14s
```

10. Finalizing the data

At some point, one must declare the results to be "good enough" and wrap things up. This point "depends".

If one is working on a very large data set, with the intent to publishing it as an atlas for other researches to use as reference, one would (presumably) invest much more effort ensuring the quality of metacells and type annotations for each of the cell types.

In contrast, if one is working on a small data set, and is investigating a specific biological question (e.g. comparing a specific cell type between wild type and knockout organisms),

one can focus the effort on specific cell types, and use coarse grained annotations for the rest (and worry much less about their quality in general).

Either way, the decision of when "enough is enough" will depend on the analyst's judgement, just like every other decision we have discussed so far.

In this specific example, the purpose of the vignette is merely to demonstrate the iterative process. We therefore decided to stop here. To remove doubt, the results presented here are **not** suitable for use in any serious analysis. This is an example of the process, nothing more.

All that said, wrapping things up requires taking a few extra steps we'll discuss below. We'll start by renaming the data to indicate it is the final result.

In [56]:
```
%%time
mc.ut.set_name(cells, "hca_bm.iterative-final.cells")
print(f"Final: {cells.n_obs} cells, {cells.n_vars} genes")
mc.ut.set_name(metacells, "hca_bm.iterative-final.cells")
print(f"Final: {metacells.n_obs} metacells, {metacells.n_vars} genes")
```

```
Final: 296693 cells, 27261 genes
Final: 3149 metacells, 27261 genes
CPU times: user 219 µs, sys: 24 µs, total: 243 µs
Wall time: 226 µs
```

10.1 Conveying type annotations

Because we will not be recomputing the metacells this time, we can apply the final type annotations from our work in MCView to both the cells (as in previous iterations) and the metacells.

In [57]:
```
%%time
metacell_types_csv = pd.read_csv("../captured/iterative.iteration-4.types.cs
assert np.all(metacell_types_csv["metacell"] == metacells.obs_names)

type_of_metacell = np.array(metacell_types_csv["cell_type"])
mc.ut.set_o_data(metacells, "type", type_of_metacell)

type_of_metacell = pd.Series(
    list(type_of_metacell) + ["Outliers"],
    index=list(metacell_types_csv["metacell"]) + ["Outliers"]
)

metacell_of_cell = cells.obs["metacell_name"]
type_of_cell = np.array(type_of_metacell[metacell_of_cell])
mc.ut.set_o_data(cells, "type", type_of_cell)
```

```
set hca_bm.iterative-final.cells.obs[type]: 3149 objects
set hca_bm.iterative-final.cells.obs[type]: 296693 objects
CPU times: user 62.4 ms, sys: 8.85 ms, total: 71.3 ms
Wall time: 82.4 ms
```

10.2 Marking essential genes

If we are preparing a reference atlas, that is, we expect to project other data into it (a process which is the subject of a separate vignette TODO), then it is recommended to designate a few genes as "essential" for each cell type.

When we will project data onto the atlas, we'll do a lot of quality assurance to ensure the projection makes biological sense. The (optional) essential genes help with this quality assurance. Specifically, if a gene is designated as essential for some cell type, and some projected data "seems like" it is of this cell type, we'll ensure that this data agrees with the atlas for the essential genes. For example, if the projected metacell does not express CD8, then we can't say it is a CD8 T-cell. It may be very *similar* to a CD8 T-cell, but we just can't reasonably that it *is* one.

Selecting the (few) genes to use as essenntial for each cell type requires, as usual, the expertise of the biologist preparing the atlas. Note that all we require is a list of genes, *not* some quantitative threshold to use as a cell type classifier. The quantitative aspect is automatically provided by the projection algorithm itself (which is outside the scope of this vignette).

In [58]:
```
#mc.ut.mark_essential_genes(
#    metacells,
#    essential_gene_names_of_types={ "CD8 T-cell" => ["CD8", ...], ...}
#)
```

10.3 Removing doublet meta/cells

Some of the metacells were annotated as doublets. In a realistic iterative process, we would exclude the cells of these metacells, re-compute the metacells, and re-annotate them. Since we artificially stopped at this point in the process, we'll skip these steps, leaving the doublets in these so-called "final" results.

10.4 Spit and polish in general

This is a good place to delete, rename, compute and in general ensure all per-gene and per-metacell properties exist, are properly named, etc. Again, how much effort (if any) is put into this varies greatly depending on the specific scennario.

Just as an example, here we'll delete the type annotations from the previous iterations from the data, because they will only confuse anyone using the final result.

In [59]:
```
%%time

for iteration_index in range(100):  # Complete overkill
    for name in (
```

```
        f"type.iteration-{iteration_index}.auto",
        f"type.iteration-{iteration_index}.manual"
    ):
        for adata in (cells, metacells):
            if name in adata.obs:
                print(f"Delete {name} from {mc.ut.get_name(cells)}")
                del adata.obs[name]
```

```
Delete type.iteration-2.manual from hca_bm.iterative-final.cells
Delete type.iteration-3.manual from hca_bm.iterative-final.cells
Delete type.iteration-4.auto from hca_bm.iterative-final.cells
CPU times: user 350 µs, sys: 2.04 ms, total: 2.39 ms
Wall time: 2.15 ms
```

10.5 Computing for MCView

We've already computed all the `MCView` data. Yes, we have changed the assignment of types to metacells since, but none of the computed data actually depends on these types, so there's no need to compute it again.

10.6 Saving the final results

We'll save the results in some location that makes it clear that this is the final data. In a real process, this would be what we'd publish out of it as a "metacells atlas". In fact, typically such an atlas wouldn't even contain the cells data, but we store it for completeness, and in case "somethings comes up" and we have to do a few more iterations.

In [60]:
```
%%time

print("# SAVE CELLS...")
cells.write_h5ad("../output/iterative/final/hca_bm.cells.h5ad")

print("# SAVE METACELLS...")
metacells.write_h5ad("../output/iterative/final/hca_bm.metacells.h5ad")

print("# IMPORT INTO MCVIEW...")
os.system("Rscript ../scripts/import_dataset.r hca_bm iterative/final 'HCABM
```

```
# SAVE CELLS...
# SAVE METACELLS...
# IMPORT INTO MCVIEW...
```

```
 creating ../mcview/iterative/final
→ You can edit the app configuration at '../mcview/iterative/final/config/co
nfig.yaml'
 Importing hca_bm-iterative-final
 Reading '../output/iterative/final/hca_bm.metacells.h5ad'
 Processing metacell matrix
 Processing 2d projection
 Calculating top genes per metacell (marker genes)
 Calculating metacell correlations of default marker genes
 Processing inner-folds matrix
 Calculating top inner-fold genes
→ Added the Inner-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/final/config/confi
g.yaml'
 Processing inner-stdev matrix
 Calculating top inner-stdev genes
→ Added the Stdev-fold tab to the config file. To change the tab order or re
move it - edit the tabs section at: '../mcview/iterative/final/config/confi
g.yaml'
 Taking cell type annotations from type field in the anndata object
 Loading cell type color annotations from '../captured/type_colors.csv'
 Clustering in order to get gene modules. k = 31
 Number of genes considered = 1000
 Loading previously calculated 30 correlated and anti-correlated genes for
each gene
✔ hca_bm-iterative-final dataset imported succesfully to '../mcview/iterativ
e/final' project
• You can now run the app using: run_app("../mcview/iterative/final")
• or create a bundle using: create_bundle("../mcview/iterative/final", name
=
"name_of_bundle")
CPU times: user 1.35 s, sys: 18.8 s, total: 20.1 s
Wall time: 2min 21s
```

Out[60]: 0

That's it. We now have our final results, both as `h5ad` files and as an `MCView` application, to use as we see fit.

To remove doubt, the results presented here are **not** suitable for use in any serious analysis. This is an example of the process, nothing more.